Dell PowerFlex 4.5.x

Install and Upgrade Guide

July 2024
Rev. 1.4

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

The Dell PowerFlex 4.5.x Install and Upgrade Guide describes how to deploy a new instance of PowerFlex 4.5.x and instructions to upgrade existing 3.6.x or 4.0.x PowerFlex deployments to 4.5.x.

Use this guide only for PowerFlex software installations and upgrades. PowerFlex rack and PowerFlex appliance customers should not use this document for installations or upgrades. For PowerFlex rack documentation, go to PowerFlex rack technical documentation. For PowerFlex appliance documentation, go to PowerFlex appliance documentation.

To deploy PowerFlex 4.5.x for the first time, go to the *Install* section and do the following:

- Select deployment type
- Configure PowerFlex Manager
- Deploy additional storage or file nodes

To upgrade an existing PowerFlex 4.0.x or 4.5 system, go to the *Upgrading PowerFlex from 4.0.2, 4.5 to 4.5.x* section and do the following:

- Upgrade PowerFlex Manager to 4.5.x
- Upgrade PowerFlex storage
- Upgrade SDCs

To upgrade an existing PowerFlex 3.6.x system, go to the *Upgrading to PowerFlex 4.5.x from PowerFlex 3.6* section and do the following:

- Deploy PowerFlex Manager
- Import existing PowerFlex 3.6.x system
- Upgrade PowerFlex system to PowerFlex 4.5.x
- Upgrade SDCs

The target audience for this document includes Dell Technologies customers and services.

For additional PowerFlex software documentation, go to PowerFlex software technical documentation.

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Revision history

Table 1. Revisions

Date	Document revision	Description of changes
July 2024	1.4	Updated: • Deploying PowerFlex file nodes
March 2024	1.3	Added new information: • Troubleshooting a PowerFlex management platform deployment Updated: • Deploying the PowerFlex management platform • Deploy the PowerFlex management platform cluster • Modify the RKE2 server service • Run script on host • Configure the PowerFlex management platform installer networking interface • Configure the NTP service • Configure the PowerFlex management platform networking interface
January 2024	1.2	Technical and editorial updates
November 2023	1.1	Technical and editorial updates
September 2023	1.0	Initial release

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Install

This section describes how to install and configure PowerFlex Manager, deploy PowerFlex storage, and deploy PowerFlex file nodes.

PowerFlex Manager can be installed in three ways:

- VMware ESXi environment: Three VMs running a management virtual machine cluster. PowerFlex Manager is deployed using a fourth VM acting as an installer VM.
- A co-residency environment: PowerFlex Manager is installed on PowerFlex storage-only nodes.
- Linux management environment: The PowerFlex management platform is installed on the Linux operating system directly.

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Installation overview and prerequisites

Installation workflow

Installing PowerFlex on physical servers involves the following activities:

1. Complete the installation checklist to prepare your environment for PowerFlex Manager.
2. Install and configure PowerFlex Manager.
3. Install the PowerFlex components.
4. For a two-layer system configuration, install storage data clients (SDCs) on external servers.
5. Install the PowerFlex license.
6. Perform any necessary post-installation tasks.
7. Move the PowerFlex system to production.

For PowerFlex custom node or VxFlex Ready Node, see the product *Hardware Configuration and Operating System Installation Guide* and *VxFlex Ready Node deployment overview* available at Dell Technologies Support site for step-by-step installation instructions on how to prepare the server and operating system before installing PowerFlex.

Supported configurations

The following PowerFlex configurations are supported:

Table 2. Supported PowerFlex configurations

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (or nodes) required	CPU resources required (per node)	Memory resources required (per node)	Disk resources required (per node)
Pre-existing VMware ESXi management infrastructure	• PowerFlex appliance • PowerFlex custom node • VxFlex Ready Node • PowerEdge servers • Non Dell Technologies hardware	PowerFlex Manager is a management cluster created on three virtual machines. Use the installer to set up management virtual machines (MVMs) Use a custom OVA file to deploy the installer and management virtual machines. Download the OVA file from the Dell Technologies Support site. The hypervisor running the VM must meet the minimum system requirements. You are responsible for the hypervisor hardware and networking requirements. Use the installer to deploy the management virtual machines. The installer can be a VM or physical node.	One installer node (VM or physical node) and three VMs for the management virtual machines	14 cores	32 GB RAM	600 GB

Table 2. Supported PowerFlex configurations (continued)

PowerFlex Manager deployment configuration	Offered on	Description	Minimum number of VMs (or nodes) required	CPU resources required (per node)	Memory resources required (per node)	Disk resources required (per node)
Co-resident on PowerFlex storage-only nodes	• PowerFlex custom node • VxFlex Ready Node • PowerEdge servers • Non Dell Technologies hardware	PowerFlex storage-only nodes in a protection domain run the system management. The eSLES image cannot be installed on the nodes and you must set up the hardware, operating system, and operating system networking.	Four physical nodes	14 cores	32 GB RAM	600 GB
Pre-existing Linux management infrastructure	• PowerFlex custom node • VxFlex Ready Node • PowerEdge servers • Non-Dell Technologies hardware	The eSLES image cannot be installed on the nodes and you must set up the hardware, operating system, and operating system networking. The hardware must meet the minimum system requirements.	Three physical nodes with a supported operating system to host the PowerFlex Management Platform	14 cores	32 GB RAM	600 GB

Installation checklist

This section contains a checklist of items required for installing PowerFlex. Ensure that you have done everything on this list before proceeding then perform the remaining preparation tasks covered in this guide.

Table 3. Checklist for installing PowerFlex

Item	Checked
General	
All servers intended for hosting PowerFlex components and PowerFlex Manager must meet system requirements. **NOTE:** The Dell-supplied hardware configurations must satisfy all the hardware requirements.	-
Secure boot on servers is not supported. Ensure that secure boot is disabled in the BIOS of each server.	-
All nodes must have unique hostnames.	-
To use secure authentication mode, ensure that the latest OpenSSL 64-bit v1.0.2 or v1.1.1 is installed on all servers in the system.	-
Ensure that you know the following: • Primary and secondary MDM IP addresses • Primary MDM admin username and password • Primary MDM operating system hostname and password • LIA password	-
VMware ESXi	
Ensure that the required network interfaces and IP addresses are configured.	-
Networking	
Follow the networking requirement covered in the Dell PowerFlex 4.5.x Technical Overview.	-

Table 3. Checklist for installing PowerFlex (continued)

Item	Checked
Powerflex Manager and its installer node	
The nodes that host PowerFlex Manager must have connectivity to both data and management PowerFlex networks. All management nodes should be on the same subnet and SSH should be enabled.	-
Prepare the following nodes for installing PowerFlex Manager ● PowerFlex Manager installer node: ○ If the PowerFlex Manager installer is on a bare-metal node, install Docker version 20.10.0 or higher. ○ If you are using PowerFlex Manager installer on an existing VMware ESXi management infrastructure, deploy the provided installer OVA (SUSE Linux-based). ● Three management nodes: ○ These nodes must have IP addresses and hostnames that are unique and are on the same subnet. ○ Hostname labels may contain only the ASCII letters 'a' through 'z' (in a case-insensitive manner), the digits '0' through '9', and the hyphen '-'. ○ The PowerFlex management platform can be deployed in the following modes: ■ One dedicated management node ■ Three dedicated management nodes with VMware ESXi or bring your own Linux dedicated management nodes ■ Co-resident with storage or Linux hyperconverged infrastructure nodes	-

Networking

PowerFlex network types and definitions:

Table 4. PowerFlex network types and definitions

Network type	Used
General-purpose LAN	To access network resources for basic networking activities
PowerFlex data (All network together)	For data traffic between all PowerFlex core components
PowerFlex data (client-traffic only)-separate client and internal storage network	For storage data clients (SDC) to storage data servers (SDS) traffic. This network can also be used for NVMe target (SDT) to NVMe over TCP host traffic. **NOTE:** It can be a separate data network than the SDS-SDC network. If you are using file this network is also used to connect the SDS to the SDCs on the file controller nodes
PowerFlex data (server-traffic only)-separate client and internal storage network	For storage data server to storage data server traffic only. The same network is used for NVMe target(SDT), SDR (Replication), MDM internal traffic.
PowerFlex management	For PowerFlex system management
PowerFlex replication	For storage data replicator (SDR) to storage data server (storage or storage data server) or system to system replication traffic For SDR to storage data clients (application or storage data clients) traffic For SDR to SDR traffic
PowerFlex file management	To support PowerFlex file management traffic
PowerFlex file data	For file data traffic between PowerFlex file server and NFS or CIFS client

Networking requirements

The following are the networking requirements for installing or upgrading PowerFlex. These requirements apply only when you are installing resources using a PowerFlex Manager template.

- When deploying PowerFlex on each node, there should be one IP address that can access PowerFlex Manager.
- If you want to separate PowerFlex management IP addresses and data networks, you can have more than one network.
- If you are deploying a server with one IP address, that IP address has to serve as both the PowerFlex management IP address and data IP address.
- If you are using more than one network, create the network that is routed and can reach PowerFlex Manager as PowerFlex management type network, and other networks as PowerFlex datatype networks. For more information about networking, see *Dell PowerFlex 4.5.x Technical Overview*.

NOTE: If bonding interfaces and using mode 6 `ALB` updelay option may need to be increased above zero (which is the default) in the bond config file.

For example:

```
MTU='9000'
BOOTPROTO='none'
STARTMODE='auto'
ZONE='public'
BONDING_MASTER='yes'
BONDING_SLAVE0='em3'
BONDING_SLAVE1='p2p1'
BONDING_MODULE_OPTS='mode=6 miimon=100 updelay=100'
```

Deploying PowerFlex Manager

Software packages

For more information about software packages, see the *Dell PowerFlex 4.5.x Release Notes* available at Dell Technologies Support site.

Download PowerFlex installation packages

Use this procedure to download and extract the installation bundle if you are deploying a PowerFlex cluster outside of PowerFlex Manager.

About this task

Based on the deployment scenario, download either of the following PowerFlex installation packages:

- Use the following packages for vSphere, co-resident, and Linux deployments:
  - `Software_Only_Complete_4.5.x.x.zip`— Contains core PowerFlex software and compliance(GPG) file
- Use the following package for VMware vSphere or virtualized deployments:
  - `PowerFlex_Manager_4.5.x.x.zip`— Contains the OVA templates for the Installer node and for the management node
- Use the following package for deploying with a bare metal Linux installation:
  - `PFMP2-4.5.x.x-249.tgz`— Contains the files and scripts for PowerFlex Manager deployment (bare-metal)

Steps

1. From the Dell Technologies Support site, download the installation bundle for this version of PowerFlex.
2. Copy the installation bundle to your personal computer or desktop.
3. Extract the installation bundle.

Calculate the required persistent memory and RAM capacity for a fine granularity storage data server

Use the following formulas to calculate the necessary capacity for NVDIMM and RAM if your system is equipped with NVDIMMs only.

Steps

1. Calculate the required persistent memory and RAM capacity using the following table, where *X* is the total capacity of storage data server devices that are used for the fine granularity storage pool, and *Y* is the total RAM needed:

   NOTE: Persistent memory refers to the following options:

   - PowerEdge servers (R650/R750) and/or PowerEdge servers (R640/R740xd/R840) with non-volatile DIMMs (NVDIMM)

     Dell PowerEdge R660 and R760 servers with software-defined persistent memory.

   Table 5. Persistent memory and RAM capacity for PowerEdge servers

Fine granularity capacity	Required persistent memory size (GiB)	Required RAM capacity (rounded)	Additional memory for MDM, LIA, and SVM operating system	Total RAM (rounded)
49 TB	32 GiB NVDIMM (in SVM it is 31 GiB)	>36 GiB	MDM: 6.5 GiB LIA: 0.35 GiB Operating system base: 1 GiB Buffer: 2 GiB CloudLink 4 GiB	50 GiB
49 TB < *X* < 96 TB	64 GiB NVDIMM (in SVM it is 62 GiB)	>41 GiB <59 GiB		49 GiB < *Y* < 73 GiB
96 TB < *X* < 128 TB (122.88 is the actual limit)	96 GiB NVDIMM (in SVM it will be 93 GiB)	>64.5 GiB <72 GiB		86 GiB

2. Alternatively, you can calculate persistent memory capacity and RAM capacity using the following formulas:

   NOTE: The calculation is in binary MiB, GiB, and TiB;

   - *persistent memory_capacity_in_GiB* = ((100 * *Number_Of_Drives*) + (700 * Capacity_in_TiB))/1024
   - *RAM_capacity_in_GiB* = 10 + ((100 * *Number_Of_Drives*) + (550 * Capacity_in_TiB))/1024

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Preparing the PowerFlex Manager environment

To deploy PowerFlex Manager, use one of the following installation options:

- Deploy into a VMware vSphere environment
- Deploy into a co-resident management environment on PowerFlex storage-only nodes
- Deploy into a Linux management environment

Deploying into a VMware vSphere environment

Use this option if the PowerFlex management virtual machines reside in an existing VMware vSphere environment.

Management VM resource requirements

Table 6. Management VM resource requirements

PowerFlex version	Storage	CPU	RAM
4.5.x	600 GB	14 vCPU	32 GB*

* To manage the PowerFlex cluster with 32 K volume, each node must have 48 GB RAM.

Deploying the PowerFlex management platform

This section describes how to install and configure the PowerFlex management platform.

This includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

The following diagram depicts the PowerFlex installer VM and management VM deployed in a virtual environment and highlights the interfaces that are needed to be configured:

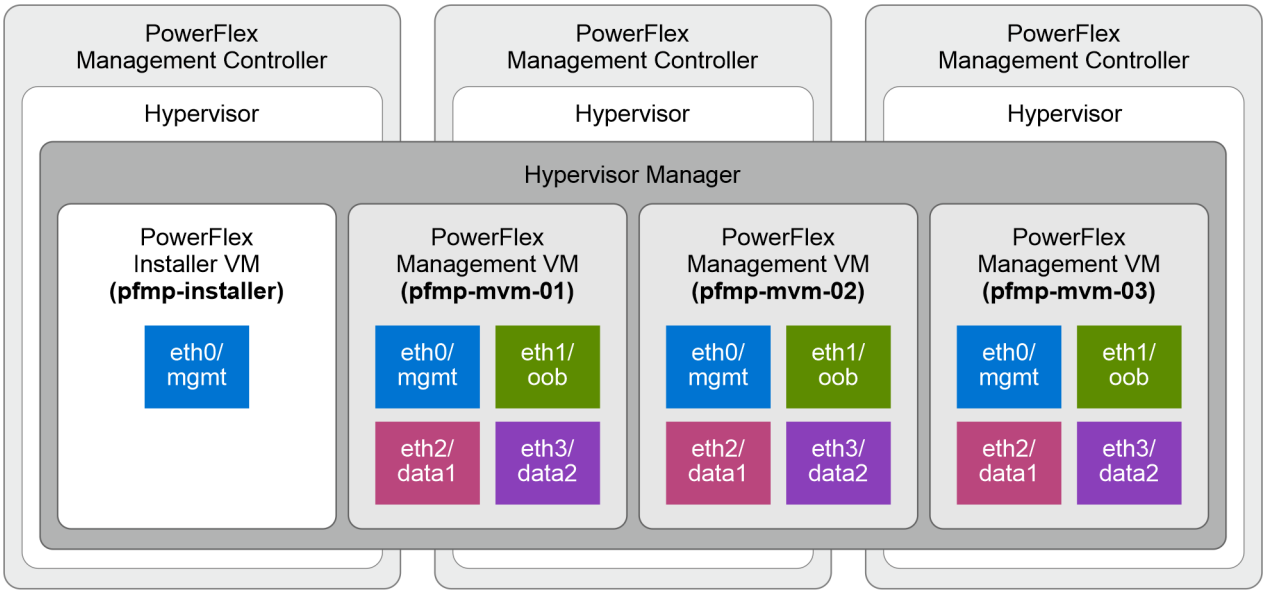


Figure 1. PowerFlex installer VM and MVMs deployed in a virtual environment

Deploying and configuring the PowerFlex management platform installer VM

Add Entrust root certificate to the VMware vCenter certificate store

If you want to use third-party certificates in your environment, you must add a third party trusted root certificate to the VMware vCenter certificate store.

Steps

1. Manually download OVCS2 certificate (OVCS2-CSBR1-crosscert) from the Entrust website: Entrust TLS certificate information center. Search for keyword "OVCS2" and click on CRT to download the certificate.
2. Upload the Entrust root certificate to VMware vCenter certificate store:
   a. Log in to the vCenter Server.
   b. Click **Administration** > **Certificates** > **Certificate Management**.
   c. Next to **Trusted Root Store**, click **Add**.
   d. Browse and select the downloaded Entrust root certificate. The certificate is added in a panel under **Trusted Root Certificates**.
3. Once the certificates are installed, OVA can be verified and deployed.

Deploy the PowerFlex management platform installer VM

Use this procedure to deploy and configure the PowerFlex management platform installer VM in a VMware environment.

Prerequisites

- For a multi-VLAN or multi-subnet configuration, pfmc-vm-mgmt-<vlanid> must be selected instead of the flex-node-mgmt-<vlanid>.
- For standard configurations without multi-networking, select either pfmc-vm-mgmt-<vlanid> (130) or flex-node-mgmt-<vlanid> (105).

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA Template**

5. Click **Open** > **Next**.
6. Enter **pfmp-installer** for the VM name.
7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.
10. Select the **PFMP datastore**.
11. Select **Virtual disk format** > **Thin provision** > **Next**.
12. Select **flex-node-mgmt-<vlanid>** > **OK** > **Next**.

    NOTE:
    - For a multi-VLAN configuration, more than one port-group for pfmc-sds-mgmt-<vlanid> exists. Select the first port-group for VM deployment.
    - For a multi-VLAN or multi-subnet configuration, the management interface of the MVMs and installer VM must be configured on VLAN 130 instead of VLAN 105.

13. Click **Finish**.
14. Right-click the VM and select **Power** > **Power On**.

Configure the PowerFlex management platform installer networking interface

Use this procedure to configure the networking for the PowerFlex management platform installer VM.

About this task

The following network adapter needs to be configured on the Powerflex management platform installer:

VLAN 105: flex-node-mgmt (eth0) or VLAN 130: pfmc-mgmt (eth0)

For a multi-VLAN or multi-subnet configuration, the PowerFlex management controller management is configured instead of the VMware ESXi management. For a standard deployment without multi-networking the network adapter can be configured on either VMware ESXi management (VLAN 105) or PowerFlex management controller management (VLAN 130).

Steps

1. Launch the web console from vCSA and log in as **delladmin**.
2. To configure the <flex-node-mgmt> interface, type **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   Example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.10
NETMASK=255.255.255.0
```

   a. To configure the default route, type **sudo vi /etc/sysconfig/network/routes. default <gateway ip> - <interface>**

      Example: **default 10.10.105.1 - eth0**

   b. To configure DNS search and DNS servers, type **sudo vi /etc/sysconfig/network/config** and modify the following: **NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>" NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"**

      Example: **NETCONFIG_DNS_STATIC_SEARCHLIST="example.com" NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"**

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs or physical nodes (co-resident or bring your own Linux PowerFlex management platform deployments).

Steps

1. To configure the NTP server, enter **`sudo vi /etc/chrony.conf`** and add the following:

   `server <ntp ip address> iburst`

   For example: **`server 10.10.105.240 iburst`**
2. To start the chronyd service, enter:

   `sudo systemctl start chronyd`
3. To enable chronyd, enter:

   `sudo systemctl enable chronyd`
4. To restart chronyd, enter:

   `sudo systemctl restart chronyd`
5. To verify that the **Network Time Protocol service** is running, enter:

   `sudo systemctl status chronyd.service`
6. Repeat all the steps for all the PowerFlex management platform VMs and the installer VM.
7. To check that the server is synced to NTP, enter:

   `chronyc tracking`

   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status

   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

   ```
   PFMP2-450-747 :~ # chronyc tracking
   Reference ID : 179628F2 (23.150.40.242)
   Stratum : 3
   Ref time (UTC) : Tue Mar 12 21:24:20 2024
   System time : 0.000531569 seconds slow of NTP time
   Last offset : -0.000336191 seconds
   RMS offset :  0.000201320 seconds
   Frequency : 6.809 ppm slow
   Skew : 0.029 ppm
   Root delay : 0.037072252 seconds
   Root dispersion : 0.004111907 seconds
   Update interval : 7242.7 seconds
   Leap status : Normal
   PFMP2-450-747 :~ #
   ```

   Unsuccessful chronyc tracking status

   - `Stratum` is 0 (not synced).
   - Leap status says `Normal`:

   ```
   node1:~ # chronyc tracking
   Reference ID : 00000000 ()
   Stratum : 0
   Ref time (UTC) : Thu Jan 01 00:00:00 1970
   System time : 0.000000000 seconds fast of NTP time
   Last offset : +0.000000000 seconds
   RMS offset : 0.000000000 seconds
   Frequency : 0.000 ppm slow
   Residual freq : +0.000 ppm
   Skew : 0.000 ppm
   ```

Deploying and configuring the PowerFlex management platform VMs

Use this procedure to complete the steps needed to deploy the PowerFlex management platform cluster VMs.

Prerequisites

The PowerFlex management platform cluster requires three VMs to be deployed.

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA**
5. Click **Open** > **Next**.
6. Enter **`pfmp-mvm-<number>`** for the VM name.
7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.
10. Review details and click **Next**.
11. Select the datastore.
12. Select **Virtual disk format** > **Thin provision** > **Next**.
13. Select **Desired Destination Network** > **OK** > **Next**.
14. Click **Finish**.
15. Repeat the above for the three management VMs.

Configure the PowerFlex management platform networking interfaces

Use the following procedure to complete the steps to configure the networking files for the PowerFlex management platform cluster VMs.

About this task

NOTE:

- For a multi-VLAN or multi-subnet configuration, the PowerFlex management controller management must be configured on PowerFlex management controller management (130).
- For standard builds without multi-VLAN or multi-subnet configuration, can be on either VMware ESXi management (105) or PowerFlex management controller management (130).

NOTE: Based on your requirement, configure only the required interfaces on the management VMs. If you have unused interfaces on the deployed management VMs, then ensure you disconnect those interfaces within the vCenter. For example, an out-of-band interface. If the unused interfaces are connected and unconfigured, then PowerFlex Manager may display unexpected errors.

Steps

1. Launch the web console from the Virtual Machine Manager, log in as **`delladmin`**.
2. Configure the flex-node-mgmt-<vlanid> eth0 interface:
   a. Type: **`sudo vi /etc/sysconfig/network/ifcfg-eth0`**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.30
NETMASK=255.255.255.0
```

b. Configure the default route. Type: **sudo vi /etc/sysconfig/network/routes**.

default <gateway ip> - <interface>

For example:

default 10.10.105.1 - eth0

c. Configure DNS search and DNS servers. Type: **sudo vi /etc/sysconfig/network/config** and modify the following:

NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"

NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"

For example:

NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"

NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"

3. Configure the flex-data1-<vlanid> eth1 interface:

a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth1**.

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

For example:

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.151.30
NETMASK=255.255.255.0
```

4. Configure the flex-data1-<vlanid> eth2 interface:

a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth2**.

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

For example:

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.152.30
NETMASK=255.255.255.0
```

5. Run the following command to update the network configuration:

```
sudo systemctl restart network
```

6. Run the following command to change the delladmin account password:

```
passwd delladmin
```

7. Change the root password with the following command:
   ```
   sudo passwd root
   ```
8. Repeat steps 1 through 8 for all PowerFlex management platform VMs.

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs or physical nodes (co-resident or bring your own Linux PowerFlex management platform deployments).

Steps

1. To configure the NTP server, enter **`sudo vi /etc/chrony.conf`** and add the following:
   `server <ntp ip address> iburst`
   For example: **`server 10.10.105.240 iburst`**
2. To start the chronyd service, enter:
   `sudo systemctl start chronyd`
3. To enable chronyd, enter:
   `sudo systemctl enable chronyd`
4. To restart chronyd, enter:
   `sudo systemctl restart chronyd`
5. To verify that the **Network Time Protocol service** is running, enter:
   `sudo systemctl status chronyd.service`
6. Repeat all the steps for all the PowerFlex management platform VMs and the installer VM.
7. To check that the server is synced to NTP, enter:
   `chronyc tracking`
   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status
   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

   ```
   PFMP2-450-747 :~ # chronyc tracking
   Reference ID : 179628F2 (23.150.40.242)
   Stratum : 3
   Ref time (UTC) : Tue Mar 12 21:24:20 2024
   System time : 0.000531569 seconds slow of NTP time
   Last offset : -0.000336191 seconds
   RMS offset :  0.000201320 seconds
   Frequency : 6.809 ppm slow
   Skew : 0.029 ppm
   Root delay : 0.037072252 seconds
   Root dispersion : 0.004111907 seconds
   Update interval : 7242.7 seconds
   Leap status : Normal
   PFMP2-450-747 :~ #
   ```

   Unsuccessful chronyc tracking status
   - `Stratum` is 0 (not synced).
   - Leap status says `Normal`:

   ```
   node1:~ # chronyc tracking
   Reference ID : 00000000 ()
   Stratum : 0
   Ref time (UTC) : Thu Jan 01 00:00:00 1970
   System time : 0.000000000 seconds fast of NTP time
   Last offset : +0.000000000 seconds
   RMS offset : 0.000000000 seconds
   ```

```
Frequency : 0.000 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
```

Deploy the PowerFlex management platform cluster

Use the following procedure to complete the steps to configure the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

Prerequisites

The following locations contain log files for troubleshooting:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

The following diagram illustrates how PFMP_config.JSON file content is applied to PowerFlex management.

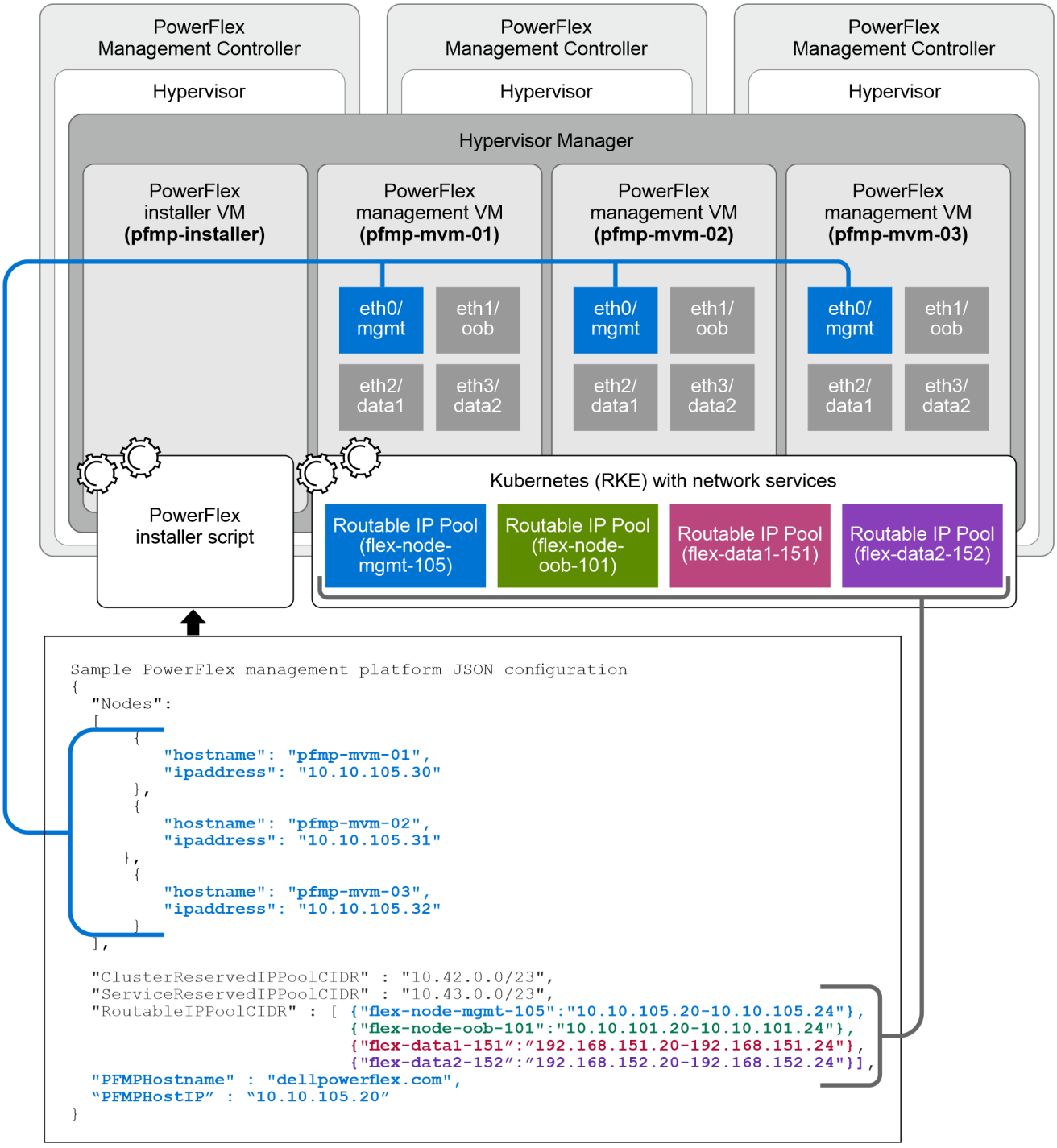


Figure 2. Relationship between the PFMP_config.JSON file and PowerFlex management

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 7. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Recommended three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these

Table 7. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
ServiceReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, SupportAssist, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Steps

1. To SSH as non-root user to the PowerFlex management platform Installer VM, run the following command: **`ssh delladmin@<pfmp installer ip>`**.
2. Go to the config directory, run the following command: **`cd /opt/dell/pfmp/PFMP_Installer/config`**
3. To configure the `PFMP_Config.json`, run the following command: **`sudo vi PFMP_Config.json`** and update the configuration parameters.

   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.105.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
```

```
                            {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                            {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                            {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE: **"RoutableIPPoolCIDR"** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address. PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}**. Use the **"PFMPHostname" : "dellpowerflex.com"** default FQDN to browse PowerFlex Manager. The name should be resolved to the **"PFMPHostIP"** IP address.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to nonrouted networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Go to the scripts folder in the working directory, type: **cd /opt/dell/pfmp/PFMP_Installer/scripts**
5. To set up the installer, run the following command: **sudo ./setup_installer.sh**.

   The **setup_installer.sh** starts a docker container with the appropriate network settings that deploys the PowerFlex Manager management virtual machine cluster when the **install_PFMP** script is run.
6. To install the PowerFlex management platform, run the following command: **sudo ./install_PFMP.sh**.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the SSH username of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **Yes** to indicate that the passwords are the same for all cluster nodes.
   - Enter the SSH password of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **No** to indicate that the nodes used for PowerFlex management platform are not co-resident nodes.

   NOTE: Ensure that you use the single SSH key for all nodes if you want to deploy a PowerFlex cluster using a CSV topology file.

   View the progress of the installation in the following logs that are in the working directory:

   - `<working directory>/PFMP_Installer/logs/`
   - `<working directory>/atlantic/logs/`

   NOTE: Installation may take more than two hours.

   - If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see Collect PowerFlex management platform support data.
   - For more information about troubleshooting, see Troubleshooting a PowerFlex management platform deployment.

Finalize firewall configuration

RKE2 uses Calico as a networking and security solution that enables Kubernetes workloads to communicate securely performing similar yet more dynamic functions as firewalld.

About this task

Firewalld can interfere with the dynamic nature of Calico, therefore, firewalld must be disabled. For more information, see Known issues and limitations with rke2.

Steps

To disable firewalld, do the following:

a. Log in to PowerFlex management platform node using SSH and run the following command:

```
systemctl stop firewalld
systemctl disable firewalld
```

b. To verify firewalld is stopped and disabled, run the following command:

```
systemctl status firewalld
```

Example output:

```
firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor
preset: disabled)
     Active: inactive (dead)
       Docs: man:firewalld(1)
     Do this on all remaining PFMP nodes
```

Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform install problems.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site. To get the utility file from the PowerFlex management platform installer, go to the `PFMP_Installer/scripts` directory. To download the data collection utility from the support site, search for **PFMP log collection Script** file on the Dell Technologies Support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as user root.

   For example, the utilitys executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as user `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
```

```
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support. By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, and you want to reset the environment use the following procedure to uninstall or reset a PowerFlex management platform cluster.

Steps

Run the following script to reset or uninstall a PowerFlex management platform cluster.

`reset_PFMP.sh`

CAUTION: This script deletes the existing docker container and reboots all cluster nodes. If the PowerFlex Manager installation fails in a co-resident deployment, contact Dell Technologies Support.

Deploying into a co-resident management environment on PowerFlex storage

In a PowerFlex co-residency management configuration, PowerFlex Manager is installed on a predeployed PowerFlex storage-only node.

NOTE: Co-resident deployments are manual in nature due to the eventual management platform residing on top of the deployed PowerFlex infrastructure. It is recommended that you create a CSV file describing the deployment for future capability and capacity expansion. For more information about deploying a PowerFlex cluster using a CSV file, see Deploying a PowerFlex cluster using a CSV topology file .

Deployment workflow

The main steps while deploying PowerFlex Manager co-resident on PowerFlex storage-only nodes are as follows:

Table 8. Deployment workflow steps

Step	Action	Reference
1	Configure BIOS settings and the BMC controller (if applicable).	*Dell Technologies PowerFlex Custom Node R650/R750/R6525 Hardware Configuration and Operating System Installation Guide for Linux* available at the Dell Technologies Support Site. *Dell VxFlex Ready Node R640-R740xd-R840 Hardware Configuration and Operating System Guide for Linux* .
2	Install the Linux operating system using the vendor's best practices.	
3	Configure the network on the nodes.	
4	Set up the `yum\zypper` repository to update the packages as needed.	See the *Adding repositories* topic in the *Managing software with command line tools | Administration Guide | SUSE Linux Enterprise Server 15 SP3*.
5	Deploy PowerFlex manually.	See Manually deploying PowerFlex storage
6	Create and map volumes to the nodes that host the Kubernetes cluster for PowerFlex Manager.	See Create and map volumes
7	Create the partitions and persistent mount point.	See Create partitions and mount points
8	Deploy the PowerFlex management platform installer and run PowerFlex management platform deployment.	See Deploying the PowerFlex management platform

Table 8. Deployment workflow steps (continued)

Step	Action	Reference
9	Log in to PowerFlex Manager and import or connect the MDM cluster.	If the MDM cluster is installed with PowerFlex 3.6.x, see Upgrade.

Co-residency management deployment prerequisites

There are certain prerequisites required by PowerFlex Manager before you can deploy a Kubernetes cluster.

- The following operating systems are supported:
  - Red Hat Enterprise Linux 7.9, 8.6, 8.8
  - SUSE Linux Enterprise Server (SLES) 15 SP3, 15 SP4
  - CentOS 7.9
- The following modules must be installed on the participating nodes before starting a co-resident deployment:

  For SUSE Linux Enterprise Server:

Table 9. SUSE Linux Enterprise Server modules for PowerFlex Manager

For PowerFlex Manager Install	Location	Repository
python3	On ISO	-
python3-pip	On ISO	-
sshpass	SUSEConnect	PackageHub/15.x/x86_64
java-11-openjdk-headless	On ISO	-
jq	On ISO	-
haproxy	SUSEConnect	sle-ha/15.x/x86_64
keepalived	SUSEConnect	sle-ha/15.x/x86_64
lvm2	On ISO	-
skopeo	SUSEConnect	sle-ha/15.x/x86_64
apparmor-parser	On ISO	-
libapr1	SUSEConnect	sle-module-basesystem/15.x/x86_64
For PowerFlex MDM certificates:		
pycparser-2.21-py2.py3 or higher	3rd Party	Python Package Index
cffi-1.15.1 or higher	3rd Party	Python Package Index
cryptography-40.0.2 or higher	3rd Party	Python Package Index

Download the appropriate versions of pycparser, cffi and cryptography for your OS release and Python version. SLES 15.4 comes with these modules standard.

For Red Hat Enterprise Linux:

Table 10. Red Hat Enterprise Linux modules for PowerFlex Manager

For PowerFlex Manager Install	Location	Repository
python3	On ISO	-
python3-pip	On Satellite	rhel-x-for-x86_64-appstream-rpms
httpd-tools	On Satellite	rhel-x-for-x86_64-appstream-rpms
libselinux-python3	On ISO	-
sshpass	On Satellite	rhel-x-for-x86_64-appstream-rpms
java-11-openjdk-headless	On Satellite	rhel-x-for-x86_64-appstream-rpms

Table 10. Red Hat Enterprise Linux modules for PowerFlex Manager (continued)

For PowerFlex Manager Install	Location	Repository
jq	On Satellite	rhel-x-for-x86_64-appstream-rpms
haproxy	On Satellite	rhel-x-for-x86_64-appstream-rpms
keepalived	On Satellite	rhel-x-for-x86_64-appstream-rpms
lvm2	On ISO	-
skopeo	On Satellite	rhel-x-for-x86_64-appstream-rpms
firewalld	On ISO	-
iptables	On ISO	-
For PowerFlex MDM certificates:		
pip-21.3.1-py3 or higher	3rd Party	Python Package Index
pycparser-2.21 or higher	3rd Party	Python Package Index
cffi-1.15.1 or higher	3rd Party	Python Package Index
cryptography-40.0.2 or higher	3rd Party	Python Package Index

Download the appropriate version of pip, pycparser, cffi and cryptography for your OS release and Python version. Python v3.11 and Python3.11-cryptography includes above mentioned modules and can be used to run the MDM certificate generation script.

- All nodes must have their clocks synchronized with a valid time source.
- Ensure that the default route is configured before installing PowerFlex Manager.
- Ensure that you disable the NetworkManager service on Red Hat based deployments. If the NetworkManager is running on the Linux operating system, the `/etc/resolv.conf` file content will be overwritten when updating the DNS records. For more information about disabling NetworkManager, refer to OS vendor guidelines.

Deployment design for co-residency management

The following deployment design principles are recommended:

- If you have a multi-rack system, place the MDMs and the PowerFlex management platform components on different racks.
- If the hardware requirements are met, any node running a SDS can also optionally run a SDR and SDT.
- The storage data client (SDC) is installed on PowerFlex storage-only nodes that are used for the co-resident management nodes used by the management layer.
- Every node has LIA installed for various maintenance use cases.

PowerFlex storage-only node and Linux hyper-converged node components layout with PowerFlex management platform

The following table details best practices in relation to component layout for PowerFlex storage-only nodes:

Table 11. Component layout for PowerFlex storage-only nodes

Number of nodes	Components that are installed on the node						
	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Node 7
Four nodes	MDM, SDS	MDM, SDS, SDC, Kubernetes for the PowerFlex management platform	Tiebreaker,SDS, SDC, PowerFlex management platform Kubernetes cluster	SDS, SDC, PowerFlex management platform Kubernetes cluster	N/A	N/A	N/A

Table 11. Component layout for PowerFlex storage-only nodes (continued)

Number of nodes	Components that are installed on the node						
	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Node 7
Five nodes		MDM, SDS	MDM, SDS, SDC, PowerFlex management platform Kubernetes cluster	Tiebreaker,SDS, SDC, PowerFlex management platform Kubernetes cluster	Tiebreaker,SDS, SDC, PowerFlex management platform Kubernetes cluster	N/A	N/A
Six nodes		MDM, SDS	MDM, SDS	Tiebreaker,SDS, SDC, PowerFlex management platform Kubernetes cluster	Tiebreaker,SDS, SDC, PowerFlex management platform Kubernetes cluster	SDS, SDC, PowerFlex management platform Kubernetes cluster	N/A
Seven nodes and above		MDM, SDS	MDM, SDS	Tiebreaker,SDS, SDC, PowerFlex management platform Kubernetes cluster	Tiebreaker,SDS, SDC, PowerFlex management platform Kubernetes cluster	SDS, SDC, PowerFlex management platform Kubernetes cluster	SDS

Manually deploying PowerFlex storage

This section describes the steps that are required to manually deploy PowerFlex storage to be used to deploy a PowerFlex management server. It is recommended to deploy a five-node management stack unless there are fewer than six nodes.

You can create the MDM cluster in the following modes:

- Three-node cluster: One primary MDM, one secondary MDM, and one tiebreaker. This cluster can sustain a single point of failure.
- Five-node cluster: One primary MDM, two secondary MDMs, and two tiebreakers. This extended cluster can sustain two points of failure.

The following are the steps for creating an MDM cluster:

- Install the MDMs with Manager role (two in a three-node cluster, or three in a five-node cluster).
- Install the MDMs with tiebreaker roles (one in a three-node cluster, or two in a five-node cluster).
- Enable scli on an MDM with Mutual Transport Layer Security (mTLS).
- Create a single-node cluster from one of the manager nodes.
- Add secondary MDMs as standbys
- Add tiebreakers to the cluster.
- Switch the cluster to three-node or five-node mode, while designating the standby secondary and standby tiebreaker names in the cluster.
- Add SDS and devices.

Creating a three-node MDM cluster

For a three-node cluster, the following sample topology is used:

Table 12. Three-node MDM cluster topology sample

Role	Management IP address	MDM IP address	Virtual IP address	Virtual IP NIC	MDM name	Cluster role
MDM 1	10.103.110.152	192.168.151.152 192.168.152.152	192.168.151.154 192.168.152.154	eth1, eth2	mdm152	Primary

Table 12. Three-node MDM cluster topology sample (continued)

Role	Management IP address	MDM IP address	Virtual IP address	Virtual IP NIC	MDM name	Cluster role
MDM 2	10.103.110.153	192.168.151.153 192.168.152.153	192.168.151.154 192.168.152.154	eth1, eth2	mdm153	Secondary
TB 1	10.103.110.154	192.168.151.154 192.168.152.154	-	-	tb154	Tiebreaker

Storage data server (SDS) and storage data client (SDC) are installed on all nodes.

Install the MDM packages (three-node cluster)

Install the PowerFlex MDM package for CentOS on SVMs that are part of a three-node cluster.

Prerequisites

The following prerequisite packages must be installed before installing the MDM for all supported operating system matrix:

Table 13. Prerequisite packages for MDM installation

Package	Installation method	Package name and reference per operating system
apr-util	rpm \ yum \ zypper \ apt-get	• apr-1.4.8-7.el7.x86_64 • apr-util-1.5.2-6.el7.x86_64 • libapr1-1.6.3-3.3.8.x86_64
Java	rpm \ yum \ zypper \ apt-get	OpenJDK 8 See:*java-openjdk-x.x.x-x.rpm* OpenJDK 11 java-11-openjdk-x.x.x-x.rpm
python3	rpm \ yum \ zypper \ apt-get	Python 3 latest build

NOTE: If there is a PowerFlex node with Red Hat Enterprise Linux, install the prerequisite packages such as, ActiveMQ from the yum configuration. Otherwise, the MDM installation may fail.

Steps

1. Copy all the RPMs to all the nodes (MDM 1, MDM 2, and tiebreaker1 (TB 1)).
2. Install the ActiveMQ package only on MDM nodes by running the following command:

   ```
   rpm -i <activemq_path.rpm>
   ```

   For example:

   ```
   EMC-ScaleIO-activemq-x.x.x-x.noarch.rpm
   ```

3. Install the MDM and configure the manager role, by running the following command on MDM 1 and MDM 2:

   ```
   MDM_ROLE_IS_MANAGER=1 rpm -i <mdm_path.rpm>
   ```

   For example:

   ```
   MDM_ROLE_IS_MANAGER=1 rpm -i EMC-ScaleIO-mdm-4.x-<build>.X.el8.x86_64.rpm
   ```

   a. Install the MDM on TB 1:

      ```
      MDM_ROLE_IS_MANAGER=0 rpm -i <mdm_path.rpm>
      ```

For example:

```
MDM_ROLE_IS_MANAGER=0 rpm -i EMC-ScaleIO-mdm-4.x-<build>.X.el8.x86_64.rpm
```

4. Install the LIA component on all the nodes with a known password.

```
TOKEN=<LIA_password> rpm -i <full rpm path to LIA file>
```

For example:

```
TOKEN=Welcome@123 rpm -i EMC-ScaleIO-lia-4.0-x.x.el8.x86_64
```

5. Install the storage data server.

```
rpm -i EMC-ScaleIO-sds-4.0-x.x.el8.x86_64
```

Generate PowerFlex MDM certificates for three-node MDM cluster

Use this procedure to generate a temporary self-signed certificate required to install the PowerFlex cluster.

Steps

1. Log in to the primary MDM, as root.
2. Go to the config folder, type the following command: `cd /opt/emc/scaleio/mdm/cfg`
3. To generate a CA certificate, type the following command:

```
python3 certificate_generator_MDM_USER.py --generate_ca mgmt_ca.pem --password <password>
```

4. To create a CLI certificate, type:

```
python3 certificate_generator_MDM_USER.py --generate_cli cli_certificate.p12 -CA
mgmt_ca.pem --password <password>
```

5. To create an MDM certificate, type the following command:

```
python3 certificate_generator_MDM_USER.py --generate_mdm mdm_certificate.pem -CA
mgmt_ca.pem
```

6. To create additional MDM certificates (each MDM needs a certificate), type the following command:

```
python3 certificate_generator_MDM_USER.py --generate_mdm sec_mdm_certificate.pem -CA
mgmt_ca.pem
```

7. Using SCP, copy the new MDM certificates and mgmt_ca.pem to the config folder to each MDM cluster member (primary and secondary).

```
scp sec_mdm_certificate.pem root@<mdm ip address>:/opt/emc/scaleio/mdm/cfg/
mdm_certificate.pem
```

```
scp cli_certificate.p12 root@<mdm ip address>:/opt/emc/scaleio/mdm/cfg/
```

```
scp mgmt_ca.pem root@<mdm ip address>:/opt/emc/scaleio/mdm/cfg/
```

8. On the primary and secondary MDMs, add the CA certificate, type the following command:

```
cd /opt/emc/scaleio/mdm/cfg; scli --add_certificate --certificate_file mgmt_ca.pem
```

9. To start the MDM service on all SVMs, type: `systemctl restart mdm.service`
10. To check the status of the MDM service on all SVMs, type: `systemctl status mdm.service`

Create the three-node MDM cluster

Use this procedure to create the three-node MDM cluster.

Prerequisites

Install or update to the latest curl version before proceeding with this procedure.

Steps

1. Create the MDM single-node cluster on MDM 1, by running the following command on MDM 1:

   NOTE: Configuring an MDM cluster with VIP is not mandatory. If using a VIP, use the relevant parameters in the general command options.

   ```
   scli --create_mdm_cluster --primary_mdm_ip <IP of MDM 1> --primary_mdm_management_ip
   <Management IP of MDM 1> --primary_mdm_name <Name to give MDM 1>
   [-- cluster_virtual_ip <IP>]  [--primary_mdm_virtual_ip_interface <INTF>] --
   accept_license --approve_certificate
   ```

   Example:

   ```
   scli --create_mdm_cluster --primary_mdm_ip 192.168.151.152,192.168.152.152
   --primary_mdm_management_ip 10.103.110.152 --primary_mdm_name
   mdm152 --cluster_virtual_ip 192.168.151.254,192.168.152.254 --
   primary_mdm_virtual_ip_interface eth1,eth2 --accept_license --approve_certificate
   ```

   NOTE: Each MDM name must be unique. Ensure that each server on which an MDM is installed has a unique hostname.

2. Log in using an existing certificate:

   ```
   scli --login --p12_path /opt/emc/scaleio/mdm/cfg/cli_certificate.p12
   ```

3. Add the secondary MDM 2 to the MDM cluster as a standby MDM, by running the following commands on MDM 1:

   ```
   scli --add_standby_mdm --new_mdm_ip <IP of MDM X> --mdm_role manager --
   new_mdm_management_ip <Management IP of MDM x> --new_mdm_virtual_ip_interface <INTF>
   --new_mdm_name <Name to give MDM X>
   ```

   For example:

   ```
   scli --add_standby_mdm --new_mdm_ip 192.168.151.153,192.168.152.153 --mdm_role manager
   --new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth1,eth2 --
   new_mdm_name mdm153
   ```

4. Add tiebreaker to the MDM cluster by running the following command on MDM 1:

   ```
   scli --add_standby_mdm --new_mdm_ip <IP of TB X> --mdm_role tb --new_mdm_name <Name
   to give TB X>
   ```

   For example:

   ```
   scli --add_standby_mdm --new_mdm_ip 192.168.151.154,192.168.152.154 --mdm_role tb --
   new_mdm_name tb154
   ```

5. Optionally query the cluster to verify that it is running in single mode, by running the following command on MDM 1:

   ```
   scli --query_cluster
   ```

   The output includes the following lines:

   ```
   Cluster: Mode: 1_node
   ```

6. Switch to three-node cluster mode by running the following command on MDM 1:

   ```
   scli --switch_cluster_mode --cluster_mode 3_node --add_secondary_mdm_name <MDM 2
   name> --add_tb_name <TB 1 name>
   ```

   For example:

   ```
   scli --switch_cluster_mode --cluster_mode 3_node --add_secondary_mdm_name mdm153 --
   add_tb_name tb155
   ```

7. Optionally query the cluster to verify that it is running in cluster mode, by running the following command on MDM 1:

```
scli --query_cluster
```

Example output:

```
# scli --query_cluster
Cluster:
    Name: block-legacy-gateway, Mode: 3_node, State: Normal, Active: 3/3, Replicas:
2/2
    Virtual IP Addresses: 192.168.151.254, 192.168.152.254
Primary MDM:
    Name: mdm152, ID: 0x0dca0c7a1efebb00
        IP Addresses: 192,168,151,152, 192,168,152.152, Port: 9011, Virtual IP
interfaces: eth1, eth2
        Management IP Addresses: 10.234.95.13, Port: 8611
        Status: Normal, Version: 4.0.1000
Secondary MDMs:
    Name: mdm153, ID: 0x0e5a86ce5cb8f301
        IP Addresses: 192.168.151.153, 192.168.152.153, Port: 9011, Virtual IP
interfaces: eth1, eth2
        Management IP Addresses: 10.234.95.14, Port: 8611
        Status: Normal, Version: 4.0.1000
Tie-Breakers:
    Name: tb154, ID: 0x625bd9c73bdd0c02
        IP Addresses: 192.168.151.154, 192.168.152.154, Port: 9011
        Status: Normal, Version: 4.0.1000
```

Add a protection domain

Use this procedure to add a protection domain.

Steps

1. Log in to the MDM by entering the following command: `scli --login --p12_path /opt/emc/scaleio/mdm/cfg/cli_certificate.p12`.
2. Create a protection domain by entering the following command: `scli --add_protection_domain --protection_domain_name PD1`.

Add a storage pool

Use this procedure to add storage pools.

Steps

1. Log in to the MDM by entering the following command:
   `scli --login --p12_path /opt/emc/scaleio/mdm/cfg/cli_certificate.p12`
2. Create a storage pool by entering the following command:
   `scli --add_storage_pool --protection_domain_name PD1 --dont_use_rmcache --media_type SSD --data_layout medium_granularity --storage_pool_name SP1`

Add storage data servers

Use this procedure to add storage data servers.

Steps

1. Log in to the MDM by entering the following command:
   `scli --login --p12_path /opt/emc/scaleio/mdm/cfg/cli_certificate.p12`
2. Add storage data servers by entering the following command:

```
scli --add_sds --sds_ip <pfmc-sds-data1-ip,pfcm-sds-data2-ip> --protection_domain_name
PD1 --storage_pool_name SP1 --disable_rmcache --sds_name PFMC-SDS-<last ip octet>
```

3. Repeat for the remaining storage nodes.

Set the spare capacity for the medium granularity storage pool

Use this procedure to set the spare capacity for the medium granularity storage pool.

Steps

1. Log in to the primary MDM by entering the following command:
   ```
   scli --login --p12_path /opt/emc/scaleio/mdm/cfg/ cli_certificate.p12
   ```
2. Modify the capacity pool by entering the following command:
   ```
   scli --modify_spare_policy --protection_domain_name PD1 --storage_pool_name SP1 --
   spare_percentage <percentage>
   ```
   NOTE: The spare percentage is 1/n (where n is the number of nodes in the cluster). For example, the spare percentage for a three-node cluster is 34%.
3. Enter `Y` to proceed.

Identify disks

Use this procedure to identify the disks on each of the PowerFlex storage-only node.

Steps

1. Log in as root to each of the storage VMs.
2. Identify all available disks on the SVM by entering the following command: `lsblk`
3. Repeat for each PowerFlex controller nodes SVMs.

Add storage devices

Use this procedure to add storage devices.

Steps

1. Log in to the MDM by entering the following command:`scli --login --p12_path /opt/emc/scaleio/mdm/cfg/cli_certificate.p12`.
2. Add SDS storage devices by entering the following command: `scli --add_sds_device --sds_name <sds name> --storage_pool_name <storage pool name> --device_path /dev/sd(x)`.
3. Repeat for all storage devices and for all PowerFlex storage-only node.

Creating a five-node MDM cluster

For a five-node cluster, the following sample topology is used:

Table 14. Five-node MDM cluster topology sample

Role	Management IP address	MDM IP address	Virtual IP address	Virtual IP NIC	MDM name	Cluster role
MDM 1	10.103.110.152	192.168.151.152 192.168.152.152	192.168.151.154 192.168.152.154	eth1, eth2	mdm152	Primary
MDM 2	10.103.110.153	192.168.151.153 192.168.152.153	192.168.151.154 192.168.152.154	eth1, eth2	mdm153	Secondary 1
MDM 3	10.103.110.154	192.168.151.154 192.168.152.154	192.168.151.154 192.168.152.154	eth1, eth2	mdm154	Secondary 2

Table 14. Five-node MDM cluster topology sample (continued)

Role	Management IP address	MDM IP address	Virtual IP address	Virtual IP NIC	MDM name	Cluster role
TB 1	10.103.110.155	192.168.151.155 192.168.152.155	-	-	tb155	tiebreaker 1
TB 2	10.103.110.156	192.168.151.156 192.168.152.156	-	-	tb156	tiebreaker 2

Storage data server (SDS) and storage data client (SDC) are installed on all nodes.

Install the MDM packages (five-node cluster)

Install the PowerFlex MDM package for CentOS on SVMs that are part of a five-node MDM cluster.

Prerequisites

The following prerequisite packages must be installed before installing the MDM for all supported operating system matrix:

Table 15. Prerequisite packages for MDM installation

Package	Installation method	Package name and reference per operating system
apr-util	rpm \ yum \ zypper \ apt-get	• apr-1.4.8-7.el7.x86_64 • apr-util-1.5.2-6.el7.x86_64 • libapr1-1.6.3-3.3.8.x86_64
Java	rpm \ yum \ zypper \ apt-get	OpenJDK 8 See:*java-openjdk-x.x.x-x.rpm* OpenJDK 11 java-11-openjdk-x.x.x-x.rpm
python3	rpm \ yum \ zypper \ apt-get	Python 3 latest build

NOTE: In the case of a PowerFlex node with Red Hat Enterprise Linux, install the prerequisite packages such as ActiveMQ from the yum configuration. Otherwise, the MDM installation may fail.

Steps

1. Copy all the RPMs to all the MDM and tiebreaker (TB) nodes (MDM 1, MDM 2, MDM 3, TB 1, and TB 2).
2. Install the ActiveMQ package by running the following command:

   ```
   rpm -i <activemq_path.rpm>
   ```

   For example:

   ```
   EMC-ScaleIO-activemq-x.x.x-x.noarch.rpm
   ```

3. Install the MDM and configure the Manager role, by running the following commands on the primary (MDM 1) and secondary MDMs (MDM 2 and MDM 3):

   ```
   MDM_ROLE_IS_MANAGER=1 rpm -i <mdm_path.rpm>
   ```

   For example:

   ```
   MDM_ROLE_IS_MANAGER=1 rpm -i EMC-ScaleIO-mdm-4.x-<build>.X.el8.x86_64.rpm
   ```

a. Install the MDM on the tiebreaker MDMs (TB 1 and TB 2):

```
MDM_ROLE_IS_MANAGER=0 rpm -i <mdm_path.rpm>
```

For example:

```
MDM_ROLE_IS_MANAGER=0 rpm -i EMC-ScaleIO-mdm-4.x-<build>.X.el8.x86_64.rpm
```

4. Install the LIA component on all the nodes with a known password.

```
TOKEN=<LIA_password> rpm -i <full rpm path to LIA file>
```

For example:

```
TOKEN=Welcome@123 rpm -i EMC-ScaleIO-lia-4.0-x.x.el8.x86_64
```

5. Install the storage data server (SDS).

```
rpm -i EMC-ScaleIO-sds-4.0-x.x.el8.x86_64
```

Generate PowerFlex MDM certificates for five-node MDM cluster

Use this procedure to generate a temporary self-signed certificate required to install the PowerFlex cluster.

Steps

1. Log in as a root user to the primary MDM.
2. Go to the config folder, type: `cd /opt/emc/scaleio/mdm/cfg`
3. To generate a CA certificate, type:
   `python3 certificate_generator_MDM_USER.py --generate_ca mgmt_ca.pem`
4. To create a CLI certificate, type:
   `python3 certificate_generator_MDM_USER.py --generate_cli cli_certificate.p12 -CA mgmt_ca.pem --password <password>`
5. To create an MDM certificate, type:
   `python3 certificate_generator_MDM_USER.py --generate_mdm mdm_certificate.pem -CA mgmt_ca.pem`
6. To create additional MDM certificates (each MDM needs a certificate), type the following command:
   `python3 certificate_generator_MDM_USER.py --generate_mdm sec_mdm_certificate.pem -CA mgmt_ca.pem`
7. Using SCP, copy the new MDM certificates and mgmt_ca.pem to the config folder to the secondary MDMs.

```
scp sec1_mdm_certificate.pem root@<sec1_mdm ip address>:/opt/emc/scaleio/mdm/cfg/mdm_certificate.pem
```

```
scp sec2_mdm_certificate.pem root@<sec2_mdm ip address>:/opt/emc/scaleio/mdm/cfg/mdm_certificate.pem
```

```
scp cli_certificate.p12 root@<sec1_mdm ip address>:/opt/emc/scaleio/mdm/cfg/
```

```
scp cli_certificate.p12 root@<sec2_mdm ip address>:/opt/emc/scaleio/mdm/cfg/
```

```
scp mgmt_ca.pem root@<sec1_mdm ip address>:/opt/emc/scaleio/mdm/cfg/
```

```
scp mgmt_ca.pem root@<sec2_mdm ip address>:/opt/emc/scaleio/mdm/cfg/
```

8. On the primary and secondary MDMs, add the CA certificate, type:

```
cd /opt/emc/scaleio/mdm/cfg; scli --add_certificate --certificate_file mgmt_ca.pem
```

9. To start the MDM service on all SVMs, type: `systemctl restart mdm.service`
10. To check the status of the MDM service on all SVMs, type: `systemctl status mdm.service`

Create the five-node MDM cluster

Use this procedure to create the five-node MDM cluster.

Prerequisites

Install or update to the latest curl version before proceeding with this procedure.

Steps

1. Create the MDM single-node cluster by running the following command on MDM 1:

   NOTE: Configuring an MDM cluster with VIP is not mandatory. If you choose to use a VIP, use the relevant parameters in the general command options.

   ```
   scli --create_mdm_cluster --primary_mdm_ip <IP of MDM 1> --primary_mdm_management_ip
   <Management IP of MDM 1> --primary_mdm_name <Name to give MDM 1>
   [-- cluster_virtual_ip <IP>]  [--primary_mdm_virtual_ip_interface <INTF>] --
   accept_license --approve_certificate
   ```

   For example:

   ```
   scli --create_mdm_cluster --primary_mdm_ip 192.168.151.1521.152,192.168.152.152
   --primary_mdm_management_ip 10.103.110.152 --primary_mdm_name mdm152 --
   cluster_virtual_ip 192.168.151.254,192.168.152.254 192.168.111.152 --
   primary_mdm_virtual_ip_interface eth1,eth24 --accept_license --approve_certificate
   ```

   NOTE: Each MDM name must be unique. Ensure that each server on which MDM is installed has a unique hostname.

2. Log in using an existing certificate:

   ```
   scli --login --p12_path /opt/emc/scaleio/mdm/cfg/cli_certificate.p12
   ```

3. Add the secondary MDMs (MDM 2 and MDM 3) to the MDM cluster as standby MDMs, by running the following commands on MDM 1:

   ```
   scli --add_standby_mdm --new_mdm_ip <IP of MDM X> --mdm_role manager --
   new_mdm_management_ip <Management IP of MDM X> --new_mdm_virtual_ip_interface <INTF>
   --new_mdm_name <Name to give MDM X>
   ```

   Examples:

   ```
   scli --add_standby_mdm --new_mdm_ip 192.168.1.153 --mdm_role manager
   --new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth4 --
   new_mdm_name mdm153
   ```

   ```
   scli --add_standby_mdm --new_mdm_ip 192.168.151.154,192.168.152.154 --mdm_role
   manager -- new_mdm_management_ip 10.103.110.154 -3-new_mdm_name mdm154
   ```

4. Add tiebreaker 1 and tiebreaker 2 to the MDM cluster by running the following commands on MDM 1:

   ```
   scli --add_standby_mdm --new_mdm_ip <IP of TB X> --mdm_role tb --new_mdm_name <Name
   to give TB X>
   ```

For example:

```
scli --add_standby_mdm --new_mdm_ip 192.168.151.155,192.168.152.155 --mdm_role tb --
new_mdm_name tb155
```

```
scli --add_standby_mdm --new_mdm_ip 192.168.151.156,192.168.152.156 --mdm_role tb --
new_mdm_name tb156
```

5. Optionally query the cluster to verify that it is running in single mode, by running the following command on MDM 1:

```
scli --query_cluster
```

Output includes the following lines:

```
Cluster: Mode: 1_node
```

6. Switch to five-node cluster mode by running the following command on MDM 1:

```
scli --switch_cluster_mode --cluster_mode 5_node --add_secondary_mdm_name <MDM 2
name>,<MDM 3 name> --add_tb_name <TB 1 name>,<TB 2 name>
```

For example:

```
scli --switch_cluster_mode --cluster_mode 5_node --add_secondary_mdm_name
mdm153,mdm154 --add_tb_name tb155,tb156
```

7. Optionally query the cluster to verify that it is running in cluster mode, by running the following command on MDM 1:

```
# scli --query_cluster Cluster:
Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3 Virtual IP Addresses: N/A
Primary MDM:
    Name: mdm152, ID: 0x0dca0c7a1efebb00
        IP Addresses: 192,168,151,152, 192,168,152.152, Port: 9011, Virtual IP
interfaces: eth1, eth2
        Management IP Addresses: 10.103.110.152, Port: 8611
        Status: Normal, Version: 4.0.1000
Secondary MDMs:
    Name: mdm153, ID: 0x0e5a86ce5cb8f301
        IP Addresses: 192.168.151.153, 192.168.152.153, Port: 9011, Virtual IP
interfaces: eth1, eth2
        Management IP Addresses: 10.103.110.153, Port: 8611
        Status: Normal, Version: 4.0.1000
    Name: mdm154, ID: 0x0e5a86ce5cb8f301
        IP Addresses: 192.168.151.154, 192.168.152.154, Port: 9011, Virtual IP
interfaces: eth1, eth2
        Management IP Addresses: 10.103.110.154, Port: 8611
        Status: Normal, Version: 4.0.1000
Tie-Breakers:
        Name: TB1, ID: 0x0a07dc126403fb04
        IP Addresses: 192.168.151.155, 192.168.152.155 Port: 9011
        Status: Normal, Version: 4.0.1000 Name: TB2, ID: 0x549e9e6641ebd603
        IP Addresses: 10.103.110.155, Port: 9011
        Status: Normal, Version: 4.0.1000
        Name: TB2, ID: 0x0a07dc126403fb05
        IP Addresses: 192.168.151.156, 192.168.152.156 Port: 9011
        Status: Normal, Version: 4.0.1000 Name: TB2, ID: 0x549e9e6641ebd604
        IP Addresses: 10.103.110.156, Port: 9011
        Status: Normal, Version: 4.0.1000
```

Install the storage data client on a Linux-based server

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- Secure boot is not supported; ensure that secure boot is disabled in the BIOS.

About this task

This section describes how to manually install an external SDC on a Linux-based server using the command line.

The steps in the procedure apply to Red Hat Enterprise Linux/CentOS/Oracle Linux operating systems.

Steps

1. Install the GPG key on every server on which SDC is installed. From the PowerFlex installation folder, run the following command on every server:

   ```
   rpm --import RPM-GPG-KEY-ScaleIO
   ```

2. From the command line, install the SDC:
   - Red Hat Enterprise Linux/CentOS /Oracle Linux/SLES

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
     ```

   - Ubuntu (Extract these files before use, as described above.)

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
     ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located
3. Verify that the driver is loaded using one of the following commands:
   - SLES 15:

     ```
     systemctl status scini
     ```

   - All other Linux:

     ```
     service scini status
     ```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Next steps

Create and map volumes

Preparing the PowerFlex cluster to host PowerFlex Manager

Use the following procedures in this section to prepare the PowerFlex cluster to host PowerFlex Manager.

Create and map volumes

Use the CLI to create a volume and map it to an SDC.

Prerequisites

Use the following reference table for volumes creation and mapping commands used below:

Table 16. Volume creation and mapping reference

Node	Number of volumes per node	Volume size	Partition layout
Node 1 running the PowerFlex management platform Kubernetes cluster	1	600 GB	• Partition 1 (350 GB) ->`/opt/platform-provisioner` • Partition 2 (100 GB) -> `/opt/platform` • Partition 3 (150 GB) -> `/var/lib/rancher`
Node 2 running the PowerFlex management platform Kubernetes cluster	1	600 GB	• Partition 1 (350 GB) -> `/opt/platform-provisioner` • Partition 2 (100 GB) ->`/opt/platform` • Partition 3 (150 GB) -> `/var/lib/rancher`
Node 3 running the PowerFlex management platform Kubernetes cluster	1	600 GB	• Partition 1 (350 GB) -> `/opt/platform-provisioner` • Partition 2 (100 GB) -> `/opt/platform` • Partition 3 (150 GB) ->`/var/lib/rancher`

About this task

This procedure involves using SCLI commands to create and map volumes. For more details on SCLI command usage, see the *Dell PowerFlex 4.5.x CLI Reference Guide*.

Steps

1. Create a volume:

```
scli --add_volume --size_gb <SIZE> --volume_name <VOL_NAME> --protection_domain_name
<PD_NAME> --storage_pool_name <SP_NAME>
```

2. Map a volume to an storage data client (SDC):

```
scli --map_volume_to_sdc --volume_name <VOL_NAME> --sdc_ip <SDC_IP>
```

Create partitions and mount points

The exposed PowerFlex volumes are connected to the servers by the network. Use this procedure to configure the mounting options of PowerFlex devices for your specific Linux-based operating system.

About this task

Use persistent device names since the `/dev/sciniX` names are not guaranteed to persist between reboots. For more information about how to use persistent devices names, see Associate PowerFlex volumes with physical disks.

For information about partitioning your hard drive, see the SUSE communities article: *Manually Partitioning Your Hard Drive with fdisk | SUSE Communities*.

Steps

1. Only one volume is mapped to each host, the volume device path is:`/dev/scinia`. To validate this device path, run the following command: **`ls -ll /dev |grep scini`**.
2. Create three partitions on each volume as specified in Create and map volumes .
3. Verify that the partition table was created as expected by completing the following:
   a. Run the following command: **`fdisk -l /dev/scinia`**
   b. Confirm the size of each partition and the device path that is assigned to it.
4. Format the partitions. For example:

   `mkfs.xfs /dev/scinia1`

   `mkfs.xfs /dev/scinia2`

   `mkfs.xfs /dev/scinia3`
5. Determine the `/dev/disk/by-id` correlation to `/dev/sciniaX` partition by running the following command:**`ls -l /dev/disk/by-id/ |grep scini`**
   Output similar to the following displays:
6. Determine the `/dev/disk/by-id` correlation to `/dev/sciniX`:

   Output similar to the following displays:

   ```
   lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f0330000000a ->
   ../../scinia
   ```
7. Create the following three directories:

   `mkdir /opt/platform-provisioner/`

   `mkdir /opt/platform/`

   `mkdir /var/lib/rancher/`
8. Use the following commands as templates and replace the device path with the actual device path in your system:

   `mount /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a-part1 /opt/platform-provisioner/`

   `mount /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a-part2 /opt/platform/`

   `mount /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a-part3 /var/lib/rancher/`
9. To make the mount command persistent, edit the `/etc/fstab` file according to the instructions for your operating system. For example:

   `/dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a-part1 /opt/platform-provisioner xfs defaults,x-systemd.after=scini.service,_netdev,noauto 0 0`

   `/dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a-part2 /opt/platform xfs defaults,x-systemd.after=scini.service,_netdev,noauto 0 0`

   `/dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a-part3 /var/lib/rancher xfs defaults,x-systemd.after=scini.service,_netdev,noauto 0 0`

   NOTE: Edit the fstab entries on each host. noauto is included so that rke2-service will not start without those partitions mounted.

   - For Red Hat Enterprise Linux 7.x, in `/etc/fstab` use a text editor to add **`_netdev`** to the PowerFlex mount lines.

     Ensure that you comply with the netdev and syntax rules for your file system, as described in the **`man`** page.
   - For SUSE Linux Enterprise Server (SLES) 15 SP3 and 15 SP4 in `/etc/fstab` use a text editor to add **`nofail`** to the end of the scini mount lines.

Next steps

Repeat the steps on all volumes for the other nodes.

Deploying the PowerFlex management platform installer

This section describes how to install and configure the PowerFlex management platform.

This includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

The following diagram depicts the PowerFlex management platform installer VM and management VM deployed in a virtual environment and highlights the interfaces that are needed to be configured:

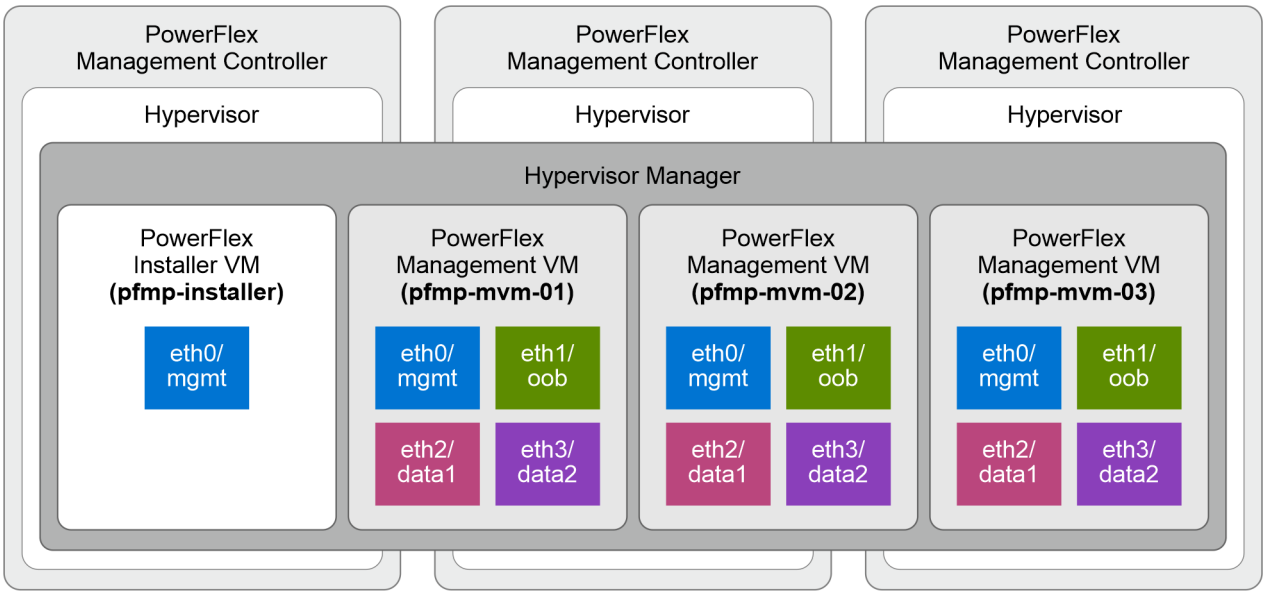


Figure 3. PowerFlex installer VM and MVMs deployed in a virtual environment

Deploy the PowerFlex management platform cluster

Use the following procedure to complete the steps to configure the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

Prerequisites

The following locations contain log files for troubleshooting:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

The following diagram illustrates how PFMP_config.JSON file content is applied to PowerFlex management.

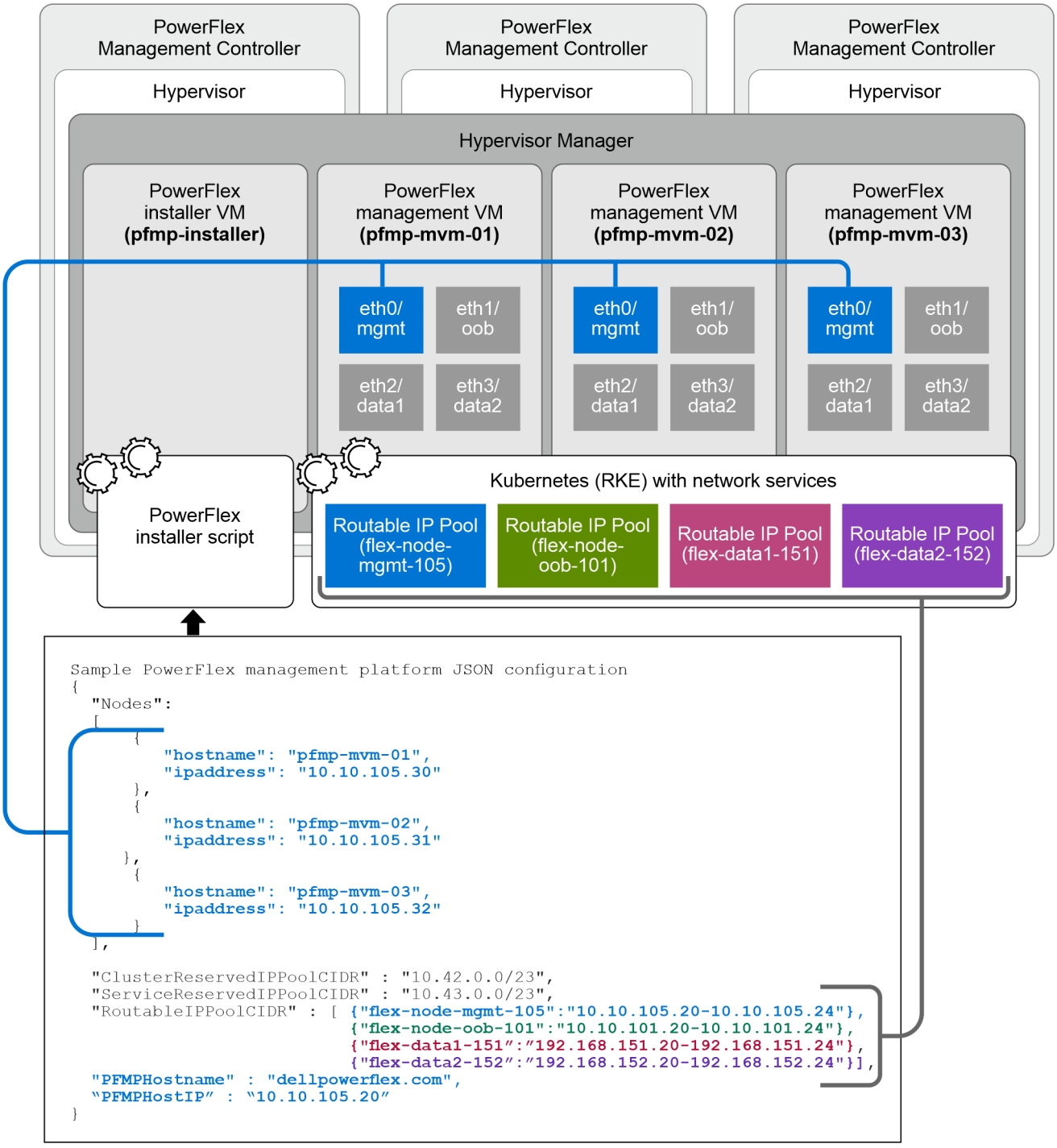


Figure 4. Relationship between the PFMP_config.json file and PowerFlex management

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 17. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Recommended three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these

Table 17. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
ServiceReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, SupportAssist, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Steps

1. To SSH as non-root user to the PowerFlex management platform Installer VM, run the following command: **`ssh delladmin@<pfmp installer ip>`**.
2. Go to the config directory and run the following command: **`cd /opt/dell/pfmp/PFMP_Installer/config`**
3. To configure the `PFMP_Config.json`, run the following command: **`sudo vi PFMP_Config.json`** and update the configuration parameters.

   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.105.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
```

```
                            {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                            {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                            {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE: **"RoutableIPPoolCIDR"** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address. PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}**. Use the **"PFMPHostname" : "dellpowerflex.com"** default FQDN to browse PowerFlex Manager. The name should be resolved to the **"PFMPHostIP"** IP address.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to nonrouted networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Go to the scripts folder in the working directory, type: **cd /opt/dell/pfmp/PFMP_Installer/scripts**
5. To set up the installer, run the following command: **sudo ./setup_installer.sh**.

   The **setup_installer.sh** starts a docker container with the appropriate network settings that deploys the PowerFlex Manager management virtual machine cluster when the **install_PFMP** script is run.
6. To install the PowerFlex management platform, run the following command: **sudo ./install_PFMP.sh**.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the SSH username of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **Yes** to indicate that the passwords are the same for all cluster nodes.
   - Enter the SSH password of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **YES** to indicate that the nodes used for PowerFlex management platform are co-resident nodes.

   NOTE: Ensure that you use the single SSH key for all nodes if you want to deploy a PowerFlex cluster using a CSV topology file.

   View the progress of the installation in the following logs that are in the working directory:

   - `<working directory>/PFMP_Installer/logs/`
   - `<working directory>/atlantic/logs/`

   NOTE: Installation may take more than two hours.

   - If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see Collect PowerFlex management platform support data.
   - For more information about troubleshooting, see Troubleshooting a PowerFlex management platform deployment.

Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform install problems.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site. To get the utility file from the PowerFlex management platform installer, go to the `PFMP_Installer/scripts` directory. To download the data collection utility from the support site, search for **PFMP log collection Script** file on the Dell Technologies Support site.

NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as user root.

   For example, the utilitys executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as user `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Modify the RKE2 server service

Steps

1. On co-resident PowerFlex management platform nodes, modify the systemd unit file to prevent the rke2-server from launching before PowerFlex storage.

   `#systemctl edit --full rke2-server`

2. In the displayed file, modify `After=network-online.target` to include scini.service. Also, add `(RequiresMountsFor= and ConditionPathIsReadWrite= lines)` at the end of the following example.
   For example,

```
[Unit]
Description=Rancher Kubernetes Engine v2 (server)
Documentation=https://github.com/rancher/rke2#readme
Wants=network-online.target
After=network-online.target scini.service
Conflicts=rke2-agent.service
RequiresMountsFor=/var/lib/rancher /opt/platform /opt/platform-provisioner
ConditionPathIsReadWrite=/var/lib/rancher
ConditionPathIsReadWrite=/opt/platform
ConditionPathIsReadWrite=/opt/platform-provisioner
```

3. Save the file.

Finalize firewall configuration

RKE2 uses Calico as a networking and security solution that enables Kubernetes workloads to communicate securely performing similar yet more dynamic functions as firewalld.

About this task

Firewalld can interfere with the dynamic nature of Calico, firewalld must be disabled. For more information, see Known issues and limitations with rke2.

Steps

1. To disable firewalld, do the following:
   a. Log in to the PowerFlex management platform node using SSH and run the following command:

   ```
   systemctl stop firewalld
   systemctl disable firewalld
   ```

   b. To verify that that firewalld is stopped and disabled, run the following command:

   ```
   systemctl status firewalld
   ```

   Example output:

   ```
   firewalld.service - firewalld - dynamic firewall daemon
        Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor
   preset: disabled)
        Active: inactive (dead)
          Docs: man:firewalld(1)
        Do this on all remaining PFMP nodes
   ```

2. Customize the Calico firewall if required.

   For example,

   If you want to open ports 90, 80, 9000, and 8000 and this is a co-resident deployment, run the following command:

   ```
   kubectl patch hr gnwpolicy -n powerflex --type merge --patch '{"spec":{"values":
   {"customtcpports":[90,80,9000,8000],"iscores":"true"}}}'
   ```

Uninstall the PowerFlex management platform cluster

If deployment fails, and you want to reset the environment use the following procedure to uninstall or reset a PowerFlex management platform cluster.

Steps

Run the following script to reset or uninstall a PowerFlex management platform cluster.

```
reset_PFMP.sh
```

CAUTION: This script deletes the existing docker container and reboots all cluster nodes. If the PowerFlex Manager installation fails in a co-resident deployment, contact Dell Technologies Support.

Collect MDM IP address and system ID

Use the following procedure to collect MDM IP address and system ID.

Steps

1. Log in to the primary MDM using the following command:

```
scli --login --p12_path /opt/emc/scaleio/mdm/cfg/cli_certificate.p12 --p12_password
password
```

2. Enter the password.

   After you log in to the primary MDM, the system ID is displayed. Record the system ID for later tasks.
3. Use the following command to display MDM cluster information:

   `scli --query_cluster`

   Note the MDM management addresses that is required in later steps.

Perform initial configuration

After installation, use the following procedure to complete the initial configuration.

About this task

When you log in to PowerFlex Manager, you are prompted with an **Initial Configuration Wizard**, which prompts you to configure the basic settings that are required to start using PowerFlex Manager.

Prerequisites

Before you begin, ensure that you have the following information:

About the type of installation you want to perform, including details about your existing PowerFlex instance, if you intend to import from another PowerFlex instance.

Steps

1. On the **Welcome** page, read the instructions, and click **Next**.
2. If you want to enable SupportAssist, see Enabling SupportAssist, and click **Next**.
3. Select **I have a powerflex cluster to import**.
4. Select **PowerFlex 4.x**.
5. Enter each MDM IP address and click **Add IP** after each one until all MDMs are displayed.
6. Enter the MDM admin username.
7. Enter the MDM admin password.
8. Enter the primary MDM operating system username.
9. Enter the primary MDM operating system password and click **Next**.
10. PowerFlex Manager validates the information and waits for validation that is passed to display, and then click **Next**.
11. Review the summary for accuracy and click **Finish**.
12. Monitor the progress on the **Jobs** page.
13. SSH to MDMs and restart activemq on MDMs by running the following command:

    `systemctl restart activemq`
14. To add certificates to enable logging in using scli, run the following command:

    `scli --add_certificate --certificate_file /opt/emc/scaleio/mdm/cfg/mgmt_ca.pem`
15. To verify if you can log in using scli, run the following command:

    `scli --login --username admin --management_system_ip <powerflex_mgmt_ip>`

    NOTE: For *<powerflex_mgmt_ip>*, use the IP or FQDN entry from `PFMP_Config.json`.

For example,

```
"PFMPHostname": "dellpowerflex.com",
        "PFMPHostIP": "10.234.95.26"
```

Deploying into a Linux management environment

In a PowerFlex, bring your own Linux management configuration, the PowerFlex management platform is installed on the Linux operating system directly.

Deployment workflow

The steps to deploy into a Linux management environment are as follows:

NOTE: The storage for PowerFlex management platform must be at least an SSD SAS level specification, it does not need to be shared storage, each node has its own partitions to use.

Table 18. Deployment workflow steps

Step	Action	Reference
1	Configure BIOS settings and the BMC controller (if applicable).	See the *Dell Technologies PowerFlex Custom Node R650/R750/R6525 Hardware Configuration and Operating System Installation Guide for Linux* available at Dell Technologies Support Site. See the *Dell VxFlex Ready Node R640-R740xd-R840 Hardware Configuration and Operating System Guide for Linux* available at Dell Technologies Support Site.
2	Install the Linux operating system using the vendor's best practices.	When deploying the operating system reserve 200 GB for /var and 600 GB for /opt
3	Configure the network on the nodes.	-
4	Set up the yum \ zypper repository to update the packages as needed.	See the *Adding repositories* topic in the *Managing software with command line tools | Administration Guide | SUSE Linux Enterprise Server 15 SP3* available at the SUSE documentation website.
5	Deploy the PowerFlex management platform installer and run PowerFlex management platform deployment.	See Deploying the PowerFlex management platform.
6	Log in to PowerFlex Manager, deploy a PowerFlex system, import, or connect the MDM cluster.	To deploy PowerFlex, see Upgrade. If the MDM cluster is installed with PowerFlex 3.6.x, see Upgrade.

Linux management deployment prerequisites

There are certain prerequisites that are required by PowerFlex Manager before you can deploy a Kubernetes cluster:

- The following operating systems are supported:
  - Red Hat Enterprise Linux 7.9, 8.6, 8.8
  - SUSE Linux Enterprise Server (SLES) 15 SP3, 15 SP4
  - CentOS 7.9
- The following modules must be installed on the participating nodes before starting a co-resident deployment:

  For SUSE Linux Enterprise Server:

Table 19. SUSE Linux Enterprise Server modules for PowerFlex Manager

For PowerFlex Manager Install	Location	Repository
python3	On ISO	-
python3-pip	On ISO	-
sshpass	SUSEConnect	PackageHub/15.x/x86_64
java-11-openjdk-headless	On ISO	-
jq	On ISO	-
haproxy	SUSEConnect	sle-ha/15.x/x86_64
keepalived	SUSEConnect	sle-ha/15.x/x86_64
lvm2	On ISO	-
skopeo	SUSEConnect	sle-ha/15.x/x86_64
apparmor-parser	On ISO	-
libapr1	SUSEConnect	sle-module-basesystem/15.x/x86_64

For Red Hat Enterprise Linux:

Table 20. Red Hat Enterprise Linux modules for PowerFlex Manager

For PowerFlex Manager Install	Location	Repository
python3	On ISO	-
python3-pip	On Satellite	rhel-x-for-x86_64-appstream-rpms
httpd-tools	On Satellite	rhel-x-for-x86_64-appstream-rpms
libselinux-python3	On ISO	-
sshpass	On Satellite	rhel-x-for-x86_64-appstream-rpms
java-11-openjdk-headless	On Satellite	rhel-x-for-x86_64-appstream-rpms
jq	On Satellite	rhel-x-for-x86_64-appstream-rpms
haproxy	On Satellite	rhel-x-for-x86_64-appstream-rpms
keepalived	On Satellite	rhel-x-for-x86_64-appstream-rpms
lvm2	On ISO	-
skopeo	On Satellite	rhel-x-for-x86_64-appstream-rpms

- Use the following reference table for volume creation and mapping commands that are used below:

Table 21. Volume creation and mapping reference

Node	Number of volumes per node	Volume size	Partition layout
Node 1 running the PowerFlex management platform Kubernetes cluster	1	600 GB	○ Partition 1 (350 GB) ->`/opt/platform-provisioner` ○ Partition 2 (100 GB) -> `/opt/platform` ○ Partition 3 (150 GB) -> `/var/lib/rancher`
Node 2 running the PowerFlex management platform Kubernetes cluster	1	600 GB	○ Partition 1 (350 GB) -> `/opt/platform-provisioner` ○ Partition 2 (100 GB) ->`/opt/platform` ○ Partition 3 (150 GB) -> `/var/lib/rancher`
Node 3 running the PowerFlex management platform Kubernetes cluster	1	600 GB	○ Partition 1 (350 GB) -> `/opt/platform-provisioner` ○ Partition 2 (100 GB) -> `/opt/platform` ○ Partition 3 (150 GB) ->`/var/lib/rancher`

- All nodes must have their clocks that are synchronized with a valid time source.

- Ensure that the default route is configured before installing PowerFlex Manager.
- Ensure that you disable the NetworkManager service on Red Hat based deployments. If the NetworkManager is running on the Linux operating system, the `/etc/resolv.conf` file content is overwritten when updating the DNS records. For more information about disabling NetworkManager, see operating system vendor guidelines.

Deployment design for Linux management

The following table details best practices in relation to component layout:

Table 22. Component layout for Linux management deployment

Number of nodes	Node 1	Node 2	Node 3
Three nodes	Kubernetes for the PowerFlex management platform	Kubernetes for the PowerFlex management platform	Kubernetes for the PowerFlex management platform

Configuring Linux management nodes

Use the procedures in this section to configure Linux management nodes.

Perform the following:

Deploy the PowerFlex management platform installer and run PowerFlex management platform deployment.

Setting up the PowerFlex management platform

Deploying the PowerFlex management platform

This section describes how to install and configure the PowerFlex management platform.

This includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

The following diagram depicts the PowerFlex installer VM and management VM deployed in a virtual environment and highlights the interfaces needed to be configured:

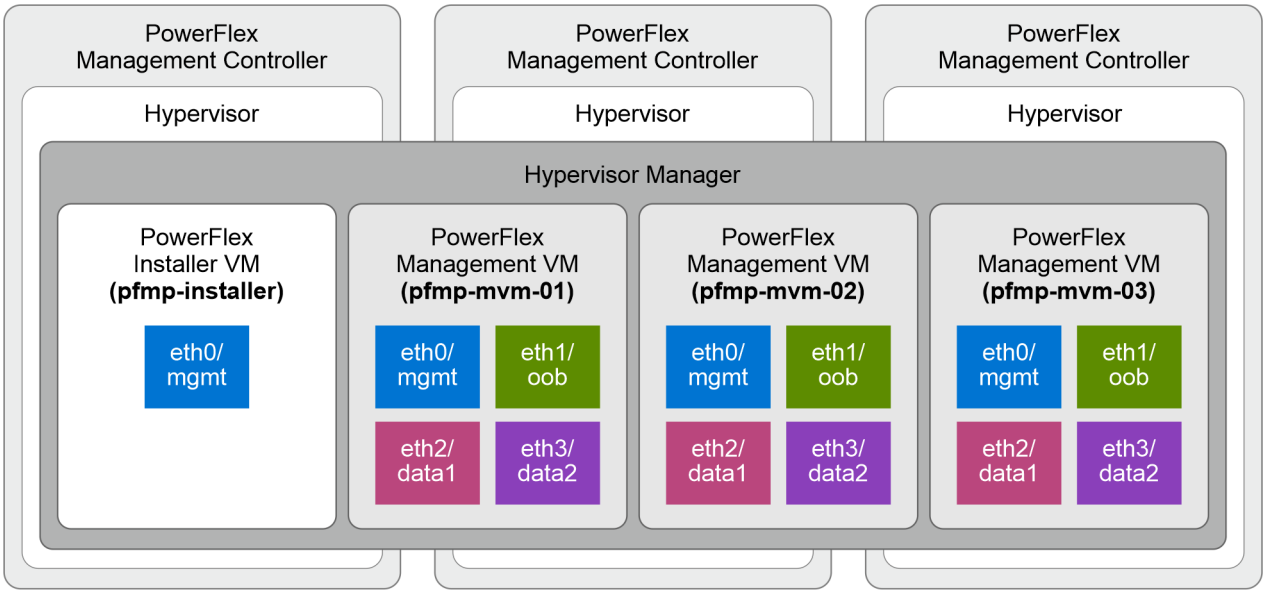


Figure 5. PowerFlex installer VM and MVMs deployed in a virtual environment

Deploy the PowerFlex management platform cluster

Use the following procedure to complete the steps to configure the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

Prerequisites

The following locations contain log files for troubleshooting:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

The following diagram illustrates how PFMP_config.JSON file content is applied to PowerFlex management.

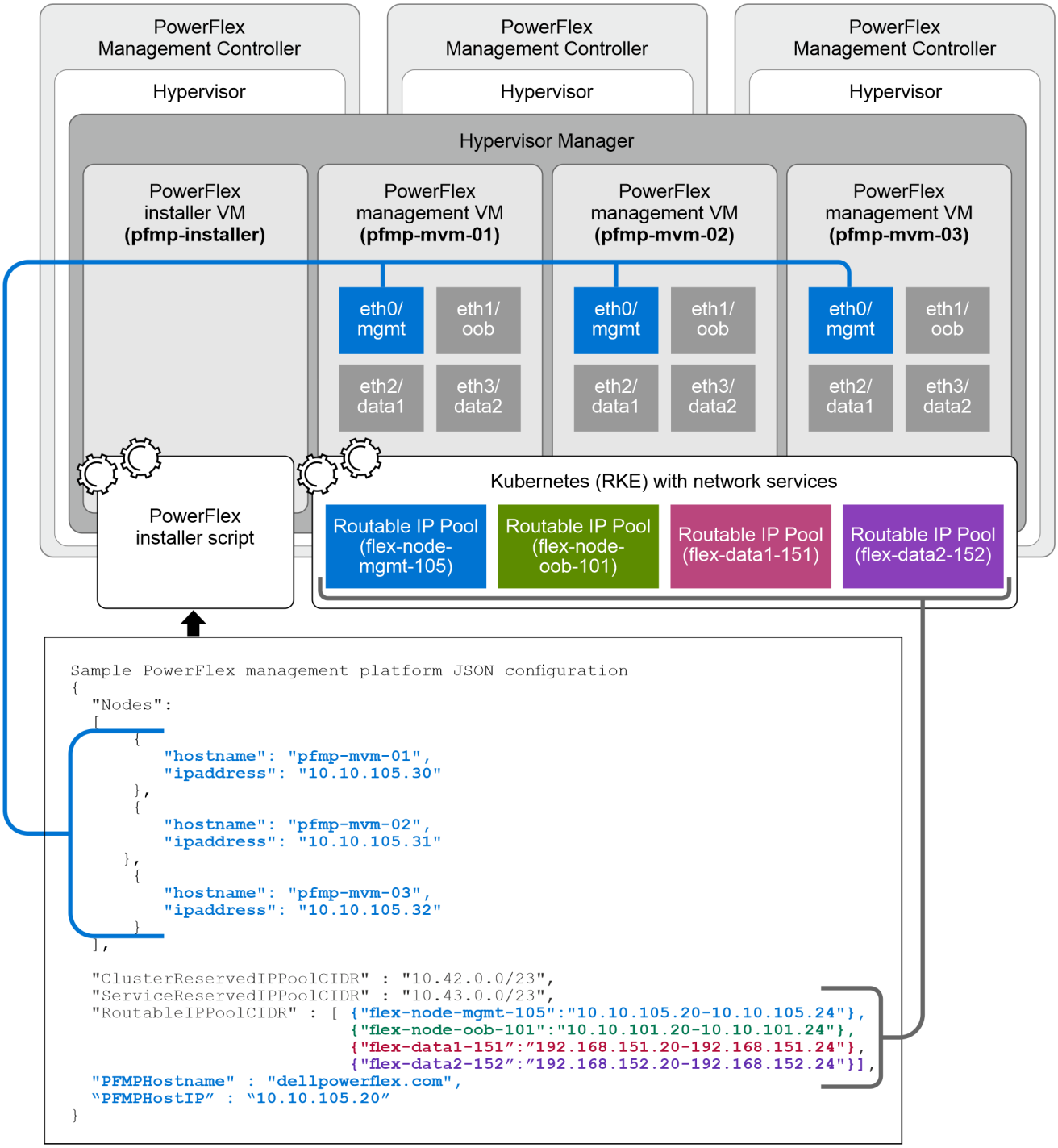


Figure 6. Relationship between the PFMP_config.JSON file and PowerFlex management

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 23. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Recommended three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
ServiceReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, SupportAssist, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Steps

1. To SSH as non-root user to the PowerFlex management platform Installer VM, run the following command: **`ssh delladmin@<pfmp installer ip>`**.
2. Go to the config directory, run the following command: **`cd /opt/dell/pfmp/PFMP_Installer/config`**
3. To configure the `PFMP_Config.json`, run the following command: **`sudo vi PFMP_Config.json`** and update the configuration parameters.

   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
```

```
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.105.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
                          {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                          {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                          {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE: **"RoutableIPPoolCIDR"** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address. PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}**. Use the **"PFMPHostname" : "dellpowerflex.com"** default FQDN to browse PowerFlex Manager. The name should be resolved to the **"PFMPHostIP"** IP address.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to nonrouted networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Go to the scripts folder in the working directory, type: **cd /opt/dell/pfmp/PFMP_Installer/scripts**
5. To set up the installer, run the following command: **sudo ./setup_installer.sh**.

   The **setup_installer.sh** starts a docker container with the appropriate network settings that deploys the PowerFlex Manager management virtual machine cluster when the **install_PFMP** script is run.
6. To install the PowerFlex management platform, run the following command: **sudo ./install_PFMP.sh**.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the SSH username of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **Yes** to indicate that the passwords are the same for all cluster nodes.
   - Enter the SSH password of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **No** to indicate that the nodes used for PowerFlex management platform are not co-resident nodes.

   NOTE: Ensure that you use the single SSH key for all nodes if you want to deploy a PowerFlex cluster using a CSV topology file.

   View the progress of the installation in the following logs that are in the working directory:

   - `<working directory>/PFMP_Installer/logs/`
   - `<working directory>/atlantic/logs/`

   NOTE: Installation may take more than two hours.

   - If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see Collect PowerFlex management platform support data.
   - For more information about troubleshooting, see Troubleshooting a PowerFlex management platform deployment.

Finalize firewall configuration

RKE2 uses Calico as a networking and security solution that enables Kubernetes workloads to communicate securely performing similar yet more dynamic functions as firewalld.

About this task

Firewalld can interfere with the dynamic nature of Calico, therefore, firewalld must be disabled. For more information, see Known issues and limitations with rke2.

Steps

To disable firewalld, do the following:

a. Log in to PowerFlex management platform node using SSH and run the following command:

```
systemctl stop firewalld
systemctl disable firewalld
```

b. To verify firewalld is stopped and disabled, run the following command:

```
systemctl status firewalld
```

Example output:

```
firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor
preset: disabled)
     Active: inactive (dead)
       Docs: man:firewalld(1)
     Do this on all remaining PFMP nodes
```

Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform install problems.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site. To get the utility file from the PowerFlex management platform installer, go to the `PFMP_Installer/scripts` directory. To download the data collection utility from the support site, search for **PFMP log collection Script** file on the Dell Technologies Support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as user root.

   For example, the utilitys executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
```

```
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as user `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support. By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, and you want to reset the environment use the following procedure to uninstall or reset a PowerFlex management platform cluster.

Steps

Run the following script to reset or uninstall a PowerFlex management platform cluster.

`reset_PFMP.sh`

CAUTION: This script deletes the existing docker container and reboots all cluster nodes. If the PowerFlex Manager installation fails in a co-resident deployment, contact Dell Technologies Support.

5

Troubleshooting a PowerFlex management platform deployment

This troubleshooting section contains verification steps and methods that can be used to identify and address misconfigurations to ensure a successful PowerFlex management platform deployment.

The PowerFlex management platform runs on Kubernetes across three virtual machines that reside on controller nodes or a customer-provided virtual environment. The PowerFlex management platform deployment process can fail for several reasons. To ensure the expected state of the PowerFlex installer and management virtual machines (MVMs) before initiating the deployment.

Verifying the network time protocol setup

Use this section to verify the network time protocol (NTP) servers are synchronized before the install is initiated.

Time synchronization across all hosts and virtual machines is required. Functioning NTP servers are required for deployment. All PowerFlex nodes and virtual machines must point to a functioning NTP server and time must be verified as synchronized before the install is attempted.

Verify the NTP setup on a vCenter Server Appliance

Use this procedure to verify the NTP is setup properly on a vCenter Server Appliance.

Steps

1. Access the vCenter Console at `https://<vcenterIP>:5480`
2. Log in as **`root`**.
3. Select **Time**.
4. Navigate to **Time Synchronization**.
5. Verify at least one of the NTP servers is set and has a green check.

Verify the NTP setup on a VMware ESXi-based PowerFlex node

Use the following procedure to verify the NTP is set up properly on a VMware ESXi-based PowerFlex node.

Steps

1. In the vSphere Client home page, go to **Home** > **Hosts and Clusters**.
2. Select a host.
3. On the **Configure** tab, select **System** > **Time Configuration**.
4. Click **Test services**.
5. Verify that the check passes and time are accurate.

Verify the NTP setup on a Linux-based PowerFlex node

Use the following procedure to verify the NTP setup on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.

2. Type the following: **chronyc tracking**.

   Output should have a valid reference ID and time should be accurate. The following gives an example output:

```
pfmp-mvm-01:~ # chronyc tracking
Reference ID    : 0AEA7C8B (dns.example.com)
Stratum         : 4
Ref time (UTC)  : Tue Jan 30 16:09:53 2024
System time     : 0.000041504 seconds slow of NTP time
Last offset     : -0.000069052 seconds
RMS offset      : 0.000188316 seconds
Frequency       : 16.336 ppm slow
Residual freq   : -0.011 ppm
Skew            : 0.077 ppm
Root delay      : 0.001050830 secondsRoot dispersion : 0.005389712 seconds
Update interval : 1034.1 seconds
Leap status     : Normal
pfmp-mvm-01:~ #
```

   If chronyc tracking returns invalid data, see the NTP setup in *Configure the NTP service*.

Verifying the DNS

PowerFlex uses mTLS and SSL certificates across all services, hosts, and virtual machines. To get proper host resolution valid DNS A-records and PTR records are required. Use this section to verify the DNS is functioning as required.

Verify DNS on vCenter Server Appliance

Use this procedure to verify DNS on vCenter Server Appliance.

Steps

1. SSH to the vCenter Appliance.
2. At the command prompt, type: **shell**.
3. Type the following: **nslookup <FQDN>**.

   Example output:

```
[root@pfmp-vcenter:~] nslookup pfmp-vcenter.example.com
Server:         10.10.105.240
Address:        10.10.105.240:53

Name:   pfmp-vcenter.example.com
Address: 10.10.105.2

[root@pfmp-vcenter:~]
```

4. Type the following: **nslookup <IP address>**.

   Example output:

```
[root@ pfmp-vcenter:~] nslookup 10.10.105.2
Server:         10.10.105.240
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-vcenter.example.com

[root@pfmp-vcenter:~]
```

Verify DNS on a VMware ESXi-based PowerFlex node

Use the following procedure to verify DNS on a VMware ESXi-based PowerFlex node.

Steps

1. SSH to the VMware ESXi-based PowerFlex node.
2. Type the following: **nslookup <FQDN>**.

   Example output:

```
[root@pfmp-r650-01:~] nslookup pfmp-r650-01.example.com
Server:         10.10.105.240
Address:        10.10.105.240:53

Name:   pfmp-r650-01.example.com
Address: 10.10.105.3

[root@pfmp-r650-01:~]
```

3. Type the following: **nslookup <IP address>**.

   Example output:

```
[root@pfmp-r650-01:~] nslookup 10.10.105.3
Server:         10.10.105.240
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-r650-01.example.com

[root@pfmp-r650-01:~]
```

Verify DNS on a Linux-based PowerFlex node

Use the procedure to verify DNS on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.
2. Type the following: **nslookup <FQDN>**.

   Example output:

```
pfmp-mvm-01:~ nslookup pfmp-mvm-01.example.com
Server:         10.10.105.240
Address:        10.10.105.240:53

Name:   pfmp-mvm-01.example.com
Address: 10.10.105.30

pfmp-mvm-01:~
```

3. Type the following: **nslookup <IP address>**.

   Example output:

```
pfmp-mvm-01:~ nslookup 10.10.105.30
Server:         10.10.105.240
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-mvm-01.example.com

pfmp-mvm-01:~
```

Check DNS on a Linux-based PowerFlex node

Use this procedure to check DNS on a Linux-based PowerFlex node.

Steps

1. SSH to the Linux-based PowerFlex node.
2. Type the following: **`nslookup <Linux IP>`**.

   Example output:

```
pfmp-mvm-01:~ nslookup 10.10.105.30
Server:         10.10.105.240
Address:        10.10.105.240:53

123.123.123.45.in-addr.arpa      name = pfmp-mvm-01.example.com

pfmp-mvm-01:~
```

Verifying the network

Use this section to verify the network. The PowerFlex management platform requires both internal and external network communication. See the *Dell PowerFlex Appliance and PowerFlex Rack with PowerFlex 4.x Cabling and Connectivity Guide* for more information.

Verify each of the three PowerFlex management virtual machines has the required interfaces:

- flex-node-mgmt-<vlanid>
- flex-oob-mgmt-<vlanid>
- flex-data1-<vlanid>
- flex-data2-<vlanid>
- flex-data3-<vlanid> (if required)
- flex-data4-<vlanid> (if required)

Verify the management virtual machine network communication

Use the following procedure to verify the management virtual machine network communication using a ping command.

Steps

1. SSH to each PowerFlex management virtual machine node.
2. For flex-node-mgmt-<vlanid> and flex-oob-mgmt-<vlanid>, type the following: **`ping <peer IP address>`**.
3. For flex-data1-<vlanid>, flex-data2-<vlanid>, flex-data3-<vlanid> (if required), and flex-data4-<vlanid> (if required), type the following: **`ping <peer IP address>`**.

   Using the following examples, verify all interfaces within the same subnet are reachable:

```
pfmp-mvm-01 mgmt/eth0 <-> pfmp-mvm-02 mgmt/eth0
pfmp-mvm-01 mgmt/eth0 <-> pfmp-mvm-03 mgmt/eth0

pfmp-mvm-01 oob/eth1 <-> pfmp-mvm-02 oob/eth1
pfmp-mvm-01 oob/eth1 <-> pfmp-mvm-03 oob/eth1

pfmp-mvm-01 data1/eth2 <-> pfmp-mvm-02 data1/eth2
pfmp-mvm-01 data1/eth2 <-> pfmp-mvm-03 data1/eth2

pfmp-mvm-01 data2/eth3 <-> pfmp-mvm-02 data2/eth3
pfmp-mvm-01 data2/eth3 <-> pfmp-mvm-03 data2/eth3

pfmp-mvm-02 mgmt/eth0 <-> pfmp-mvm-03 mgmt/eth0
pfmp-mvm-02 oob/eth1 <-> pfmp-mvm-03 oob/eth1
```

```
pfmp-mvm-02 data1/eth2 <-> pfmp-mvm-03 data1/eth2
pfmp-mvm-02 data2/eth3 <-> pfmp-mvm-03 data2/eth3
```

Example output:

```
pfmp-mvm-01:~ # ping 10.10.105.31
PING pfmp-mvm-02 (10.10.105.31) 56(84) bytes of data.
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from pfmp-mvm-02 (10.10.105.31): icmp_seq=3 ttl=64 time=0.086 ms

--- from pfmp-mvm-02 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~
```

Verify routing

Use the following procedure to verify the routing of networks.

About this task

Verify the routing for the following networks:

- flex-node-mgmt-<vlanid> route to flex-stor-mgmt-<vlanid>
- flex-node-mgmt-<vlanid> route to pfmc-sds-mgmt-<vlanid>
- pfmc-vm-mgmt-<vlanid> route to pfmc-sds-mgmt-<vlanid> (optional, used for multi-VLAN configurations)

Steps

1. SSH to each PowerFlex management virtual machine on flex-node-mgmt-<vlanid>.
2. Type the following: **`ping <pfmc-sds-mgmt-<vlandid> target ip address>`**.
3. Type the following: **`ping <flex-stor-mgmt-<vlanid> target ip address>`**.

   Example output:

```
pfmp-mvm-01:~ # ping 10.10.150.10
PING 10.10.150.10 (10.10.150.10) 56(84) bytes of data.
64 bytes from 10.10.150.10: icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from 10.10.150.10: icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from 10.10.150.10: icmp_seq=3 ttl=64 time=0.086 ms

--- from 10.10.150.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~


pfmp-mvm-01:~ # ping 10.10.140.10
PING 10.10.140.10 (10.10.140.10) 56(84) bytes of data.
64 bytes from 10.10.140.10: icmp_seq=1 ttl=64 time=0.112 ms
64 bytes from 10.10.140.10: icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from 10.10.140.10: icmp_seq=3 ttl=64 time=0.086 ms

--- from 10.10.140.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.077/0.378/1.240/0.497 ms
pfmp-mvm-01:~
```

Verifying the PFMP_config.json file

Use this section to verify the PFMP_config.json file.

The PowerFlex management platform deployment process requires a configuration file to be filled out with customer network information. The `install_pfmp.sh` script uses the data in PFMP_config.json to deploy all the required packages, install Kubernetes, and deploy all the pods that are required to run PowerFlex Manager.

The template file is found on the installer VM under: `/opt/dell/pfmp/PFMP_Installer/config/PFMP_Config.json`.

Verify the PFMP_config.JSON key input restrictions

Use this procedure to verify the key input restrictions for the PFMP_config.JSON file.

Steps

Verify the following PFMP_config.JSON key input restrictions:

- Ensure that all hostnames and network names are in lowercase.
- Management virtual machine hostname values must NOT be fully qualified (FQDN).
- ClusterReserved and ServiceReserved Pools are left at default if not used.
- RoutableIPPoolCIDR requires five unused contiguous IP addresses in each range.
- The PFMPHostname value must be fully qualified (FQDN).
- PFMPHostIP value must be in the flex-node-mgmt-<vlanid> range.

6

Configuring PowerFlex Manager

After installation, log in to PowerFlex Manager and complete the initial configuration procedures.

The remaining PowerFlex installation and configuration tasks are then performed using PowerFlex Manager.

NOTE: To deploy the PowerFlex management platform, see Preparing the PowerFlex Manager environment.

Log in to PowerFlex Manager

Log in to PowerFlex Manager to continue configuration activities and deploy storage resources.

Prerequisites

- Ensure that you have access to a web browser that has network connectivity with PowerFlex Manager.
- Ensure that you know the address that was configured for accessing PowerFlex Manager. This address was configured for *PFMPHostname* in the JSON file, during PowerFlex Manager installation.
- Prepare a new password for accessing PowerFlex Manager. The default password cannot be considered as a new password. For more information about default credentials, see the Dell PowerFlex 4.5.x Security Configuration Guide.
- Password rules are:
  - Contains fewer than 32 characters
  - Contains only alphanumeric and punctuation characters

Steps

1. Log in to PowerFlex Manager using the default credentials. The default credentials are provided in the Dell PowerFlex 4.5.x Security Configuration Guide.
2. Change the default password.

   Make a note of the password for future use. This password is for the SuperUser who is performing the initial configuration activities. Additional users and passwords can be configured later in the process.
3. Click **Submit**.
   You are now logged into the system. The initial configuration wizard is displayed. Proceed with initial configuration activities.

Enabling SupportAssist

SupportAssist is a secure support technology for the data center. SupportAssist refers to Secure Connect Gateway, which is used for call home functionality and remote connectivity. You can enable SupportAssist, as part of the initial configuration wizard. Alternatively, you can enable it later by adding it as a destination to a notification policy in Events and Alerts.

About this task

PowerFlex Manager provides support through integration with the secure connect gateway ensuring better alignment with Dell Technologies services initiatives to enhance the user experience. SupportAssist is the functionality that enables this connection.

When you enable SupportAssist, you can take advantage of the following benefits, depending on the service agreement on your device:

- Automated issue detection - SupportAssist monitors your Dell Technologies devices and automatically detects hardware issues, both proactively and predictively.
- Automated case creation - When an issue is detected, SupportAssist automatically opens a support case with Dell Technologies Support.
- Automated diagnostic collection - SupportAssist automatically collects system state information from your devices and uploads it securely to Dell Technologies. Dell Technologies Support uses this information to troubleshoot the issue.
- Proactive contact - A Dell Technologies Support agent contacts you about the support case and helps you resolve the issue.

The first time that you access the initial configuration wizard, the connection status displays as not configured.

Prerequisites

Ensure you have the details about your SupportAssist configuration.

About this task

If you do not enable SupportAssist in the initial setup, you can add it later as a destination when adding a notification policy through **Settings** > **Events and Alerts** > **Notification Policies**.

Steps

1. Click **Enable SupportAssist**.
2. From the **Connection Type** tab, there are two options:

If you want to:	Do the following:
Connect directly to Dell Technologies.	Click **Connect Directly**.
Connect to a SupportAssist gateway server that connects to Dell Technologies by remote access.	Click **Connect via Gateway Server** and provide the following details: • The IP and port number that is assigned to the primary and secondary gateway servers. The port number is auto populated to **9443**. • Optionally, click **Add Gateway** to include an additional gateway server. The maximum number of gateways is eight. • If your environment has a proxy server, enter the IP address, port number, username, and password for the proxy server. • Enter the access key and PIN for authentication to SupportAssist. • Optionally, click **Test connection** to test the connection status. • Optionally, click **Send Test Alert** to send a test alert.

3. From the **Support Contacts** tab, identify the primary support contact by providing their first name, last name, email, and phone details. Optionally, you can add up to two additional support contacts.
4. From the **Device Registration** tab, register your device by completing the following steps:
   a. Select the type of device registration from the **Type** menu.
   b. Depending on the device registration type, you may have to enter information:

Table 24. Device registration types and required information

Device registration type	Information required
• PowerFlex rack • PowerFlex appliance	• Enterprise License Management Systems (ELMS) Software Unique ID (eSWUID) • Solution serial number
PowerFlex software	N/a

5. From the **Customize Connection to SupportAssist** tab, you can:
   a. Click **Connect to CloudIQ** to enable Dell Technologies to send telemetry data, alerts, and analytics through SupportAssist.
   b. Click **Enable Remote Support** to enable authorized Dell Technologies support engineers to troubleshoot your system remotely.
6. From the **Re-authenticate SupportAssist** tab, click **request a New Access Key and PIN** to generate an access key and pin. The **AccessKey Portal** opens.
   a. From the menu, select **SERIALNUMBER**.
   b. Enter the serial number of your device or PowerFlex software and click **Submit**.
   c. Click **Generate New Access key**.
7. From the **Re-authenticate SupportAssist** tab, enter the access key and pin.
8. Click **Next**.

Next steps

After you complete the initial configuration, go to the **Settings** page and verify that SupportAssist has been turned on. You may need to configure the SupportAssist policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.

Getting started

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

If you would like to use the legacy PowerFlex Installer to provide an installation topology file, click **Deploy With Installation File**. See Deploying a PowerFlex cluster using a CSV topology file for more information.

This is a list of options available in the getting started screen. Not all are required for all system types as noted.

Table 25. Getting started options

Step	Description
Add compliance file	Provide compliance file location and authentication information for use within PowerFlex Manager. The compliance file defines the specific hardware components and software version combinations that are tested and certified by Dell for hyperconverged infrastructure and other Dell products. This step enables you to choose a default compliance version for compliance or add new compliance versions. You can also click **Settings** > **Repositories** > **Compliance Versions**. Before you make the software catalog the default compliance version, upload a suitable compatibility matrix file under **Settings** > **Repositories** > **Compatibility Management**. If you have more than one compliance version, set one of them as default.
Define networks	Enter detailed information about the available networks in the environment. This information is used later during deployments to configure nodes to have the right network connectivity. PowerFlex Manager uses the defined networks in templates to specify the networks or VLANs that are configured on PowerFlex nodes for your resource groups. This step is enabled immediately after you perform an initial configuration for PowerFlex Manager. You can also click **Settings** > **Networking** > **Networks**. **NOTE:** This step is not applicable for co-resident deployments.
Discover resources	Grant PowerFlex Manager access to resources (nodes) in the environment by providing the management IP address and credential for the resources to be discovered. You can also click **Resources** > **Discover Resources**. **NOTE:** This step is not applicable for co-resident deployments.
Manage deployed resources (Optional)	Add an existing resource group for a cluster that is already deployed and manage the resources within PowerFlex Manager. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Resource Groups** > **Add Existing Resource Group**. **NOTE:** This step is not applicable for co-resident deployments.
Deploy resources	Create a template with requirements that must be followed during a deployment. Templates enable you to automate the process of configuring

Table 25. Getting started options (continued)

Step	Description
	and deploying infrastructure and workloads. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and make modifications as needed. Choose the sample template that is most appropriate for your environment. For example, for a PowerFlex hyperconverged deployment, clone one of the PowerFlex hyperconverged templates. For a PowerFlex two-layer deployment, clone the compute-only templates. Then clone one of the storage templates. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Templates**. **NOTE:** This step is not applicable for co-resident deployments.
Manage PowerFlex license	Configure licensing for PowerFlex. You can also click **Settings** > **License Management**.

To revisit the **Getting Started** page, click **Getting Started** on the help menu.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

Steps

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from SupportAssist (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
     - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to SupportAssist.
3. If you selected **Download from SupportAssist (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from SupportAssist, you must configure SupportAssist. To configure the SupportAssist, you must enable SupportAssist in the **Initial Configuration** wizard.

   If you are downloading a compliance file from SupportAssist, the file type is a ZIP or TAR.GZ file.

   NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: **`\\host\share\filename.zip`**
     - CIFS share for TAR.GZ file: **`\\host\share\filename.tar.gz`**
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Mark the target upgrade catalog as the default version for compliance checking.

   Download the 3.6.x and the 4.5.x software catalogs. PowerFlex requires the catalog for the version that the system is on and the target version.

5. Click **Save**.

PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

If the compliance file contains an embedded ISO file for an operating system image (such as for an VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Upload the compatibility matrix file

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if the software catalog versions that are currently loaded are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from SupportAssist or upload it from a local directory path. The file has a GPG extension and an associated compatibility version number.

Prerequisites

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using SupportAssist, ensure that SupportAssist has been properly configured.

Steps

1. On the menu bar, click **Settings** and click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies SupportAssist** if you are using SupportAssist.
4. Click **Upload from Local** to use a local file. Then, click **Choose File** to select the GPG file.

7

Deploying PowerFlex storage

The next stage of installation involves defining infrastructure and resources, and then deploying PowerFlex storage.

Deploy the PowerFlex data storage as a resource group by using one of the following methods:

- Deploy a PowerFlex cluster using REST API commands
- Deploying a PowerFlex cluster using PowerFlex Manager templates and resource groups
- Deploying a PowerFlex cluster using a CSV topology file from PowerFlex Manager

NOTE: This is an advanced deployment method for deploying block storage. If you use this method, ensure that you are familiar with setting up the CSV file, as described in detail in this guide.

Operating system requirements

The following table lists the operating system requirements for PowerFlex 4.5.x. A valid time source should be configured on all nodes prior to PowerFlex deployment.

For the most updated list of operating systems supported by PowerFlex, see *Dell PowerFlex 4.5.x Support Matrix* at Dell Technologies Support site for the supported versions.

Table 26. Operating system requirements

Operating system	Requirement
Linux	Packages required for all components, all Linux types: • numactl • libaio • wget • apr • libapr • python2-rpm or python3-rpm • yum-utils Additional packages required for MDM components: • bash-completion (for SCLI completion) • Latest version of Python 2.X or Python 3.X (for newer operating system such as CentOS or RHEL 8.x) • binutils • java-11-openjdk-headless Additional packages required for storage data server: • smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils Additional packages required for diagnostic tools: • sar • sysstat Additional packages required for supported storage controllers: • DELL: perccli

Table 26. Operating system requirements (continued)

Operating system	Requirement
	• LSI raid controllers: storcli • HP raid controllers: hpssacli Additional packages required for SDS components that contain NVDIMMs: • ndctl • jq • daxio • libpmem To use the secure authentication mode, ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system. • `libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm` • `openssl1-1.0.1g-0.40.1.x86_64.rpm` To use LDAP, ensure that OpenLDAP 2.4 is installed on all servers. Additional packages needed for LIA: pciutils
Windows	• Only SDC is a supported component on Windows. • To install SDC, ensure that Security Update for Windows 7 for x64-based Systems (KB3033929) is installed. To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1, however, is not supported) • Visual C++ redistributable 2010 package, 64-bit The Microsoft Windows *Storage Spaces* feature (available from Windows 8.1 and later) is not supported in Microsoft failover cluster running on PowerFlex volumes.
VMware ESXi	Compute-only (SDC) support: VMware ESXi operating system: 7.0 (SDC only)

Preparing the storage node

For large-scale systems with up to 2048 storage data client, use the following procedure to prepare the operating systems of servers that host PowerFlex metadata manager, storage data server, and storage data replication components.

Prepare Linux-based server operating systems for support of large-scale systems (up to 2048 SDCs)

For large-scale systems with up to 2048 storage data client, use this procedure to prepare the Linux-based operating systems of servers that host PowerFlex metadata manager, storage data server, and storage data replication components.

Prerequisites

Superuser permissions are required to configure the servers.

About this task

Configure the servers hosting MDM, SDS, and SDR components to support large-scale systems with up to 2048 SDCs.

- A system setting must be validated and configured on servers hosting MDM, SDS, and SDR.
- For all servers hosting MDM, SDS, and SDR components, certain threshold parameters must be configured. This change enables a larger number of records in the Address Resolution Protocol (ARP) cache table of the operating system.

Steps

1. On servers hosting PowerFlex metadata manager, storage data server, and storage data replication components, do the following:
   a. Run this command:

      ```
      cat /proc/sys/fs/file-max
      ```

      The output for the command is the number of files that the operating system can open.
   b. If the number is 32768 or higher, go to step 2. If the number is less than 32768, go to the next substep.
   c. Open the file `/etc/sysctl.conf` using the vi editor.
   d. If the following line exists, edit it with the vi editor. If it does not exist, add the full line with the following setting:

      ```
      fs.file-max = 32768
      ```

2. On servers hosting MDM, SDS, or SDR components, if the following lines exist in the file `/etc/sysctl.conf`, edit them with the vi editor. If they do not exist, add the full lines with the following settings:

   ```
   net.ipv4.neigh.default.gc_thresh1 = 8192
   net.ipv4.neigh.default.gc_thresh2 = 16384
   net.ipv4.neigh.default.gc_thresh3 = 32768
   ```

3. Save the file.
4. Apply the changes using the command:

   ```
   sysctl -p
   ```

5. Repeat the steps on the next server, until all servers are configured.

Prepare the DAX devices

If you are using NVDIMM devices for acceleration, perform the following procedures to prepare the NVDIMMs as DAX acceleration devices. This step is not required if you are deploying with PowerFlex Manager as a resource group.

NVDIMM information table

Before you prepare the NVDIMM as a DAX device, use the following NVDIMM information table. Use it to record the information that you discover while preparing an NVDIMM as a DAX device. You need the information to add the acceleration device to PowerFlex.

Table 27. NVDIMM information table

Item	Replacement NVDIMM
Serial Number	-
Device name (NMEM)	-
Namespace	-
DAX device name (chardev)	-
Acceleration device path	-

Prepare an NVDIMM as a DAX (acceleration) device

Prepare a new or replacement NVDIMM as a DAX or acceleration device before adding it to the SDS in a Linux system.

Steps

1. Using SSH, log in to the Linux server.

2. View information for the new or replacement NVDIMM:

```
dmidecode --type memory | grep "Non-" -B 3 -A 3 | grep -E 'Locator|Serial' | grep -v
Bank
```

Output similar to the following appears:

```
Locator: A7
Serial Number: 17496594
Locator: B7
Serial Number: 174965AC
```

3. Find the serial number in the output and record it in the NVDIMM information table.
4. Display the correlation between the ID and NMEM device name of each NVDIMM mounted on the server:

```
ndctl list -Dvvv | jq '.[].dimms[]'
```

Output similar to the following appears:

```
{
  "dev": "nmem1",
  "id": "802c-0f-1722-174965ac",
  "handle": 4097,
  "phys_id": 4370,
  "health": {
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}
{
  "dev": "nmem0",
  "id": "802c-0f-1711-16492521",
  "handle": 1,
  "phys_id": 4358,
  "health": {
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}
```

5. In the output from the previous step, find the device (dev) with the `id` that partially correlates with the serial number you discovered previously for the failed device.

For example:

- The NVDIMM output displays serial number 16492521 for the NVDIMM device.
- In the previous step, the output displays the ID of device nmem0 as 802c-0f-1711-16492521.

Table 28. NVDIMM serial number and device ID correlation

Serial number	Device ID

`Locator: A7`
`Serial Number: 16492521`
`Locator: B7`
`Serial Number: 1649251B`
	`"dev": "nmem0",`
	`"id": "802c-0f-1711-16492521",`

In the above example, the NMEM name is nmem0.

6. Record the NMEM name in the `Device name` row of the NVDIMM information table.
7. Correlate between the NMEM DIMM and the namespace/DAX device (acceleration device):

```
ndctl list -Dvvv | jq '.[].dimms[]'
```

Output similar to the following appears:

```
{
  "dev": "nmem1",
  "id": "802c-0f-1722-174965ac",
  "handle": 4097,
  "phys_id": 4370,
  "health": {
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}
{
  "dev": "nmem0",
  "id": "802c-0f-1722-17496594",
  "handle": 1,
  "phys_id": 4358,
  "health": {
    "health_state": "ok",
    "temperature_celsius": 255,
    "life_used_percentage": 30
  }
}

ndctl list -Dvvv | jq '.[].regions[].namespaces[]'

{
  "dev": "namespace1.0",
  "mode": "raw",
  "size": 17179869184,
  "sector_size": 512,
  "blockdev": "pmem1",
  "numa_node": 1
}
{
  "dev": "namespace0.0",
  "mode": "raw",
  "size": 17179869184,
  "sector_size": 512,
  "blockdev": "pmem0",
  "numa_node": 0
}
```

8. To identify the new NVDIMM. You can run the command below to find the NVDIMM region with no associated namespace. For example, region1 use to have namespace1.0 until the NVDIMM was replaced.

```
ndctl list --regions --verbose
{
  "regions":[
    {
      "dev":"region1",
      "size":17179869184,
      "available_size":17179869184,
      "max_available_extent":17179869184,
      "type":"pmem",
      "numa_node":1,
      "iset_id":-4649799281728009192,
      "persistence_domain":"unknown"
    },
    {
      "dev":"region0",
      "size":17179869184,
      "available_size":0,
      "max_available_extent":0,
      "type":"pmem",
      "numa_node":0,
      "iset_id":4335710697273129127,
      "persistence_domain":"unknown",
      "namespaces":[
        {
```

```
            "dev":"namespace0.0",
            "mode":"devdax",
            "map":"dev",
            "size":16909336576,
            "uuid":"265ef925-571d-4a0a-bb96-dca4a3873000",
            "raw_uuid":"ff7b05b9-510e-4604-8329-7f3dccf10300",
            "daxregion":{
              "id":0,
              "size":16909336576,
              "align":4096,
              "devices":[
                {
                  "chardev":"dax0.0",
                  "size":16909336576
                }
              ]
            },
            "align":4096,
            "name":"",
            "numa_node":0
          }
        ]
      }
    ]
}
```

The NVDIMM that was replaced or sanitized will reuse the same region[x] but will not have a configured namespace.

On storage-only nodes, the region carries over from the old NVDIMM to the newly installed or sanitized NVDIMM. This is the device that you want to create a namespace for.

For example, the new NVDIMM shows up in the command above as **`region1`**. The command below will create a dax device and namespace based on the region.

9. Run this command to create the namespace, and DAX on the new NVDIMM labeled region1.

```
ndctl create-namespace --mode=devdax --region=region<X> --align=4k
```

For ndctl build 61.2 or later, run:

```
ndctl create-namespace --mode=devdax --region=region<X> --align=4k --no-autolabel
```

NOTE: `--no-autolabel` is only required for Red Hat Enterprise Linux 7.6.

For example,

```
ndctl create-namespace --mode=devdax --region=region1 --align=4k
{
  "dev":"namespace1.0",
  "mode":"devdax",
  "map":"dev",
  "size":"15.75 GiB (16.91 GB)",
  "uuid":"e47c3ce0-8a98-4256-8fed-ff2bacead155",
  "daxregion":{
    "id":1,
    "size":"15.75 GiB (16.91 GB)",
    "align":4096,
    "devices":[
      {
        "chardev":"dax1.0",
        "size":"15.75 GiB (16.91 GB)"
      }
    ]
  },
  "align":4096
}
```

10. Run this command again to validate the namespace was created. In this example, you can see namespace1.0 and dax1.0 was created for region1 NVDIMM.

```
ndctl list --regions --verbose
{
  "regions":[
    {
      "dev":"region1",
      "size":17179869184,
      "available_size":17179869184,
      "max_available_extent":17179869184,
      "type":"pmem",
      "numa_node":1,
      "iset_id":-4649799281728009192,
      "persistence_domain":"unknown"
    },
  {
    "dev":"namespace1.0",
    "mode":"devdax",
    "map":"dev",
    "size":16909336576,
    "uuid":"e47c3ce0-8a98-4256-8fed-ff2bacead155",
    "chardev":"dax1.0",
    "align":4096
  },
    {
      "dev":"region0",
      "size":17179869184,
      "available_size":0,
      "max_available_extent":0,
      "type":"pmem",
      "numa_node":0,
      "iset_id":4335710697273129127,
      "persistence_domain":"unknown",
      "namespaces":[
        {
          "dev":"namespace0.0",
          "mode":"devdax",
          "map":"dev",
          "size":16909336576,
          "uuid":"265ef925-571d-4a0a-bb96-dca4a3873000",
          "raw_uuid":"ff7b05b9-510e-4604-8329-7f3dccf10300",
          "daxregion":{
            "id":0,
            "size":16909336576,
            "align":4096,
            "devices":[
              {
                "chardev":"dax0.0",
                "size":16909336576
              }
            ]
          },
          "align":4096,
        }
      ]
    }
```

11. Record the acceleration device name in the NVDIMM information table.
12. Find the full acceleration device path:

```
    ls -ltr /dev/ | grep dax
```

The following output should appear:

```
/dev/daxX.X
```

For example:

```
/dev/dax0.0
```

13. Record the acceleration device path in the NVDIMM information table.

Deploying PowerFlex using a non-root user

Install or expand PowerFlex in Linux environments using a nonroot sudo user in noninteractive mode. This option is used only when a CSV file is used to install PowerFlex.

Sudo is a program that allows you to run or install a program as the root user. Create a sudo user to install PowerFlex.

To deploy PowerFlex with a nonroot user, the nonroot user must meet the following conditions:

- The username in the CSV file must already exist.
- A nonroot user must be a sudo user.
- Nonroot user must be in noninteractive mode.
- The requirement for TTY must be disabled.

In the CSV file used for installation, you must indicate that you are intending to use a sudo nonroot username by appending the string **(sudo)** to the username in the **Username** field. For example, if you are using a nonroot user with the username *non_root*, enter the string **non_root(sudo)** in the username field of the CSV file.

Configure a non-root non-interactive sudo user

In Linux, you can install or extend PowerFlex with a nonroot user when installing PowerFlex using a CSV file. You must configure a nonroot sudo user in noninteractive mode.

About this task

The following procedure describes one method for configuring a nonroot noninteractive sudo user. Perform the commands from the operating system console of the node where PowerFlex will be installed.

Steps

1. Create a user group named **admin**.

```
groupadd admin
```

2. Create a user name **non_root** and add it to the admin group.

```
useradd -G admin non_root
```

3. Change the password of the non_root user.

```
passwd non_root
```

When prompted, enter the new password and then confirm it by entering it again.

4. Open the sudoers `/etc/sudoers` file for editing.

```
vi /etc/sudoers
```

5. Search the sudoers file for **## Same thing without a password**.

```
:s/## Same thing without a password
```

6. In the line below the search result, add the text `%admin ALL=(ALL) NOPASSWD: ALL` to the file.
7. Search the sudoers file for **Defaults requiretty**, and replace it with `Defaults ! requiretty`.
8. Exit the vi editor by typing the following command to exit: `:wq!`

9. Create a hidden directory in the non_root user's home directory to store the SSH configuration.

```
mkdir /home/non_root/.ssh
```

10. Copy the SSH configuration from the root user to the non_root user's directory.

```
cp -rf /root/.ssh/* /home/non_root/.ssh/
```

Deploying a PowerFlex cluster using a CSV topology file

This section describes how to prepare a CSV topology file, upload the CSV topology file using REST API commands or PowerFlex Manager, and then deploy block storage using REST API commands or PowerFlex Manager.

If you are using the legacy PowerFlex to provide an installation topology file using PowerFlex Manager, click **Deploy With Installation File**.

NOTE: This is an advanced deployment method for deploying block storage. If you use this method, ensure that you are familiar with setting up the CSV file, as described in detail in this guide.

Consider the following:

- While deploying PowerFlex using PowerFlex installer, if you see an error indicating that the channel cannot be opened, then verify if PowerFlex is already deployed.
- While deploying additional servers with SDTs on an existing system with components such as SDTs, you may see errors listed in the **Events** tab. These errors are for the already existing components, such as, SDTs in the cluster.
- Ensure that you use the single SSH key for all nodes if you want to deploy a PowerFlex cluster using a CSV topology file.

Prepare the CSV topology file

Create a CSV topology file when a CSV file and REST API commands are used to deploy block storage. Follow the instructions that match your environment and the PowerFlex support matrix for your version. CSV file templates are provided as part of the software download.

Prerequisites

Ensure that the settings you enter comply with PowerFlex product limits. See the *Dell PowerFlex 4.5.x Technical Overview* for product limits.

Ensure that you have a copy of the following CSV templates. Select a template that is most suitable for your requirements. In this explanation, we refer to and illustrate the CSV file as a spreadsheet.

Table 29. CSV templates

CSV template	Description
`PowerFlex_config_example_all fields.csv` (All fields)	This spreadsheet template contains all available fields, both required and optional. In particular, it contains an optional section for storage pool configuration, which is relevant for multiple nodes.
`PowerFlex_basic_example_minimum fields.csv` (Minimum fields)	This spreadsheet template contains only the required fields. The optional fields (those that are in the all fields spreadsheet, but not in the minimum fields spreadsheet) will be assigned default values.

About this task

NOTE: During CSV file preparation, note that there are important differences between hyperconverged and two-layer topologies.

- In hyperconverged topologies, each row in the upper section represents a server that can host MDM, SDS, and SDC components
- In two-layer topologies where front-end servers are Linux or Windows-based, SDCs must be represented by separate, additional rows. The values for SDC-specific rows are described in *CSV topology for two-layer deployment*. Alternatively, omit the SDCs from the CSV file, and install them manually after deploying the backend.

NOTE: Save a copy of the CSV file for future use. It is recommended to save the file in a secure encrypted folder on your operating system, after deployment. When you add nodes to the system in the future, you need the CSV file that you used for initial deployment of the system.

You can edit a CSV file with Microsoft Excel or other file-editing software.

Steps

1. In the CSV file, enter your site-specific information in the appropriate places. You can overwrite the default information provided in the file.

   You must only use one spreadsheet for the installation, as follows:

   - To manually enter all configuration details, use the all fields spreadsheet.
   - To use default values for the nonmandatory fields, use the minimum fields spreadsheet.
   - To configure nondefault values for columns that are not in the minimum fields spreadsheet, either use the all fields spreadsheet, or copy the column heading from there into the minimum fields spreadsheet, and enter your custom values into the minimum fields spreadsheet.

   NOTE: The examples provided here do not show all possible parameters that are configurable in a PowerFlex system.

   The following figure shows part of a sample an all fields CSV file used to deploy Linux nodes in a hyperconverged system:

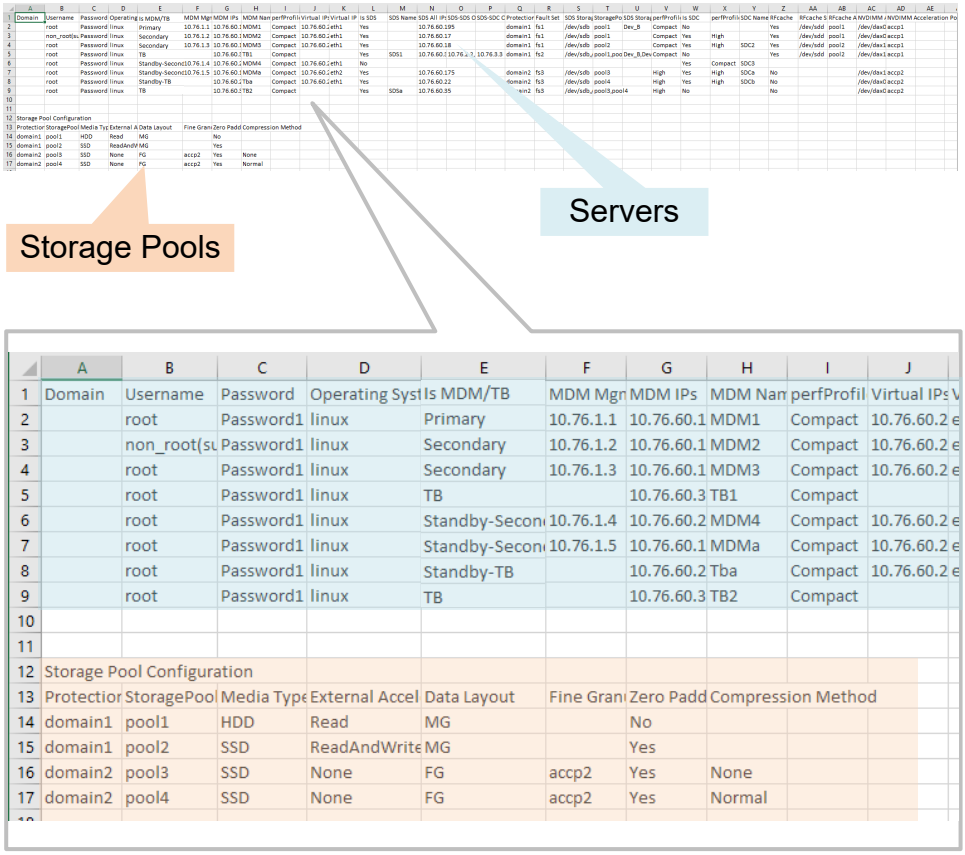


Figure 7. Sample CSV file fields used to deploy Linux nodes in a hyperconverged system

The following figure shows a sample an all fields CSV file, where SDCs are defined below the rows for MDMs and SDSs. In a two-layer configuration, the Linux servers form the MDM cluster, and each of those servers will also be SDSs. SDCs can be added here, in separate rows, or added to the system later.

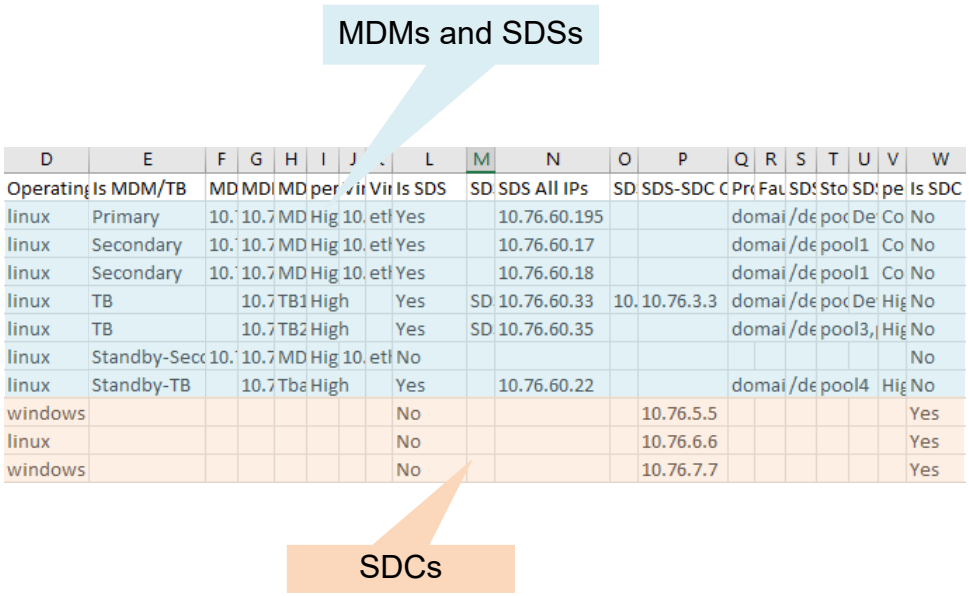


Figure 8. Sample on CSV file fields where SDCs are defined below rows for MDMs and SDSs

2. For Linux nodes, the **Domain** column is not relevant. Leave the column blank or remove the column.
   The following table describes the fields in the spreadsheets. The required fields appear in both all fields and minimum fields spreadsheets. Field names are case-sensitive; the order of the columns is not significant.

Table 30. CSV topology fields

Field	Description	Required
Domain	If using a domain user, the name of the domain (not relevant for Linux)	-
Username	The name of the user. • Linux nodes: Enter either `root` or a nonroot sudo user, indicated by appending **`(sudo)`** to the end of the username (default: `root`). For more information about deploying with a nonroot user, see *Deploying PowerFlex using a non-root user*. • Windows nodes: Enter a user with administrator rights (default: `administrator`).	-
Password	Password used to log in to the node. This should be the password of the user entered in the **Username** column.	Yes
Operating system	The operating system of the server: `linux` or `windows`	Yes
IPs	IP address to be used for multiple purposes. Use this field to designate one IP address that will be assigned to all the following: MDM IP, MDM Mgmt IP and SDS All IP. This option is provided for use cases where separate networks for data and management are not required.	-
Is MDM/TB	The MDM role to deploy on this node: `Primary`, `Secondary`, `TB`, `Standby-Secondary`, `Standby-TB`, or blank (if not an MDM). For more information, see the *Dell PowerFlex 4.5.x Technical Overview* for information about the MDM cluster.	Yes
MDM Mgmt IP	The IP address for the management-only network. The management IP address is not required for: tiebreaker, standby tiebreaker, and any host that is not an MDM. In such cases, leave this field blank.	-
MDM IPs	MDM IP addresses used to communicate with other PowerFlex components in the storage network. This is required for all MDMs, tiebreakers, and standbys. Leave this field blank for hosts that are not part of the MDM cluster.	-

Table 30. CSV topology fields (continued)

Field	Description	Required
	Maximum of eight addresses, comma-separated, no spaces.	
MDM Name	The name to be assigned to an MDM node, up to 31 characters in length. MDM names must be unique and may not contain spaces. Ensure that each server on which MDM is installed has a unique hostname.	-
perfProfileForMDM	Optional performance profile to set for MDMs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature.	-
Virtual IPs	A virtual IP address (VIP) for each possible manager MDM.] This virtual IP address can be used for communication between the MDM cluster and SDCs on the PowerFlex data networks (not applicable for the management network). Only one virtual IP address can be mapped to each NIC, with a maximum of four virtual IP addresses per system.	-
Virtual IP NICs	The NIC to which the virtual IP addresses are mapped. Maximum length of 11 characters.	-
Is SDS	Deploy the SDS component on this node: `Yes` or `No`	Yes
SDS Name	The name for the SDS node, up to 31 characters in length. The name must not contain spaces. Ensure that each server on which SDS is installed has a unique hostname.	-
SDS All IPs	SDS IP addresses to be used for communication among all nodes. Maximum of eight addresses, comma-separated, no spaces. **NOTE:** When using this option, all roles use the same IP addresses, and there is no need to fill the SDS\SDC only columns.	-
SDS-SDS Only IPs	SDS IP addresses to be used for communication among SDS nodes. When the replication feature is used, these addresses are also used for SDS-SDR and SDT-SDT communication. Maximum of eight addresses, comma-separated, no spaces.	-
SDS-SDC Only IPs	SDS IP addresses to be used for communication among SDS and SDC nodes only. Maximum of eight addresses, comma-separated, no spaces. For SDC-only nodes, enter the IP address in this column.	-
Protection Domain	The protection domain to which to assign this SDS or SDR. The name can be up to 31 characters in length.	-
Fault set	The fault set for an SDS. **NOTE:** When using fault sets, you must assign a minimum of three. See the *Dell PowerFlex 4.5.x Technical Overview* for guidelines on fault sets.	-
SDS Storage Device List	Storage devices to be added to an SDS. For more than one device, use a comma-separated list, with no spaces. Ensure that the device names and paths do not contain any special characters, such as: `?, `, !, @, #` See the *Dell PowerFlex 4.5.x Technical Overview* to configure direct attached storage (DAS). Device name format on Linux: `/dev/sdb,/dev/sdc`	-

Table 30. CSV topology fields (continued)

Field	Description	Required

	When specifying the SDS device path on a Linux node, use the path according to how it is listed in `cat/proc/partitions` (and not according to the output of `fdisk -l`). Use persistent device names.
	
	For example:
	
	`fdisk` output: `/dev/mapper/samah-lv1, /dev/sdb`
	
	`cat /proc/partitions` output: `dm-3, sdb`
	
	Use these values in the CSV file: `/dev/dm-3, /dev/sdb`
	
	To enable volume creation, you must add (at least) one device to (at least) three SDSs, where each SDS is in a separate fault set. Each device has a minimum of 130 GB free storage capacity (an SDS which has not been assigned to a fault set is treated as a fault set of its own). You can do that using the CSV file, or at a later stage. For the maximum number of devices per SDS, see product limits in the *Dell PowerFlex 4.5.x Technical Overview*.
	
	Device data is erased when devices are added to an SDS. When adding a device to an SDS, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is displayed for each device.
	
	PowerFlex might not perform optimally if there are large differences between the sizes of the devices in the storage pool—for example, if one device is as big as the rest of the devices. After adding devices, you can define how much of the device capacity is available to PowerFlex by using the SCLI command `modify_sds_device_capacity`.
	
	For optimal performance, try to balance the number of devices of a storage pool, and the capacity of those devices among the relevant SDSs.	

StoragePool List	Sets storage pool names, up to 31 characters in length. Use this option in one of the following ways to assign storage pools to the devices in the **SDS Storage Device List**: • A comma-separated list of storage pool names; the length of the list must be the same length as the list of devices in the **SDS Storage Device Names** list. In this case, the storage pools are mapped to the devices in that list, respectively. The list must be comma-separated, with no spaces. • If one storage pool is entered here, the same storage pool are mapped to all the devices in the **SDS Storage Device List**. • If no storage pool is listed, a storage pool is automatically created during installation, named `default`.	-
SDS Storage Device Names	Sets names for devices. Assigns device names to the devices in the **SDS Storage Device List**, respectively. The number of names must equal the number of devices. Comma-separated, no spaces.	-
perfProfileForSDS	Optional performance profile to set for SDSs: `High` (default) or `Compact` When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see *Configure performance profile during deployment*.	-
Is SDC	`Yes` or `No`.	Yes
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see *Configure performance profile during deployment*.	-
SDC Name	Sets a name for the SDC, which can be up to 31 characters in length. Ensure that each server on which SDC is installed has a unique hostname.	-

Table 30. CSV topology fields (continued)

Field	Description	Required
NVDIMM Acceleration Device List	List of NVDIMM devices used to provide acceleration for storage pools using fine granularity data layout. Up to six devices, comma-separated, with no spaces. Ensure that the NVDIMM devices are configured for DAX mode.	-
NVDIMM acceleration pool List	Sets NVDIMM acceleration pool names. Assigns NVDIMM acceleration pools to the devices in the **NVDIMM Acceleration Device List**, respectively. The number of NVDIMM acceleration pools must equal the number of devices. Comma-separated, no spaces.	-
Is SDR	Deploy the SDR component on this node: `Yes` or `No` (default). A minimum of two SDRs per peer system must be deployed for the replication feature. SDRs must be deployed on nodes that also contain the SDS component.	-
SDR Name	The name of the SDR node, up to 31 characters in length. The name must not contain spaces. Default: empty field.	-
SDR Port	The port used by the SDR. Default: 11088	-
SDR Application IPs	The IP addresses through which the SDC communicates with the SDR. When an application writes data, it works with the SDR through these IP addresses. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for Application, Storage, External. IPv6 is supported.	-
SDR Storage IPs	The IP addresses through which the SDR communicates with the MDM for server-side control communications. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for application, storage, external. IPv6 is supported.	-
SDR External IPs	The IP addresses through which the SDR communicates with peer systems' SDRs. Maximum of eight addresses, comma-separated, no spaces. The same IP addresses can be used for Application, Storage, External. IPv6 is supported.	-
SDR All IPs	SDR IP addresses to be used for communication among all nodes (including all three roles) Maximum of eight addresses, comma-separated, no spaces. This is the default setting.	-
perfProfileForSDR	Optional performance profile to set for SDRs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see Configuring performance profile during deployment.	-
Storage pool Configuration section	Optional section of the CSV file that lets you configure properties for each of the storage pools to which the SDS devices are assigned.	-
ProtectionDomain	The protection domain to which the storage pool belongs.	-
StoragePool	The storage pool name	-
Media Type	The expected device media type in the storage pool: `HDD` or `SDD`	-

Table 30. CSV topology fields (continued)

Field	Description	Required
External Acceleration	External acceleration type, if used: ● `Read`—All devices are accelerated by a non- PowerFlex read cache ● `Write`—All devices are accelerated by a non- PowerFlex write cache ● `ReadAndWrite`—All devices are accelerated by both non- PowerFlex read and write cache	-
Data Layout	Data layout type: ● `FG`—Fine granularity layout ● `MG`—Medium granularity layout	-
Fine granularity ACCP	The name of the fine granularity acceleration pool associated with the storage pool.	-
Zero Padding	Zero padding enabled on the storage pool (mandatory for fine granularity): ● `Yes`—zero padding enabled ● `No`—zero padding disabled	-
Compression Method	The compression method used in the fine granularity storage pool. Compression methods might affect performance. ● `None`—no compression is used ● `Normal`—compression is enabled	-
Replication journal capacity percentage	The percentage of capacity dedicated to the replication journal: a value between 1–100 The percentage is calculated from the net capacity, and not from the total capacity. Use the information provided in the system limits section of system documentation to calculate the percentage of capacity that is required. This calculation also applies to expansion use cases where SDR is being added to an existing system. Each protection domain that contains SDRs should have at least one storage pool with replication journal capacity higher than 0%. Each SDR needs at least 108 GB of replication journal capacity in order to work. Take this into consideration when calculating capacity requirements.	-
Is SDT	Deploy the SDT component on this node: `Yes` or `No` SDT is an NVMe target, which is a front end component that translates NVMe over Fabric protocol into internal protocols.	-
SDT Name	The name for the SDT node, up to 31 characters in length. The name must not contain spaces.	-
SDT All IPs	SDT IP addresses used for both hosts communication and MDM communication (including both roles). Maximum of eight IP addresses, comma-separated, no spaces.	-

3. Save the CSV file and retain it for future use.

Prepare the CSV topology file for two-layer deployment

When deploying SDC nodes on a Linux server in a two-layer environment, add rows to the CSV file specifically for SDCs, that only contain the values listed in this table.

Table 31. CSV topology fields for two-layer deployment

Field	Description	Required
Domain	If using a domain user, the name of the domain (not relevant for Linux)	-
Username	The name of the user Linux nodes: Enter either `root` or a nonroot sudo user, indicated by appending "(sudo)" to the end of the username (default: `root`). For more information about deploying with a nonroot user, see Deploying PowerFlex using a non-root user.	-
Password	Password used to log in to the node. This should be the password of the user entered in the **Username** column.	Yes
Operating System	The server's operating system: `linux` or `windows`	Yes
SDS-SDC Only IPs	SDC IP addresses used for communication among SDS and SDC nodes only. Comma-separated, no spaces. **NOTE:** For SDC-only nodes, enter the IP address in this column.	-
Is SDC	`Yes`	Yes
perfProfileForSDC	Optional performance profile to set for SDCs: `High` (default) or `Compact`. When this field is left empty, the default option is applied. High performance is required for the fine granularity data layout feature. For more information, see Configuring performance profile during deployment.	-
SDC Name	Name for the SDC.	-

CSV file examples for typical configurations

Refer to the examples provided in this section to see how the CSV file is used to configure various system topologies and features.

NOTE: These examples represent a subset of all possible topologies and features that can be configured.

Four data networks: two frontends and two backends, no fault sets

This example shows a CSV topology file for four data networks, where two are front end networks, and two are backend networks. No fault sets are configured.

Four data networks are configured as follows:

- Two front end networks for SDCs
- Two backend networks for MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- No fault sets

To download the example `BE-FE-5_node.xlsx` CSV file, see the *Downloading the example CSV files Reference Guide* available at Dell Technologies Support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 32. CSV parameters for four data networks with two frontends and two backends

Parameter	Example
Username	root

Table 32. CSV parameters for four data networks with two frontends and two backends (continued)

Parameter	Example
Password	Scaleio123
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.153.10,192.168.154.10
Virtual IP NICs	ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS-SDS Only IPs	192.168.151.11,192.168.152.11
SDS-SDC Only IPs	192.168.153.11,192.168.154.11
Protection Domain	PD01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	No
perfProfileForSDC	high
SDC Name	SDC1
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01

Table 32. CSV parameters for four data networks with two frontends and two backends (continued)

Parameter	Example
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Four data networks: two frontend and two backend, with fault sets

This example shows a CSV topology file for four data networks, where two are frontend networks, and two are backend networks. Fault sets are also configured.

Four data networks are configured as follows:

- Two frontend networks for SDCs
- Two backend networks for MDMs, SDSs, and SDRs (when replication is required)
- Two external networks for replication (when applicable)
- Fault sets

To download the example `BE-FE-6_node_FaultSet.xlxs` CSV file, see the *Downloading the example CSV files Reference Guide* available at Dell Technologies Support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 33. CSV parameters for four data networks with fault sets

Parameter	Example
Username	root
Password	Scaleio123
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, tiebreaker (TB)
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.153.10,192.168.154.10
Virtual IP NICs	ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS-SDS Only IPs	192.168.151.11,192.168.152.11
SDS-SDC Only IPs	192.168.153.11,192.168.154.11
Protection Domain	PD01
Fault Set	FS01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	Yes

Table 33. CSV parameters for four data networks with fault sets (continued)

Parameter	Example
perfProfileForSDC	high
SDC Name	SDC1
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Four data networks: Frontend and backend on all four networks, no fault sets

This example shows a CSV topology file for four data networks, where all four are both front end and backend networks. No fault sets are configured.

Four data networks are configured as follows:

- Four networks for both front-end and backend: SDCs, MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- No fault sets

To download the example `SDS-SDC-Same-Network-5_node.xlsx` CSV file, see the *Downloading the example CSV files Reference Guide* available at Dell Technologies Support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 34. CSV parameters for four data networks with frontend and backend on all

Parameter	Example
Username	root
Password	Scaleio123

Table 34. CSV parameters for four data networks with frontend and backend on all (continued)

Parameter	Example
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.151.10,192.168.152.10,192.168.153.10,192.168.154.10
Virtual IP NICs	ens151,ens152,ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS All IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
Protection Domain	PD01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	No
perfProfileForSDC	high
SDC Name	SDC1
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
SDR All IPs	-
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01
Zero Padding	Yes

Table 34. CSV parameters for four data networks with frontend and backend on all (continued)

Parameter	Example
Compression Method	Normal
Replication journal capacity percentage	18

Four data networks: frontend and backend on all four networks, with fault sets

This example shows a CSV topology file for four data networks, where all four are both frontend and backend networks. Fault sets are also configured.

Four data networks are configured as follows:

- Four networks for both frontend and backend: SDCs, MDMs, SDSs, and SDRs (for replication only)
- Two external networks for replication (if applicable)
- Fault sets

To download the example `SDS-SDC-Same-Network-6_node_withFaultSet.xlsx` CSV file, see the *Downloading the example CSV files Reference Guide* available at Dell Technologies Support site.

The following table lists the parameters that must be configured for this use case. Other optional parameters in the CSV file should be configured as necessary.

Table 35. CSV parameters for four data networks with frontend and backend on all and fault sets

Parameter	Example
Username	root
Password	Scaleio123
Operating System	linux
Is MDM/TB	Primary Other options: Secondary, TB
MDM Mgmt IP	192.168.150.11
MDM IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
MDM Name	MDM1
perfProfileForMDM	high
Virtual IPs	192.168.151.10,192.168.152.10,192.168.153.10,192.168.154.10
Virtual IP NICs	ens151,ens152,ens153,ens154
Is SDS	Yes
SDS Name	SDS1
SDS All IPs	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
Protection Domain	PD01
Fault Set	FS01
SDS Storage Device List	/dev/sdb,/dev/sdc,/dev/sdd,/dev/sde
StoragePool List	SP01,SP01,SP02,SP02
perfProfileForSDS	high
Is SDC	No
perfProfileForSDC	high
SDC Name	SDC1

Table 35. CSV parameters for four data networks with frontend and backend on all and fault sets (continued)

Parameter	Example
NVDIMM Acceleration Device List	/dev/dax0.0
NVDIMM Acceleration Pool List	accp01
Is SDR (optional—for replication only)	Yes
SDR Name (optional—for replication only)	SDR1
SDR Application IPs (optional—for replication only)	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
SDR Storage IPs (optional—for replication only)	192.168.151.11,192.168.152.11,192.168.153.11,192.168.154.11
SDR External IPs (optional—for replication only)	192.168.161.11,192.168.162.11
Storage Pool Configuration (this section is used to configure data layout and replication)	
ProtectionDomain	PD01
StoragePool	SP01
Media Type	SSD
External Acceleration	None
Data Layout	FG
Fine Granularity ACCP	accp01
Zero Padding	Yes
Compression Method	Normal
Replication journal capacity percentage	18

Configuring performance profile during deployment

During deployment, system performance is enhanced by default implementation of the high performance profile for the MDM, SDC, SDS, and SDR components. Most servers support the requirements for this configuration and can benefit considerably from its implementation.

The high performance profile is mandatory for fine granularity data layout. However, for medium granularity data layout, you can choose to use the compact performance profile instead.

High performance profile should be enabled on servers equipped with processors based on Haswell-equivalent or newer architectures and 48 GB of RAM. Other configurations may benefit from this setting as well.

The performance profile is set to High by default, but if you want to change the profile to Compact in the CSV file, use the `perfProfileFor<COMPONENT>` columns.

Adding new components or features by editing the existing CSV file

Use your PowerFlex system's existing CSV topology file to add components or new features to your system.

The CSV file that is used to deploy a system can be used later, in the following scenarios, for adding features or components to an existing PowerFlex system:

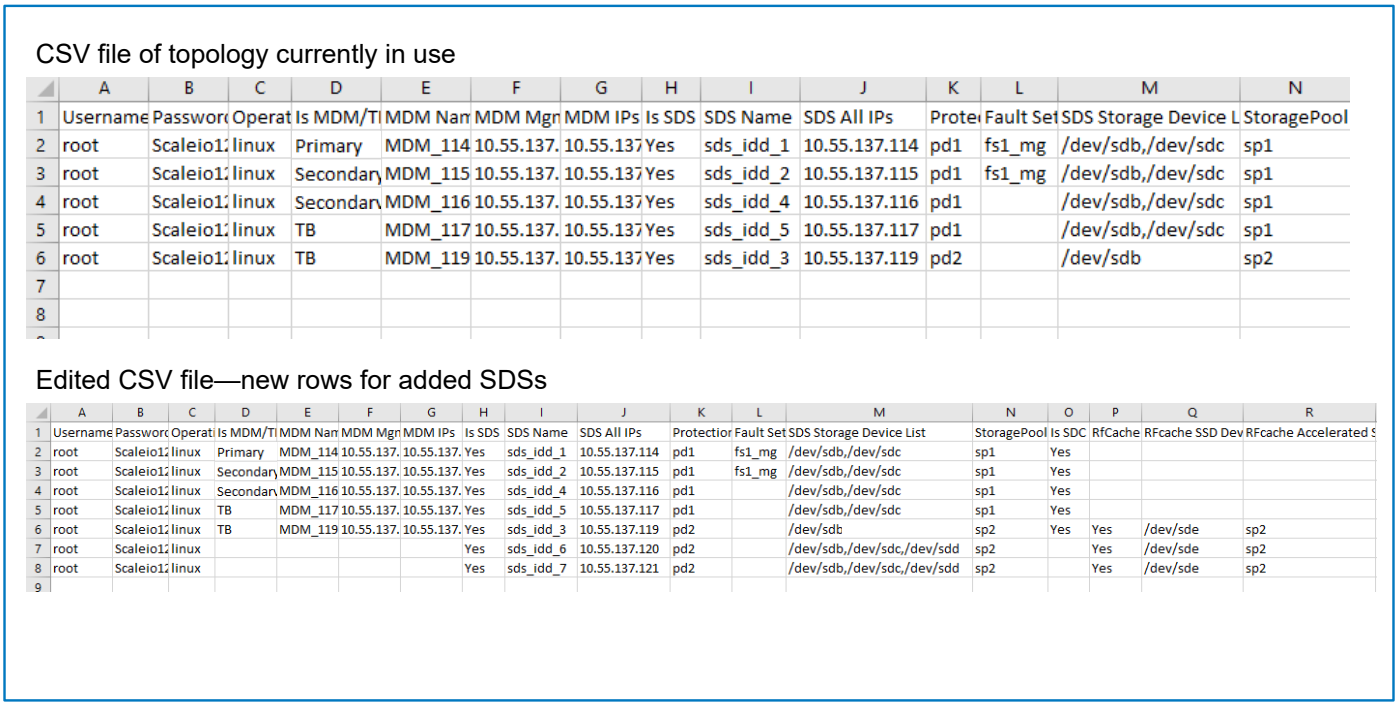
- Add SDSs, with or without new storage and acceleration devices
- Add MDMs or tiebreakers. The PowerFlex installer adds them to the MDM cluster and configure them automatically.
- Add SDRs for the replication feature

- Add SDCs
- Add SDTs
- Configure storage pools
- Configure PowerFlex system features which can be configured using the CSV file, such as performance profiles, acceleration, and so on.

To add and configure components and features, make a copy of your system's latest CSV topology file. Add rows or columns, or edit existing rows and columns. For new features, you can copy columns from the latest complete CSV template, and paste them into the copy of your system's CSV file. Get the latest complete template by downloading it.

The following examples show how you can edit the CSV topology file which can be edited for various system modification scenarios.

Add storage and acceleration:



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Username	Passwor	Operat	Is MDM/T	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Prote	Fault Set	SDS Storage Device L	StoragePool
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
3	root	Scaleio1	linux	Secondar	MDM_115	10.55.137.	10.55.137	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
4	root	Scaleio1	linux	Secondar	MDM_116	10.55.137.	10.55.137	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb	sp2
7														
8														

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Username	Passwor	Operat	Is MDM/T	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectio	Fault Set	SDS Storage Device List	StoragePool	Is SDC	RfCache	RFcache SSD Dev	RFcache Accelerated S
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes			
3	root	Scaleio1	linux	Secondar	MDM_115	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes			
4	root	Scaleio1	linux	Secondar	MDM_116	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1	Yes			
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137.	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1	Yes			
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137.	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb	sp2	Yes	Yes	/dev/sde	sp2
7	root	Scaleio1	linux					Yes	sds_idd_6	10.55.137.120	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2		Yes	/dev/sde	sp2
8	root	Scaleio1	linux					Yes	sds_idd_7	10.55.137.121	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2		Yes	/dev/sde	sp2
9																		

Figure 9. Add storage and acceleration

Add an MDM and a tiebreaker:

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Username	Password	Operating	Is MDM/TI	MDM Nam	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Storag	StoragePc	Is SDC	Is SDR
2	root	Scaleio123	linux	Primary	MDM_118	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
3	root	Scaleio123	linux	Secondary	MDM_120	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
4	root	Scaleio123	linux	TB	MDM_121	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.	pd1	/dev/sdb,	sp1	Yes	Yes
5															

Edited CSV file—new rows added for an additional MDM and TB

	A	B	C	D	E	F	G	H	I	J
1	Username	Password	Operating	Is MDM/TI	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs
2	root	Scaleio123	linux	Primary	MDM_118	10.55.137.118	10.55.137.118	Yes	sds_idd_1	10.55.137.
3	root	Scaleio123	linux	Secondary	MDM_120	10.55.137.120	10.55.137.120	Yes	sds_idd_2	10.55.137.
4	root	Scaleio123	linux	TB	MDM_121	10.55.137.121	10.55.137.121	Yes	sds_idd_4	10.55.137.
5	root	Scaleio123	linux	Secondary	MDM_122	10.55.137.122	10.55.137.122			
6	root	Scaleio123	linux	TB	MDM_123	10.55.137.123	10.55.137.123			
7										

Figure 10. Add an MDM and a tiebreaker

Add storage data servers (SDRs) for replication:

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Username	Password	Opera	Is MDM/TI	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Storag	StoragePc	Is SDC			
2	root	Scaleio123	linux	Primary	MDM_118	10.55.137.118	10.55.137.118	Yes	sds_idd_1	10.55.137.118	pd1	/dev/sdb,	sp1	Yes			
3	root	Scaleio123	linux	Secondary	MDM_120	10.55.137.120	10.55.137.120	Yes	sds_idd_2	10.55.137.120	pd1	/dev/sdb,	sp1	Yes			
4	root	Scaleio123	linux	TB	MDM_121	10.55.137.121	10.55.137.121	Yes	sds_idd_4	10.55.137.121	pd1	/dev/sdb,	sp1	Yes			
5																	

Edited CSV file—new columns added for SDRs

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Username	Password	Opera	Is MDM/TI	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	SDS Storag	StoragePc	Is SDC	Is SDR	SDR Name	SDR Port	SDR Application IPs	SDR Storage IPs	SDR External IPs	SDR All IPs	
2	root	Scaleio123	linux	Primary	MDM_118	10.55.137.118	10.55.137.118	Yes	sds_idd_1	10.55.137.118	pd1	/dev/sdb,	sp1	Yes	Yes	SDR1		10.55.137.118	10.55.137.118	10.55.137.118	10.55.137.118	
3	root	Scaleio123	linux	Secondary	MDM_120	10.55.137.120	10.55.137.120	Yes	sds_idd_2	10.55.137.120	pd1	/dev/sdb,	sp1	Yes	Yes	SDR2		10.55.137.120	10.55.137.120	10.55.137.120	10.55.137.120	
4	root	Scaleio123	linux	TB	MDM_121	10.55.137.121	10.55.137.121	Yes	sds_idd_4	10.55.137.121	pd1	/dev/sdb,	sp1	Yes	Yes	SDR3		10.55.137.121	10.55.137.121	10.55.137.121	10.55.137.121	
5																						

	N	O	P	Q	R	S	T	U
	gePc Is SDC	Is SDR	SDR Name	SDR Port	SDR Application IPs	SDR Storage IPs	SDR External IPs	SDR All IPs
	Yes	Yes	SDR1		10.55.137.118	10.55.137.118	10.55.137.118	10.55.137.118
	Yes	Yes	SDR2		10.55.137.120	10.55.137.120	10.55.137.120	10.55.137.120
	Yes	Yes	SDR3		10.55.137.121	10.55.137.121	10.55.137.121	10.55.137.121

Figure 11. Add SDRs for replication

Add storage data clients (SDCs):

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Username	Passwor	Operating	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	SDS-SDC Only IPs	Protectior	Fault Set	SDS Stora	StoragePc	Is SDC	RfCache	RFcache S	RFcache A	Is SDR
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.114		pd1	fs1_mg	/dev/sdb,	sp1	Yes				Yes
3	root	Scaleio1	linux	Secondary	MDM_115	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.115		pd1	fs1_mg	/dev/sdb,	sp1	Yes				Yes
4	root	Scaleio1	linux	Secondary	MDM_116	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.116		pd1		/dev/sdb,	sp1	Yes				Yes
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137.	Yes	sds_idd_5	10.55.137.117		pd1		/dev/sdb,	sp1	Yes				
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137.	Yes	sds_idd_3	10.55.137.119		pd2		/dev/sdb	sp2	Yes	Yes	/dev/sdc	sp2	
7	root	Scaleio1	linux					Yes	sds_idd_6	10.55.137.120		pd2		/dev/sdb,	sp2		Yes	/dev/sdc	sp2	
8	root	Scaleio1	linux					Yes	sds_idd_7	10.55.137.121		pd2		/dev/sdb,	sp2		Yes	/dev/sdc	sp2	
9																				

Edited CSV file—new rows added for SDCs

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Username	Passwor	Operating Sys	Is MDM/TI	MDM Nan	MDM Mgn	MDM IPs	Is SDS	SDS Name	SDS All IPs	SDS-SDC Only IPs	Protectior	Fault Set	SDS Stora	StoragePc	Is SDC	RfCache
2	root	Scaleio1	linux	Primary	MDM_114	10.55.137.	10.55.137.	Yes	sds_idd_1	10.55.137.114		pd1	fs1_mg	/dev/sdb,	sp1	Yes	
3	root	Scaleio1	linux	Secondary	MDM_115	10.55.137.	10.55.137.	Yes	sds_idd_2	10.55.137.115		pd1	fs1_mg	/dev/sdb,	sp1	Yes	
4	root	Scaleio1	linux	Secondary	MDM_116	10.55.137.	10.55.137.	Yes	sds_idd_4	10.55.137.116		pd1		/dev/sdb,	sp1	Yes	
5	root	Scaleio1	linux	TB	MDM_117	10.55.137.	10.55.137.	Yes	sds_idd_5	10.55.137.117		pd1		/dev/sdb,	sp1	Yes	
6	root	Scaleio1	linux	TB	MDM_119	10.55.137.	10.55.137.	Yes	sds_idd_3	10.55.137.119		pd2		/dev/sdb	sp2	Yes	Yes
7	root	Scaleio1	linux					Yes	sds_idd_6	10.55.137.120		pd2		/dev/sdb,	sp2		Yes
8	root	Scaleio1	linux					Yes	sds_idd_7	10.55.137.121		pd2		/dev/sdb,	sp2		Yes
9	root	Scaleio1	windows					No			10.55.5.1					Yes	
10	root	Scaleio1	windows					No			10.55.5.2					Yes	
11	root	Scaleio1	windows					No			10.55.5.3					Yes	

Figure 12. Add SDCs

Configure storage pools:

CSV file of topology currently in use

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Username	Password	Operating S	Is MDM/TB	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	Fault Set	SDS Storage Device List	StoragePool	Is SDC
2	root	Scaleio123	linux	Primary	MDM_114	10.55.137.114	10.55.137.	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes
3	root	Scaleio123	linux	Secondary	MDM_115	10.55.137.115	10.55.137.	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1	Yes
4	root	Scaleio123	linux	Secondary	MDM_116	10.55.137.116	10.55.137.	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1	Yes
5	root	Scaleio123	linux	TB	MDM_117	10.55.137.117	10.55.137.	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1	Yes
6	root	Scaleio123	linux	TB	MDM_119	10.55.137.119	10.55.137.	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2	Yes
7	root	Scaleio123	linux					Yes	sds_idd_6	10.55.137.120	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2	
8	root	Scaleio123	linux					Yes	sds_idd_7	10.55.137.121	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2	

Edited CSV file—new rows for Storage Pool configuration

	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	Username	Password	Operating S	Is MDM/TB	MDM Name	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS All IPs	Protectior	Fault Set	SDS Storage Device List	Stora
2	root	Scaleio123	linux	Primary	MDM_114	10.55.137.114	10.55.137.	Yes	sds_idd_1	10.55.137.114	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
3	root	Scaleio123	linux	Secondary	MDM_115	10.55.137.115	10.55.137.	Yes	sds_idd_2	10.55.137.115	pd1	fs1_mg	/dev/sdb,/dev/sdc	sp1
4	root	Scaleio123	linux	Secondary	MDM_116	10.55.137.116	10.55.137.	Yes	sds_idd_4	10.55.137.116	pd1		/dev/sdb,/dev/sdc	sp1
5	root	Scaleio123	linux	TB	MDM_117	10.55.137.117	10.55.137.	Yes	sds_idd_5	10.55.137.117	pd1		/dev/sdb,/dev/sdc	sp1
6	root	Scaleio123	linux	TB	MDM_119	10.55.137.119	10.55.137.	Yes	sds_idd_3	10.55.137.119	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2
7	root	Scaleio123	linux					Yes	sds_idd_6	10.55.137.120	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2
8	root	Scaleio123	linux					Yes	sds_idd_7	10.55.137.121	pd2		/dev/sdb,/dev/sdc,/dev/sdd	sp2
9														
10	Storage Pool Configuration													
11	Protectior	StoragePo	Media Type	External Accele	Data Layout	Fine Granularity ACCP	Zero Padd	Compression Method	Replication journal capacity percentage					
12	pd1	sp1	HDD	ReadAndWrite	MG		Yes		50					
13	pd2	sp2	SSD	Read	FG	accp2	Yes	Normal	20					
14														

Figure 13. Configure storage pools

Deploy a PowerFlex cluster using a CSV topology file from PowerFlex Manager

Use this procedure to deploy a PowerFlex cluster using a CSV topology file.

Steps

1. Log in to PowerFlex Manager.
2. Go to **Settings** > **Getting Started**.
3. Click **Deploy With Installation File** to provide an installation topology file for deployment of nodes.
4. In the **PowerFlex Installer** window, click **Browse** to upload the appropriate artifacts according to the operating system of your PowerFlex operating system machines.
5. Click **Upload Files** and click **Next**.
6. Click **Browse** to upload an installation CSV topology file that describes the desired topology of PowerFlex.

   Example: PowerFlex_config_example_all fields.csv (all fields) and PowerFlex_basic_example_minimum fields.csv (minimum fields).
7. Click **Upload Installation CSV**.
8. Select the **I accept the terms of the End User License Agreement (mandatory)** check box.
9. Click **Next**.
10. In the **Advanced Settings** tab, you can select how you would like to install PowerFlex 4.5.x. You can skip options or force an overwrite.
11. Select **Force overwrite of existing data on devices used for PowerFlex storage**.

    NOTE: If you select this option, data is erased.
12. Click **Next**.
13. In the **Summary** tab, verify all the configurations and click **Finish**.
14. Click the **Go to Events Page** button to track the installation progress.

Deploy a PowerFlex cluster using REST API commands

After you have prepared the CSV topology file, use the REST API to parse it and upload the configuration to PowerFlex Manager, and then deploy block storage resources using REST API commands. This procedure is only recommended for advanced users familiar with working with the REST API.

Prerequisites

- Ensure that you have prepared the CSV topology file.
- Ensure that you know the load balancer IP address and user credentials of PowerFlex Manager. All the steps performed in this procedure must be performed using the load balancer IP address. The load balancer IP address is the same IP address you are using to access PowerFlex Manager.
- Ensure that you have the software packages of the PowerFlex core components that you want to deploy, and the ActiveMQ package.
- Refer to the Dell Technologies Developer portal for information about working with the REST API.
- Log in using the REST API, and make a note of the Keycloak token that was returned from `/auth/login` or from `/api/gatewayLogin`.

About this task

The following procedure provides examples using cURL.

Steps

1. Get the configuration from the CSV file, by parsing it using the PowerFlex REST API. Use the command `POST /im/types/Configuration/instances/actions/parseFromCSV`, where the body is the CSV file reference. For example:

```
curl -v -k -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
${token}" -F "file=<PATH_TO_CSV_FILE>" https://<IP_ADDRESS>/im/types/Configuration/
instances/actions/parseFromCSV
```

   where: *{token}* is the Keycloak token, *<PATH_TO_CSV_FILE>* is the path to the CSV topology file, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.
   The command returns a JSON file containing the required cluster configuration. The installation command requires this file in order to install resources. Save this file.
2. Add packages:
   a. Copy the installation packages of the required core components (including ActiveMQ RPM) to: "`/usr/local/tomcat/temp/scaleio`".
   b. Open the shell, and run the following command:

```
kubectl get pods -n powerflex | grep block-legacy
```

   c. Make a note of the block-legacy-gateway ID that is returned in the first line of the command output. For example, in this sample output, the ID is `76594f9459-mqgxj`:

```
block-legacy-gateway-76594f9459-mqgxj    1/1    Running    0    24h
block-legacy-gateway-mds-6696f6dd4c-s9qp8    1/1    Running    0    24h
```

   d. In the shell, run the following command:

```
kubectl exec --stdin --tty <ID> -n powerflex -- /bin/sh
```

   where *<ID>* is the block-legacy-gateway ID obtained in the previous step.
   e. In the shell, run the following commands:

```
apt-get update
apt-get install openssh-client
scp root@<RPM_MACHINE_IPS>:<RPMS_LOCATION>* temp/scaleio
```

   where *<RPM_MACHINE_IPS>* is the IP address where the packages are saved, and *<RPMS_LOCATION>* is the path to the packages' location.
3. Install block storage resources using the PowerFlex REST API. Use the command `POST /im/types/Configuration/actions/install`, where the body is the configuration that was returned from parsing the CSV file.

   If the parameter *systemId* (type string) is added to the configuration, the cluster is created with the given ID. Otherwise, the gateway generates an ID.

   Optional parameters:

   - noUpload—Skip upload package phase (true/false, default if false)
   - noInstall—Skip install package phase (true/false, default if false)
   - noConfigure—Skip configure phaseand repository files (true/false, default if false)
   - noSecurityBootstrap—Skip security bootstrap (part of install phase) (true/false, default if true)
   - noLinuxDevValidation (true/false, default if true)
   - sdsInstances (number, optional)
   - extend—extend an existing system (true/false, default if false)
   - globalZeroPadPolicy (true/false, default if true)
   - forceDeviceTakeover—Clean devices before installing SDS (without it, installation will fail when it run after uninstall; must not be used in production) (true/false, optional)

   For example:

```
curl -v -s -k -i -X POST -H "Content-Type:application/json" -H
"Authorization: Bearer ${token}" "https://<IP_ADDRESS>/im/types/Configuration/actions/
install?noSecurityBootstrap=false&forceDeviceTakeover=true" -d @<PATH>/config.json
```

where: *{token}* is the Keycloak token, *<IP_ADDRESS>* is the PowerFlex Manager IP address, and *<PATH>* is the path to the JSON configuration file that was output in step 1.

4. Installation is asynchronous and consists of four stages: query, upload, install, and configure. Manually monitor and move from one phase to the next one, using the commands `queryPhaseState` and `moveToNextPhase`. You can also monitor the progress of each stage by viewing events in PowerFlex Manager.
   The following is an example of an installation event:

   ```
   COMMAND_STARTED_ON_OPERATION_INSTALL_AND_PHASE_QUERY
   ```

5. After installation is complete, clear the gateway state. Use the command `POST im/types/Command/instances/actions/clear`.
   For example:

   ```
   curl -v -s -k -i -X POST -H "Content-Type:application/json" -H "Authorization: Bearer ${token}" "https://<IP_ADDRESS>/im/types/Command/instances/actions/clear"
   ```

   where: *{token}* is the Keycloak token, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.

Deploying a PowerFlex cluster using PowerFlex Manager templates and resource groups

Use PowerFlex Manager to create and publish templates that define infrastructure and resources, and then deploy a resource group.

Configuring networking

Adding the details of an existing network enables PowerFlex Manager to automatically configure nodes that are connected to the network.

Defining a network

Adding the details of an existing network enables PowerFlex Manager to automatically configure nodes that are connected to the network.

Prerequisites

Ensure that the following conditions are met before you define the network:

- PowerFlex Manager can communicate with the out-of-band management network.
- PowerFlex Manager can communicate with the operating system installation network in which the appliance is deployed.
- PowerFlex Manager can communicate with the hypervisor management network.
- The DHCP node is fully functional with appropriate PXE settings to boot PXE images from PowerFlex Manager in your deployment network.

Steps

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Click **Define**. The **Define Network** page opens.
4. In the **Name** field, enter the name of the network. Optionally, in the **Description** field, enter a description for the network.
5. From the **Network Type** drop-down list, select one of the following network types:
   - **General Purpose LAN**
   - **Hypervisor Management**
   - **Hypervisor Migration**
   - **Hardware Management**
   - **PowerFlex Data**

- **PowerFlex Data (Client Traffic Only)**
- **PowerFlex Data (Server Traffic Only)**
- **PowerFlex Replication**
- **NAS File Management**
- **NAS File Data**
- **PowerFlex Management**

For a PowerFlex configuration that uses a hyperconverged architecture with two data networks, you typically have two networks that are defined with the PowerFlex data network type. The PowerFlex data network type supports both client and server communications. The PowerFlex data network type is used with hyperconverged resource groups.

For a PowerFlex configuration that uses a two-layer architecture with four dedicated data networks, you typically have two PowerFlex (client traffic only) VLANs and two PowerFlex data (server traffic only) VLANs. These network types are used with storage-only and compute-only resource groups.

6. In the **VLAN ID** field, enter a VLAN ID between **1** and **4094**. VLANS are not required, but do get prompted for when creating networks. If no VLAN is associated with the network, enter **1** as the VLAN ID, to satisfy the required field. This VLAN value will not be used.

   NOTE: PowerFlex Manager uses the VLAN ID to configure I/O modules that enable network traffic to flow from the node to configured networks during deployment.

7. Click **Save**.

(Optional) Edit a network

If a network is not associated with a template or resource group, you can edit the network name, the VLAN ID, or the IP address range.

Steps

1. On the menu bar, click **Settings** > **Networking** and click **Networks**.
2. Select the network that you want to modify, and click **Modify**.
3. Edit the information in any of the following fields: **Name**, **Network Type**, and **VLAN ID**.
4. Click **Save**.

(Optional) Deleting a network

You cannot delete a network that is associated with a template or resource group.

Steps

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Click the network that you want to delete, and then click **Delete**.
4. Click **OK** when the confirmation message is displayed.

Discover a resource

A resource must be discovered in PowerFlex Manager in order for PowerFlex Manager to manage it.

Prerequisites

Before you start discovering a resource, complete the following:

- Gather the IP addresses and credentials that are associated with the resources.
- Ensure that both the resources and PowerFlex Manager are connected to the network.

About this task

A resource in PowerFlex Manager is categorized into one of the following groups: Element Manager, Node (Hardware/Software Management), Switch, VM manager, PowerFlex gateway, Node (Software Management) or PowerFlex System.

Steps

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources** on the **All Resources** tab.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select a resource that you want to discover.
      - **Element Manager**, for example, CloudLink Center.
      - **Node (Hardware / Software Management)**
      - **Switch**
      - **VM Manager**
      - **PowerFlex Gateway**
      - **Node (Software Management)**
      - **PowerFlex System**

      If you want to discover a software-only node or management VM (MVM) node with a credential that uses SSH key pairs, you must add the public keys to the nodes manually before performing the discovery. If you do not add the public keys first, the discovery fails.

      The **PowerFlex System** resource type is used to discover an MDS gateway.

      For a PowerFlex 4.x system, indicate whether the PowerFlex instance will be used for **Production Storage** or a **Management Cluster**.

   b. Enter the management IP address (or hostname) of the resources that you want to discover in the **IP/Hostname Range** field.

      To discover one or more nodes by IP address, select **IP Address** and provide a starting and ending IP address.

      To discover one or more nodes by hostname, select **Hostname** and identify the nodes to discover in one of the following ways:

      - Enter the fully qualified domain name (FQDN) with a domain suffix.
      - Enter the FQDN without a domain suffix.
      - Enter a hostname search string that includes one of the following variables:

      Table 36. Hostname search variables

Variable	Description
${num}	Produces an automatically generated unique number.
${num_2d}	Produces an automatically generated unique number that has two digits.
${num_3d}	Produces an automatically generated unique number that has three digits.

      If you use a variable, you must provide a start number and end number for the hostname search.

   c. Select one of the following options from the **Resource State** list:

      Table 37. Resource State options

Option	Description
Managed	Select this option to monitor the firmware version compliance, upgrade firmware, and deploy resource groups on the discovered resources. A managed state is the default option for the switch, vCenter, element manager, and PowerFlex gateway resource types.

Table 37. Resource State options (continued)

Option	Description
	Resource state must be set to **Managed** for PowerFlex Manager to send alerts to SupportAssist.
Unmanaged	Select this option to monitor the health status of a device and the firmware version compliance only. The discovered resources are not available for a firmware upgrade or deploying resource groups by PowerFlex Manager. This option is the default for the node resource type. If you have yet not uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, **Unmanaged** is the only option available.
Reserved	Select this option to monitor firmware version compliance and upgrade firmware. The discovered resources are not available for deploying resource groups by PowerFlex Manager.

d. To discover resources into a selected node pool instead of the global pool (default), select an existing or create a node pool from the **Discover into Node Pool** list. To create a node pool, click the **+** sign to the right of the **Discover into Node Pool** box.

e. Select an existing or create a credential from the **Credentials** list to discover resource types. To create a credential, click the **+** sign to the right of the **Credentials** box. PowerFlex Manager maps the credential type to the type of resource that you are discovering. The credential types are as follows:

- Element Manager
- Node (Hardware/Software Management)
- Switch
- VM Manager
- PowerFlex Gateway
- Node (Software Management)
- PowerFlex System

The default node (Hardware/Software Management) credential type is **Dell PowerEdge iDRAC Default**.

f. If you want PowerFlex Manager to automatically reconfigure the iDRAC nodes it finds, select the **Reconfigure discovered nodes with new management IP and credentials** check box. This option is not selected by default, because it is faster to discover the nodes if you bypass the reconfiguration.

g. To have PowerFlex Manager automatically configure iDRAC nodes to send alerts to PowerFlex Manager, select the **Auto configure nodes to send alerts to PowerFlex Manager** check box.

4. Click **Next**.

   You might have to wait while PowerFlex Manager locates and displays all the resources that are connected to the managed networks.

   To discover multiple resources with different IP address ranges, repeat steps 2 and 3.

5. On the **Discovered Resources** page, select the resources from which you want to collect inventory data and click **Finish**. The discovered resources are listed on the **Resources** page.

Cloning or creating a template

Use this section to clone an existing template or create a new template.

Add a template

About this task

The **Create** feature allows you to create a template, clone the components of an existing template into a new template, or import a pre-existing template.

For most environments, you can simply clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

Steps

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click **Create**.
3. In the **Create** dialog box, select one of the following options:
   - **Clone an existing PowerFlex Manager template**
   - **Upload External Template**
   - **Create a new template**

   If you select **Clone an existing PowerFlex Manager template**, select the **Category** and the **Template to be Cloned**. The components of the selected template are in the new template. For software-only file storage, ensure that you select **PowerFlex-File-SW only** template.
4. Enter a **Template Name**.
5. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
6. Enter a **Template Description** (optional).
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the check box next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
9. Click **Next**.
10. In **Additional Settings** provide the new values for the following settings:
    - **Network**
    - **PowerFlex Gateway**
    - **Node Pool**
11. Click **Finish**.

Clone a template

About this task

The **Clone** feature allows you to copy an existing template into a new template. A cloned template contains the components that existed in the original template. You can edit it to add additional components or modify the cloned components.

For most environments, you can simply clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment. If you already created a template using the Add a template option, skip this task.

To clone an existing template:

Steps

1. On the menu bar, click **Templates**.
2. Click **Sample Template** > **Select**, click the **PowerFlex-File-SW Only** template from the list.

   You can also click **Clone** on the **My Templates** page if you want to clone one of your own templates, rather than one of the sample templates.
3. In the right pane, click **More Actions** > **Clone**.

4. In the **Clone Template** dialog box, enter a template name in the **Template Name** box.
5. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
6. In the **Template Description** box, enter a description for the template.
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the check box next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
9. Click **Next**.
10. On the **Additional Settings** page, provide new values for the **Network Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
11. Click **Finish**.
12. See the *Edit a Template* topic, in related information, for modifying the cluster components: Node (Software only), PowerFlex cluster, and PowerFlex file cluster.

Build and publish a template

After creating a template using the **Create** feature, you use the template builder page to build and publish the customized template. Publishing a template indicates that a template is ready for deployment.

Steps

1. Click **Modify Template**.
2. To add a component type to the template, click **Add Node**, **Add Cluster**, or **Add VM** at the top of the template builder.
   The corresponding **<component type> component** dialog box appears.
3. If you are adding a cluster, in the **Select a Component** box, choose one of the following cluster types:
   - **PowerFlex Cluster**
   - **VMware Cluster**
   - **PowerFlex File Cluster**
4. If you are adding a VM in the **Select a Component** box, choose one of the following cluster types:
   - **CloudLink Center**
   - **PowerFlex Gateway**
5. Under **Related Components**, perform one of the following actions:
   - To associate the component with all existing components, click **Associate All**.
   - To associate the component with only selected components, click **Associate Selected** and then select the components to associate.

   Based on the component type, specific settings and properties appear automatically that are required and can be edited.
6. Click **Save** to add the component to the template builder.
7. Repeat steps 1 through 6 to add additional components.
8. After you finish adding components to your template, click **Publish Template**.
   A template must be published to be deployed. It remains in draft state until published.
   After publishing a template, you can use the template to deploy a resource group on the **Resource Groups** page.

Deploying a resource group

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

Steps

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups**. Then, click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates**. Then, click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:
   a. From the **Select Published Template** list, select the template to deploy a resource group.
   b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
   c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

      You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

      PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the ESXi version is greater than the vCenter version, PowerFlex Manager blocks the resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade vCenter first, or use a different compliance version that is compatible with the installed vCenter version.

      NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

   d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
      - To restrict access to super users, select **Only PowerFlex SuperUser**.
      - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
        i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
        ii. Select which users will have access to this resource group.
        iii. To delete a user from the list, select the user and click **Remove User(s)**.
        iv. After adding the users, select or clear the check box next to the users to grant or block access.
      - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
3. Click **Next**.
4. On the screens that follow the **Deployment Settings** page, configure the settings, as needed for your deployment.
5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.
7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

Create the configuration file manually

Create or edit the `driver_sync.conf` file manually using a text editor. This topic is relevant for deployments on Linux types other than Red Hat.

About this task

If PowerFlex was deployed with the PowerFlex gateway, the file already exists in the `etc` folder, with dummy values, and you must edit that file, and then copy it to the `etc` folder on every SDC in the system. The `etc` folder is located in: SDC

Table 38. Location of driver_sync.conf file

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

The configuration of the `driver_sync` script is read from `driver_sync.conf` (an example appears below), which contains the following parameters:

Table 39. driver_sync.conf configuration parameters

Parameter	Description
repo_address	Encodes the chosen method and address
repo_user	Identifies the username used when logging in to the drivers server. Required for SFTP and FTP methods.
repo_password	The FTP server login password (for user repo_user). Required for FTP method.
user_private_rsa_key	Contains the path of the private RSA key file used to SFTP connection. This key file (provided by PowerFlex on installation) enables SFTP connection without a password.
local_dir	The local directory in which the downloaded SDC/RF drivers are stored. For SDC/RF clients, this value is set up during installation and does not normally require a change.
repo_public_rsa_key	The path of the public RSA key of the repository server machine (also known as the host public key).
module_sigcheck	Determines whether to verify (1) or not to verify (0) the downloaded kernel drivers.
emc_public_gpg_key	A key that is used to verify the signatures of downloaded drivers. Enter the path to the public GPG signing key of PowerFlex (just like the one that comes with PowerFlex LIA). This only needs to be set if module_sigcheck was set to 1.
The following parameters are used when `driver_sync.sh` is used to synchronize a local repository against a remote one.	
sync_pattern	A regular expression that can be used to fetch several drivers at once. This parameter is not normally needed, and becomes relevant only when `driver_sync.sh` is used by your own private repository to fetch from a remote (for example, Dell) repository.
is_local_repo	Set to 0 (default) or omit, for SDC/RF clients. Set to 1 when `driver_sync.sh` is used to synchronize a local repository against a remote repository. Setting to 1 will cause the fetched tar files not to be extracted.

For example:

```
################################################
#driver_sync Configuration file
#Everything after a '#' until the end of the line is ignored
################################################
#Repository address, prefixed by protocol
repo_address = sftp://localhost/path/to/repo/dir
#repo_address = ftp://localhost/path/to/repo/dir
#repo_address = file://local/path/to/repo/dir
# Repository user (valid for ftp/sftp protocol)
repo_user = scini
# Repository password (valid for ftp protocol)
repo_password = scini
# Local directory for modules
```

```
local_dir = /bin/emc/scaleio/scini_cache/
# User's RSA private key file (sftp protocol)
user_private_rsa_key = /bin/emc/scaleio/scini_key
# Repository host public key (sftp protocol)
repo_public_rsa_key = /bin/emc/scaleio/scini_repo_key.pub
# Should the fetched modules' signatures be checked [0, 1]
module_sigcheck = 1
# EMC public signature key (needed when module_sigcheck is 1)
emc_public_gpg_key = /bin/emc/scaleio/emc_key.pub
# Sync pattern (regular expression) for massive retrieve
sync_pattern = .*
```

Update the PowerFlex signature key

If the PowerFlex package-signing key expires, you must update the configuration for each storage data client (SDC).

About this task

When a PowerFlex package-signing key expires, a new public key is issued. You must update `driver_sync.conf` to use the new public key.

Steps

1. Download the new public key from the Dell Technologies Support site.
2. On each SDC, edit `driver_sync.conf` so that the value of `emc_public_gpg_key` is the path to the new key file.

   The expired public key will still be in the gpg keyring, and older packages that were signed with the old key can still be verified.

   Location of `driver_sync.conf`:

   - SDC

   Table 40. Location of driver_sync.conf file

Host	Path
Ubuntu/OL	`/etc/emc/scaleio/scini_sync/driver_sync.conf`

Create a mirror repository

For Ubuntu or Oracle Linux (OL, also known as OEL), set up a local mirror repository (optional).

This section describes how to set up a local repository. This is necessary for scenarios when connecting all storage data client (SDC) nodes to the PowerFlex repository is not possible, due to security considerations or other limitations.

A common workflow would be to configure your mirror repository to synchronize with the PowerFlex repository, while your storage data client (SDC) nodes fetch drivers (synchronize) from the mirror repository.

Driver repository directory hierarchy

The directory hierarchy in the repository consists of the following levels:

<Distro>/<PowerFlex Version>/<Kernel Version>/

where:

- *<Distro>* is either `Ubuntu` or `OL` (for Oracle Linux)
- *<PowerFlex Version>* is encoded as `x.x.x.x`
- *<Kernel Version>* is the output of `uname -r`.

The files themselves are TAR files named `scini.tar` and `xcache.tar`.

For example:

<Repo_base>`/OL//X.X.X-OL/scini.tar` is a TAR file of the SDC driver for PowerFlex version for Oracle Linux with kernel version X.X.X-OL.

Synchronize the mirror repository

Synchronizing is performed by calling `driver_sync.sh sync`.

In normal workflow, you do not have to (and should not) modify the directory structure manually. Instead, synchronizing is done by simply calling `driver_sync.sh sync`.

In the `driver_sync.conf` file, you must set the `is_local_repo` parameter to **1**. This instructs `driver_sync.sh` not to untar the downloaded files.

Dell Technologies recommends that you change the `sync_pattern` parameter, a regular expression, to something more specific than **".*"**. Otherwise, the sync operation mirrors the entire Dell Technologies driver repository every time.

SFTP-based repository setup script

To save time in setting up a new SFTP-based driver repository, use the `driver_repo_wizard.sh` script.

The `driver_repo_wizard.sh` script can be found in the following locations on Ubuntu or Oracle Linux-based servers:

- `/bin/emc/scaleio/scini_sync`
- `/bin/emc/scaleio/xcache_sync`

The script does the following:

1. Creates a new user with a name you provide. The user will be created shell-less.
2. Creates a pair of RSA keys that will be used for passwordless SSH connection. You must provide the created private key to all your client nodes, while the public key will be added to the `authorized_keys` file of the created user.
3. Updates the SSH service configuration file (`/etc/ssh/ssh_config`) for the new user, instructing to `chroot` the user to its home directory, which should also be the repository base directory.

Configure the OpenStack Nova Compute service

If your system operates in an OpenStack environment, configure the openstack-nova-compute service to start after the scini service.

About this task

The openstack-nova-compute service must start after the scini service, to ensure that all volumes are recognized and configured by the storage data client (SDC) before the compute service starts, and the VM instances are started.

Steps

1. In the command line on the Nova Compute server, run the following command:

```
systemctl status openstack-nova-compute
```

2. Identify the service file that is currently loaded.

   For example, in the following output, the file is: `/usr/lib/systemd/system/openstack-nova-compute.service`

```
Loaded: loaded (/usr/lib/systemd/system/openstack-nova-compute.service; enabled;
vendor preset: disabled)
```

3. Open the file in a text editor, and insert the following lines under the `[Unit]` section:

```
[Unit]
...
After=scini.service
```

4. Save the file.

Results

The openstack-nova-compute service starts after the scini service.

8

Deploying PowerFlex file nodes

File storage is managed through NAS servers, which must be created before creating file systems. NAS servers can be created to support the SMB protocol, NFS protocol, or both. After the NAS servers are created, you can create file systems as containers for your SMB shares for Windows users, or NFS exports for UNIX users.

The following table provides definitions for some PowerFlex file terms.

Table 41. PowerFlex file terminology

Term	Definition
PowerFlex file services	A virtualized network-attached storage server that uses the SMB, NFS, FTP, and SFTP protocols to catalog, organize, and transfer files within file system shares and exports. PowerFlex file service is responsible for the configuration parameters on the set of file systems that it serves.
PowerFlex file node	On which the PowerFlex file node software is running.
NAS Server	A Virtual Data Mover which provides data services to users.
Virtual Data Mover (VDM)	A single point of access to a remote resource.
File system	A storage resource that can be accessed through the file sharing protocols such as SMB or NFS.
Network file system (NFS)	An access protocol that enables users to access files and folders on a network. NFS is typically used by Linux or UNIX hosts.
Server message block (SMB)	An access protocol that allows remote file data access from clients to hosts on a network. SMB is typically used in Microsoft Windows environments.
Snapshot	A point-in-time view of data stored on a storage resource. A user can recover files from a snapshot or restore a storage resource from a snapshot.

PowerFlex file node definition

- PowerFlex file services solution that is offered with PowerFlex R650 node and PowerFlex R660 nodes only.
- Single PowerFlex file cluster supports maximum of 16 nodes and a minimum of two nodes.
- Expansion can be incremented in one or multiple up to a maximum of 16 nodes.

Node configurations

Dell Technologies recommends using the following configurations for the best performance and high availability of data.

Table 42. PowerFlex R650 node file node configurations

Config	Cores	RAM (GB)	NICs (GB)	Local storage (GB)
Small	2 x 12 (24)	128	4 * 25 GbE	BOSS 480 GB * 2 M.2
Medium	2 x 16 (32)	256	4 * 25 GbE	BOSS 480 GB * 2 M.2
Large	2 x 28 (56)	256	4 * 25 / 100 GbE	BOSS 480 GB * 2 M.2

Table 43. PowerFlex R660 nodes file node configurations

Config	Cores	RAM (GB)	NICs (GB)	Local storage (GB)
Small	2 x 12 (24)	128	4 * 25 GbE	BOSS-N1 480 GB * 2 M.2
Medium	2 x 16 (32)	256	4 * 25 GbE	BOSS-N1 480 GB * 2 M.2
Large	2 x 32 (64)	384	4 * 25 / 100 GbE	BOSS-N1 480 GB * 2 M.2

Prerequisites for PowerFlex file services

Install PowerFlex file services with the following prerequisites.

- PowerFlex block storage pool with 5.5 TB free space available during the install.
- Required license.
- Supported operating system: SUSE Linux Enterprise Server 15 SP3 or SP4.
- Supported only port channel with LACP networking.
- Two*NICs with four ports (bonded interfaces with LACP enabled)
- Install the following additional packages on operating systems:
  - docker
  - sudo
  - python3-paramiko
  - python3-requests
  - inotify-tools
- Enable and start the docker service with the following commands:
  - `#sudo systemctl enable docker`
  - `#sudo systemctl start docker`
- Configure the PowerFlex file nodes to use NTP time synchronization.
- Configure the following networks on the nodes and switches:
  - **Node Management**: PowerFlex management network or any other L3 routable network must be configured. This network will be used for SSH communication from management and operations.
  - **PowerFlex data (SDS-SDC communication)**: Two or four PowerFlex data networks (same configuration as PowerFlex block deployment)
- Configure the following networks on the switch only.
  - **NAS Management**: Manages the NAS Management traffic. Configuring the IP address is not needed in the operating system. To define the IP address, see the *Define networks* topic in related information.
  - **NAS File Data**: Access PowerFlex file data from the client through NFS and SMB. Configuring the IP address is not needed in the operating system for the **NAS File Data**.
  - Create a bonded interface on the PowerFlex file node. No sub-interfaces or IP addresses should be assigned to this bonded interface dedicated to NAS traffic.

Verify connectivity

Use the following procedure to verify connectivity between SDS, SDC, and management nodes before starting the PowerFlex file deployment.

Steps

1. Verify the SDS, the block storage node, and SDC, PowerFlex file node, connections.
   a. Log in to the SDS node and check that the MTU 9000 jumbo frames are working correctly with the following command:
      `# ping -M do -s 8972 destinationIP`, where the destinationIP is data 1 or data 2 IP addresses of the PowerFlex file node.
   b. Repeat the steps for all the PowerFlex file nodes.
2. Verify the connection of the PowerFlex management platform management node and the PowerFlex file node.
   a. Log in to one of the management virtual machines (MVMs) and ping the PowerFlex file node management IP address.
      `# ping <PowerFlex management IP address of the file node>`
   b. Ensure that the SSH access is working correctly.

```
#ssh <PowerFlex management IP address of the file node>
```

NOTE: If the customer is using the non-root user for deployment, ensure that `sudo` access is working.

Sample configurations

A sample network configuration:

```
eth0 - PowerFlex Management (Node management)
eth1 - PowerFlex Data
eth2 - PowerFlex Data
bond0 (there should not be any tagged interface under bond0)
```

A sample IP address configuration on the PowerFlex file node:

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:50:56:b0:95:da brd ff:ff:ff:ff:ff:ff
    altname enp11s0
    altname ens192
    inet 10.1.1.226/20 brd 10.234.223.255 scope global eth0
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:50:56:b0:bb:8c brd ff:ff:ff:ff:ff:ff
    altname enp19s0
    altname ens224
    inet 172.16.221.226/16 brd 172.16.255.255 scope global eth1
       valid_lft forever preferred_lft forever
    inet6 fe80::250:56ff:feb0:bb8c/64 scope link
       valid_lft forever preferred_lft forever
4: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:50:56:b0:3f:09 brd ff:ff:ff:ff:ff:ff
    altname enp27s0
    altname ens256
    inet 172.17.221.226/16 brd 172.17.255.255 scope global eth2
       valid_lft forever preferred_lft forever
    inet6 fe80::250:56ff:feb0:3f09/64 scope link
       valid_lft forever preferred_lft forever
5: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group
default
    link/ether 02:42:14:01:33:16 brd ff:ff:ff:ff:ff:ff
    inet 172.29.200.1/24 brd 172.29.200.255 scope global docker0
       valid_lft forever preferred_lft forever
6: eth3: <BROADCAST,MULTICAST,SECONDARY,UP,LOWER_UP> mtu 1500 qdisc mq primary bond0
state UP group default qlen 1000
    link/ether 00:50:56:b0:65:04 brd ff:ff:ff:ff:ff:ff
    altname enp12s0
    altname ens193
7: eth4: <BROADCAST,MULTICAST,SECONDARY,UP,LOWER_UP> mtu 1500 qdisc mq primary bond0
state UP group default qlen 1000
    link/ether 00:50:56:b0:65:04 brd ff:ff:ff:ff:ff:ff
    altname enp4s0
    altname ens161
8: bond0: <BROADCAST,MULTICAST,PRIMARY,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
group default qlen 1000
    link/ether 00:50:56:b0:65:04 brd ff:ff:ff:ff:ff:ffa
```

NOTE: The switch port connected to interfaces that are assigned to NAS bond interface should be configured to accept NAS management VLAN as untagged and NAS data VLANs as tagged.

A sample Cisco Nexus switch configuration:

```
interface ethernet1/1/4
  no shutdown
  switchport mode trunk
  switchport access vlan 100
  switchport trunk allowed vlan 200,201
  mtu 9216
  flowcontrol receive off
  channel-group 8 mode active
interface port-channel10
  switchport mode trunk
  switchport trunk native vlan 100
  switchport trunk allowed vlan 200.201
  mtu 9216
  no lacp suspend-individual
  lacp vpc-convergence vpc 8
```

Resource group deployment

Define networks

Adding the details of an existing network enables PowerFlex Manager to automatically configure nodes that are connected to the network.

About this task

Enter detailed information about the available networks in the environment. This information is used later during deployments to configure nodes and switches to have the right network connectivity. PowerFlex Manager uses the defined networks in templates to specify the networks or VLANs that are configured on nodes and switches for your resource groups. This step is enabled immediately after you perform an initial setup for PowerFlex Manager.

Steps

1. On the menu bar, click **Settings** and click **Networks**.
   The **Networks** page opens.
2. Click **Define**. The **Define Network** page opens.
3. In the **Name** field, enter the name of the network. Optionally, in the **Description** field, enter a description for the network.
4. From the **Network Type** drop-down list, select one of the following network types:
   - **PowerFlex Management** - Depends on number of nodes in the PowerFlex file cluster.
   - **PowerFlex Data** - Define the number of data networks depends on the number of data networks configured on the PowerFlex block storage.
   - **NAS File Management** – Always define one additional IP address for NAS cluster which means if you have three Powerflex file nodes in the cluster, define four IP addresses (three IP addresses for nodes and one IP address for cluster). Ensure that you configured untagged VLAN in switch side for NAS File Management network.
   - **NAS File Data** - Define the VLAN ID, which is the VLAN used for configuring the NAS server interface after deployment. Reserving static IP addresses for PowerFlex file node data is optional.
5. In the **VLAN ID** field, enter a VLAN ID between 1 and 4094.

   NOTE: PowerFlex Manager uses the VLAN ID to configure I/O modules to enable network traffic to flow from the node to configured networks during deployment.

6. Optionally, select the **Configure Static IP Address Ranges** check box, and then do the following:
   a. In the **Subnet** box, enter the IP address for the subnet. The subnet is used to support static routes for data and replication networks.
   b. In the **Subnet Mask** box, enter the subnet mask.
   c. In the **Gateway** box, enter the default gateway IP address for routing network traffic.

d. Optionally, in the **Primary DNS** and **Secondary DNS** fields, enter the IP addresses of primary DNS and secondary DNS.
e. Optionally, in the **DNS Suffix** field, enter the DNS suffix to append for hostname resolution.
f. To add an IP address range, click Add **IP Address Range**. In the row, indicate the role in PowerFlex nodes for the IP address range and then specify a starting and ending IP address for the range. For the **Role**, select **Client Only**. The range is assigned to the client role on PowerFlex file nodes.

NOTE: IP address ranges cannot overlap. For example, you cannot create an IP address range of 10.10.10.1–10.10.10.100 and another range of 10.10.10.50–10.10.10.150.

7. Click **Save**.

(Optional) Edit a network

If a network is not associated with a template or resource group, you can edit the network name, the VLAN ID, or the IP address range.

Steps

1. On the menu bar, click **Settings** > **Networking** and click **Networks**.
2. Select the network that you want to modify, and click **Modify**.
3. Edit the information in any of the following fields: **Name**, **Network Type**, and **VLAN ID**.
4. Click **Save**.

(Optional) Deleting a network

You cannot delete a network that is associated with a template or resource group.

Steps

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Click the network that you want to delete, and then click **Delete**.
4. Click **OK** when the confirmation message is displayed.

Discover a resource

A resource must be discovered in PowerFlex Manager in order for PowerFlex Manager to manage it.

Prerequisites

Before you start discovering a resource, complete the following:

- Gather the IP addresses and credentials that are associated with the resources.
- Ensure that all the resources are connected to the network.

About this task

A resource in PowerFlex Manager is categorized into one of the following groups:

- Element Manager
- Node (Hardware or Software Management)
- Power Distribution Unit (PDU)
- Switch
- VM manager
- PowerFlex gateway
- PowerFlex system
- Node (Software Management)

Discover only **Node (Software Management)** for PowerFlex file deployment. This assumes that the PowerFlex file gateway was automatically added as part of the PowerFlex management platform deployment and that PowerFlex block system has already been deployed.

Steps

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources** on the **All Resources** tab.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select **Node (Software Management)**.
   b. Enter the management IP address (or hostname) of the operating system (SLES 15 SPx) that you want to discover in the **IP/Hostname Range** field.

      To discover one or more nodes by IP address, select **IP Address** and provide a starting and ending IP address.

      To discover one or more nodes by hostname, select **Hostname** and identify the nodes to discover in one of the following ways:

      - Enter the fully qualified domain name (FQDN) with a domain suffix.
      - Enter the FQDN without a domain suffix.
      - Enter a hostname search string that includes one of the following variables:

      Table 44. Hostname search variables

Variable	Description
${num}	Produces an automatically generated unique number.
${num_2d}	Produces an automatically generated unique number that has two digits.
${num_3d}	Produces an automatically generated unique number that has three digits.

      If you use a variable, you must provide a start number and end number for the hostname search.

   c. Select the **Managed** option from the **Resource State** list:
   d. To discover resources into a selected node pool instead of the global pool (default), select an existing or create a node pool from the **Discover into Node Pool** list. To create a node pool, click the **+** sign to the right of the **Discover into Node Pool** box.
   e. Select an existing or create a credential from the **Credentials** list to discover resource types. To create a credential, click the **+** sign to the right of the **Credentials** box. PowerFlex Manager maps the credential type to the type of resource that you are discovering.
   f. In **Create Credentials** select the **Credentials type** from the following available options: **OS Admin** or **OS User**.
      - If the deployment is using root user, then select **OS Admin**.
      - If the deployment is using non-root user, then select **OS User**.
   g. Provide the required information and click **Save**.
4. Click **Next** and in the **Discovery Wizard** click **Finish**.

   You might have to wait while PowerFlex Manager locates and displays all the resources that are connected to the managed networks.

   To discover multiple resources with different IP address ranges, repeat step 2 and 3.

5. The discovered resources are listed on the **Resources** page.

Add a template

About this task

The **Create** feature allows you to create a template, clone the components of an existing template into a new template, or import a pre-existing template.

For most environments, you can simply clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

Steps

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click **Create**.
3. In the **Create** dialog box, select one of the following options:
   - **Clone an existing PowerFlex Manager template**
   - **Upload External Template**
   - **Create a new template**

   If you select **Clone an existing PowerFlex Manager template**, select the **Category** and the **Template to be Cloned**. The components of the selected template are in the new template. For software-only file storage, ensure that you select **PowerFlex-File-SW only** template.
4. Enter a **Template Name**.
5. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
6. Enter a **Template Description** (optional).
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the check box next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
9. Click **Next**.
10. In **Additional Settings** provide the new values for the following settings:
    - **Network**
    - **PowerFlex Gateway**
    - **Node Pool**
11. Click **Finish**.

Edit a template

You can edit an existing template to change its draft state to published for deployment, or to modify its components and their properties.

Steps

1. On the menu bar, click **Lifecycle** > **Templates**.
2. Open a template, and click **Modify Template**.
3. Make changes as needed to the settings for components within the template. Based on the component type, required settings and properties are displayed automatically and can be edited.
   a. To edit PowerFlex cluster settings, select the **PowerFlex Cluster** component and click **Modify**. Make the necessary changes, and click **Save**.
   b. To edit PowerFlex File cluster settings, select the **PowerFlex File** component and click **Modify**. Make the necessary changes, and click **Save**.
   c. To edit node settings, select the **Node** component and click **Modify**. Make the necessary changes, and click **Save**.
4. Optionally, click **Publish Template** to make the template ready for deployment.

(Optional) Edit template information

About this task

Optionally, edit the information for a template, with the following procedure:

Steps

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click the template that you want to edit and click **Modify Template** in the right pane.
3. On the template builder page, in the right pane, click **Modify**.
4. In the **Modify Template Information** dialog box, enter a template name in the **Template Name** field.
5. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
6. In the **Template Description** box, enter a description for the template.
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriverReplacer users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the check box next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
9. Click **Save**.

Clone a template

About this task

The **Clone** feature allows you to copy an existing template into a new template. A cloned template contains the components that existed in the original template. You can edit it to add additional components or modify the cloned components.

For most environments, you can simply clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment. If you already created a template using the Add a template option, skip this task.

To clone an existing template:

Steps

1. On the menu bar, click **Templates**.
2. Click **Sample Template** > **Select**, click the **PowerFlex-File-SW Only** template from the list.

   You can also click **Clone** on the **My Templates** page if you want to clone one of your own templates, rather than one of the sample templates.

3. In the right pane, click **More Actions** > **Clone**.
4. In the **Clone Template** dialog box, enter a template name in the **Template Name** box.
5. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
6. In the **Template Description** box, enter a description for the template.
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.

- To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
  a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
  b. Select which users will have access to this resource group.
  c. To delete a user from the list, select the user and click **Remove User(s)**.
  d. After adding the users, select or clear the check box next to the users to grant or block access.
- To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.

9. Click **Next**.
10. On the **Additional Settings** page, provide new values for the **Network Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
11. Click **Finish**.
12. See the *Edit a Template* topic, in related information, for modifying the cluster components: Node (Software only), PowerFlex cluster, and PowerFlex file cluster.

Component types

Components (physical or virtual or applications) are the main building blocks of a template. Use the following section for reference purposes only, since you have already created or cloned your templates with the required information.

PowerFlex Manager has the following component types: Specific to the PowerFlex file template PowerFlex Manager has three component types:

- PowerFlex cluster
- PowerFlex file cluster
- Nodes

Cluster component settings

The following table describes the cluster component settings:

Table 45. Cluster component settings

Field name	Description
Select a Component	Select **PowerFlex Cluster** or **PowerFlex File Cluster**.
Component name	Indicates the cluster component name.
Related components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
PowerFlex cluster	
Target PowerFlex gateway	Select a target PowerFlex gateway, which acts an endpoint for PowerFlex API calls. During the provisioning process, PowerFlex Manager connects to the PowerFlex gateway and uses its APIs to configure all SDS and SDC parameters.
PowerFlex File cluster	
PowerFlex File gateway	Select PowerFlex file gateway from drop down list. PowerFlex file gateway would be discovered automatically once you have successful PowerFlex Manager platform deployment.
Number of protection domains	Determines the number of protection domains that are used for PowerFlex File configuration data. Control volumes will be created automatically for every node in the PowerFlex File cluster, and spread across the number of protection domains specified for improved cluster resiliency. To add data volumes, you need to use the tools provided on the **File** tab. You can have between one and four protection domains.

Table 45. Cluster component settings (continued)

Field name	Description
Protection domain <n>	Includes a separate section for each protection domain used in the template.
Storage pool <n>	Includes a separate section for each storage pool used in the template.

Node (software only) settings

The following table describes the following node settings: operating system and network:

Table 46. Node settings

Setting	Description
Number of Instances	Enter the number of instances that you want to add. If you select more than one instance, a single component representing multiple instances of an identically configured component is created. Edit the component to add extra instances. If you require different configuration settings, you can create multiple components.
Component Name	Indicates the node component.
Related Components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
OS Settings	
Operating System	Selected by default. There is only one option available for PowerFlex file which is SUSE Enterprise Linux.
Use Node for Dell PowerFlex	Selected by default.
Enable PowerFlex file	Always true, since the template is created for PowerFlex file.
PowerFlex role	Select the **Compute Only** option always.
Node Settings	
Node Pool	Specifies the pool from which nodes are selected for the deployment.
Network Settings	
Add New Interface	Click **Add New Interface** to create a network interface in a template component. Under this interface, all network settings are specified for a node. This interface is used to find a compatible node in the inventory. For example, if you add a **`Two Port, 100 gigabit`** to the template, when the template is deployed PowerFlex Manager matches a node with a two-port 100 GB network card as its first interface. To add one or more networks to the port, select **Add Networks to this Port**. Then, choose the networks to add, or mirror network settings defined on another port. To see network changes that are previously made to a template, you can click **View/Edit** under **Interfaces**. Or, you can click **View All Settings** on the template, and then click **View Networks**. To see network changes at resource group deployment time, click **View Networks** under **Interfaces**.
Validate Settings	Click **Validate Settings** to determine what can be chosen for a deployment with this template component.

Table 46. Node settings (continued)

Setting	Description
	The **Validate Settings** wizard displays a banner to when one or more resources in the template do not match the configuration settings that are specified in the template. The wizard displays the following tabs: • **Valid (*number*)** lists the resources that match the configuration settings. • **Invalid (*number*)** lists the resources that do not match the configuration settings.

Deploying a resource group

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

Steps

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups**. Then, click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates**. Then, click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:
   a. From the **Select Published Template** list, select the template to deploy a resource group.
   b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
   c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

      You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

      PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the ESXi version is greater than the vCenter version, PowerFlex Manager blocks the resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade vCenter first, or use a different compliance version that is compatible with the installed vCenter version.

      NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

   d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
      - To restrict access to super users, select **Only PowerFlex SuperUser**.
      - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
        i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
        ii. Select which users will have access to this resource group.
        iii. To delete a user from the list, select the user and click **Remove User(s)**.
        iv. After adding the users, select or clear the check box next to the users to grant or block access.
      - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
3. Click **Next**.
4. On the screens that follow the **Deployment Settings** page, configure the settings, as needed for your deployment.
5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.

7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

Verify resource group status

Use this procedure after the resource group is successfully deployed, to verify the status of the resource group in **Resource Groups** page.

Steps

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, ensure **Status** and **Deployment state** shows as **Deployed**.
   See the following table for various states of resource group:

Table 47. Resource group states

State	Description
Deployed	The resource group is successfully deployed and healthy.
Warning	One or more resources in the resource group requires corrective action.
Critical	The resource group is in a severely degraded or nonfunctional state and requires attention.
Pending	The deployment is scheduled for a later time or date.
In Progress	The resource group deployment is in progress, or has other actions currently in process, such as a node expansion or removal.
Cancelled	The resource group deployment has bee stopped. You can update the resources or retry the deployment, if necessary.

The **Resource Groups** page displays the resource groups that are in the following states in both **Tile** and **List** view.

9

Post-installation activities

Perform the procedures listed in the post-installation checklist after system installation to get your PowerFlex storage up and running in an optimal manner.

See the post-installation checklist to determine the procedures that are relevant for your system. Some procedures are performed using PowerFlex Manager, and some are performed using command line. PowerFlex Manager activities are described in detail in the online help system.

Post-installation checklist

The following table lists the configuration tasks that are performed after installing PowerFlex software.

The table also indicates whether tasks are required or optional. It also indicates where you can find documentation for each task.

Table 48. Post-installation checklist

Item	Required or optional?	Follow the procedures in this post-installation section	Follow PowerFlex Manager online help procedures
Enable OpenSSL FIPS compliance	Optional	X	-
Check port usage	Required	X	-
Configure SDC driver for security-enhanced Linux (SELinux)	Optional	X	-
Mount a PowerFlex volume	Optional	X	-
Configure NVMe targets and hosts for block storage if they were not configured by the installation template.	Required	-	X
Add block storage if it is not configured by the installation template, in this order: Add protection domains > Add storage pools > Add acceleration pools (for fine granularity data layout only) > Add devices to SDS (including mapping them to storage pools) > Add volumes > Map volumes to hosts	Required	-	X
Create and map volumes for block storage if not configured by the installation template	Required	-	X
Associate PowerFlex volumes with physical disks	Optional	X	-
Configure local protection for block storage (local snapshots)	Optional	-	X
Configure remote protection for block storage (asynchronous replication to a peer system) if not configured by the installation template, in this order: Install the remote PowerFlex system > Extract certificates on both systems and copy to peers > Configure journal capacity > ensure	Optional	-	X

Table 48. Post-installation checklist (continued)

Item	Required or optional?	Follow the procedures in this post-installation section	Follow PowerFlex Manager online help procedures
SDR is deployed and added > Define peer systems > Create and activate RCG			
Set up file storage services	Optional	-	X
Configure the SDS for optimal memory usage	Required	X	-
Configure the SDS for NVDIMM health awareness	Required	X	-
Configure local and LDAP users	Required	-	X
Fine tune block storage system performance	Optional	X	-
Configure event and alert notifications to external systems	Optional	-	X
Install the system license	Required	-	X
Configure SDC access to the MDM	Optional	X	-

Configuring NVMe initiators

Use the following procedures to configure the NVMe initiators on hosts for Linux and VMware ESXi-based systems.

Configure NVMe initiators on hosts for Linux-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. This procedure is provided as an example, only. Consult your operating system's documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

Prerequisites

Ensure that you have the login credentials for each host.

About this task

When NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running Linux operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex 4.5.x Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Ensure that the NVMe modules are loaded on the NVMe initiator:
   a. Log in to the host.
   b. In the command line, run the command:

```
lsmod | grep nvme
```

   c. Ensure that `nvme_tcp` and `nvme_fabrics` modules are loaded.
   If they are loaded, go to step 2. If they are not loaded, continue to the next substep.

d. Perform one of the following procedures:
   - For manual loading of modules (perform after every reboot), run:

     ```
     modprobe nvme_tcp
     modprobe nvme_fabrics
     ```

   - For automatic loading of NVMe modules, run:

     ```
     echo nvme_tcp >> /etc/modules-load.d/nvme_tcp.conf
     echo nvme_fabrics >> /etc/modules-load.d/nvme_tcp.conf
     ```

2. Ensure that the NVMe CLI is installed on the initiator:
   a. Run the following command:

      ```
      nvme
      ```

      If the CLI responds, go to step 3. If it does not, go to the next substep.
   b. Run the command:

      ```
      dnf install nvme-cli
      ```

3. Ensure that a unique host NQN and unique host ID are configured in the initiator.
   a. To verify that a host NQN is present, run the command:

      ```
      nvme show-hostnqn
      ```

      Example of a host NQN:

      ```
      nqn.2014-08.org.nvmexpress:uuid:4c4c4544-0051-5610-8046-b5c04f484b32
      ```

   b. If no host NQN exists, generate one by running the command:

      ```
      nvme gen-hostnqn > /etc/nvme/hostnqn
      ```

      NOTE: Make a note of each PowerFlex node's host NQN. You need this information when configuring the NVMe targets with PowerFlex Manager.

   c. To verify that a host ID is present, run the command:

      ```
      cat /etc/nvme/hostid
      ```

      Example of a host ID:

      ```
      4c4c4544-0051-5610-8046-b5c04f484b32
      ```

      Last 36 characters of a nvme gen-hostqn name can be used. The format must be precise, otherwise the connection fails.
4. Perform the following actions for each required protection domain:
   a. Add an NVMe host to PowerFlex.
   b. Create volume.
   c. Map volume

   For more information, see the *Dell PowerFlex 4.5.x Administration Guide* and *Dell PowerFlex 4.5.x CLI Reference Guide*.
5. Discover and connect to the target PowerFlex system. For more information, see Discover and connect to the target PowerFlex system.
6. Configure the type of multipath used:

Table 49. Multipath type configuration

Multipath type	Procedure

Native NVMe Multipath	Check if the native NVMe multipath is enabled, using the command:
	`cat /sys/module/nvme_core/parameters/multipath`
	If `Y` is returned, native NVMe multipath is already enabled.
	To enable it, run:
	`grubby --update-kernel=ALL --args="nvme_core.multipath=Y"`
	`grub2-mkconfig -o /boot/grub2/grub.cfg`
	`reboot`
	To enable active/active protocol which remains persistent after reboots, run:
	`echo 'ACTION=="add", SUBSYSTEM=="nvme-subsystem", ATTR{iopolicy}="round-robin"' > /lib/udev/rules.d/71-nvme-iopolicy.rules`

7. Configure a persistent initiator. This initiator automatically forms NVMe connections after reboots.
   a. Create a service that runs `nvme connect-all` on reboots:

```
vi /etc/systemd/system/nvme_fabrics_persistent.service
[Unit]
Description=NVMf auto discovery service
Requires=network.target
After=systemd-modules-load.service network.target
[Service]
Type=oneshot
ExecStart=/usr/sbin/nvme connect-all
StandardOutput=journal
[Install]
WantedBy=multi-user.target timers.target
```

   b. Start a new service, and enable it:

```
systemctl start nvme_fabrics_persistent.service
systemctl enable nvme_fabrics_persistent.service
```

8. Run the following command to update the udev rule to make the recovery delay 15 seconds:

```
echo ACTION!=\"remove\", KERNELS==\"ctl\", SUBSYSTEMS==\"nvme-fabrics\", ATTR{transport}==\"tcp\", ATTR{model}==\"powerflex\", ATTR{recovery_delay}=\"15\" >> /etc/udev/rules.d/71-nvme-recovery-delay.rules
```

NOTE: This step is required only for SUSE Linux Enterprise Server (SLES) 15.5 version.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **NVMe Targets**.

2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To perform the discovery to check that there is an NVMe connection, use the following command:

   ```
   #nvme discover --transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
   trsvcid=8009
   ```

   192.168.163.17 is the NVMe over TCP controller IP address (data 1 VLAN) and **192.168.163.251** is the NVMe over TCP host IP address (data 1 VLAN).

```
Discovery Log Number of Records 4, Generation counter 2
=====Discovery Log Entry 0======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  1
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.163.19
eflags:  not specified
sectype: none
=====Discovery Log Entry 1======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  17
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.163.18
eflags:  not specified
sectype: none
=====Discovery Log Entry 2======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  32
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.161.20
eflags:  not specified
sectype: none
=====Discovery Log Entry 3======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  49
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.163.17
eflags:  not specified
sectype: non
```

5. On the NVMe over TCP client, connect the client to the PowerFlex NVMe over TCP target system.
6. Configure the `/etc/nvme/discovery.conf file` to add the parameters used in the NVMe discover command and use `-p` at the end of the command for persistent controller discovery.

   Persistent discovery controller will:

   a. Notify a host of fabric changes after initial discovery.
   b. Host retains connection to discovery controller.

   NOTE: Add an entry for all the SDT IP addresses discovered in the previous step:

```
echo "--transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.18 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
```

```
echo "--transport=tcp --traddr=192.168.163.19 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.161.20 --host-traddr=192.168.161.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
```

Where

- `--traddr=192.168.161.xx`— Data 1 IP address configured in SDT.
- `--traddr=192.168.163.xx`—Data 2 IP address configured in SDT.
- `--host-traddr=192.168.163.251`—Data 1 IP address configured in the NVMe host.
- `--host-traddr=192.168.161.251`—Data 2 IP address configured in the NVMe host.

7. Connect the NVMe over TCP host to the controller system.

   ```
   # nvme connect-all
   ```

8. Verify that the NVMe over TCP host can access the namespace:

   ```
   # nvme list-subsys
   ```

Configure NVMe initiators on hosts for VMware ESXi-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. This procedure is provided as an example only. See VMware documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

About this task

When NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running VMware ESXi operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex 4.5.x Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Configure at least three vSwitches: one for management, and two for data (for example: data1 and data2)
2. Configure the VMkernel on each data vSwitch:
   a. In the command line, add the **NVMeTCP** tag to the data VMkernels. For example:

      ```
      esxcli network ip interface tag add -t NVMeTCP -i vmk1
      esxcli network ip interface tag add -t NVMeTCP -i vmk2
      ```

   b. Verify that the **NVMeTCP** tag was added:

      ```
      esxcli network ip interface tag get -i vmk1
      ```

      The returned output should include the following:

      ```
      Tags: NVMeTCP
      ```

   c. Enable NVMeTCP on the data vSwitch uplink network adapters:

      ```
      esxcli nvme fabrics enable -p TCP -d vmnic4
      esxcli nvme fabrics enable -p TCP -d vmnic5
      ```

   d. Verify that NVMeTCP was enabled on the network adapters:

      ```
      esxcli nvme adapter list
      ```

Output similar to the following should be returned:

```
Adapter  Adapter Qualified Name           Transport Type  Driver   Associated
Devices

-------  -------------------------------  --------------  -------
------------------

vmhba64  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic4
vmhba65  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic5
```

3. Get the ESXi NQN of each ESXi node, and make a note of each one. You need this information when configuring the NVMe targets with PowerFlex Manager.

```
esxcli nvme info get
```

Output similar to the following should be returned:

```
Host NQN:nqn.2014-08.com.xyz.lab.sio:nvme:sio-dcoe-116AW-13
```

4. Perform the following actions:
   a. Add NVMe host to PowerFlex.
   b. Create volume.
   c. Map volume.
   For more information, see the *Dell PowerFlex 4.5.x Administration Guide* and *Dell PowerFlex 4.5.x CLI Reference Guide*.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To discover and connect the NVMe over TCP Target PowerFlex system, perform the following:

   NOTE: Use the esxcli command to perform the operation, there is issue with persistent discovery when perform the steps using UI.

   a. Log in to VMware ESXi server using SSH.
   b. Run a discovery query on each adapter:

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009
```

   Where 6x is the first NVMe over TCP software adapter, and 6y is the second NVMe over TCP software adapter.

   Where 192.168.x.x is the first IP address and 192.168.y.y is the second IP address of any one of the SDTs.

   c. Connect to the PowerFlex system (append **-c** on the discovery query command).

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009 -c
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009 -c
```

   d. Get the controller list. Verify the connected controllers:

```
#esxcli nvme controller list
```

5. Perform a rescan of the storage.

a. In the vSphere Client, browse to a host, a cluster, a data center, or a folder that contains hosts.
b. From the right-click menu, select **Storage** > **Rescan Storage**.
c. Select **Scan for new Storage Devices** and click **OK**.

6. Create a new VMFS datastore.
   a. Log in to vCenter
   b. Select **Home/Inventory** and select the storage icon.
   c. In the left right, click the PowerFlex compute-only cluster, and select **Storage/New Datastore**.
   d. Select **VMFS** and click **NEXT**.
   e. Enter the name of the datastore and select a host from the list.
      You should see a new disk called NVMe TCP Disk (eui.######)
   f. Select the disk and click **NEXT**.
   g. Select **VMFS 6** and click **NEXT**.
   h. Leave the default configuration unless instructed by the customer and click **NEXT**.
   i. Review your selections and click **FINISH**.

SDC deployment preparation tasks for SLES, Oracle Linux, and Ubuntu

Use this procedure to deploy SDC on SLES, Oracle Linux (OL), or Ubuntu servers.

Ensure that the PowerFlex SDC is compatible with the SLES/Ubuntu/OL kernels and that the SDC driver is compatible with the SLES kernel.

NOTE: All references to Ubuntu in this section apply to supported versions of SLES and OL.

To ensure that a client node always has an SDC kernel driver that matches the client kernel, the SDC startup script uses `driver_sync.sh`, a synchronization utility that fetches SDC kernel drivers from a configured repository. This repository can be either the Dell Technologies repository, or a local repository in your private network.

For Ubuntu nodes with PowerFlex SDC drivers:

- The `driver_sync.sh` script is located, by default, at `/bin/emc/scaleio/`.
- In the repository, a driver package tarball is located at `<base_dir>/Ubuntu/<PowerFlex_version>/<kernel_version>/scini.tar`.

It may be necessary to manually restart the driver/s and services on hosts/servers hosting SDCs.

In a normal work flow, an SDC driver is loaded on boot. If there is a mismatch between the version of the running kernel and the kernel against which the driver was compiled, the `driver_sync.sh` script is run automatically to fetch the appropriate driver. If the script fails for any reason (for example, due to incorrect configuration) after fixing the issue with the `driver_sync.sh` script, the driver services must be restarted manually, as described below. Root permission is required to run the commands.

To manually restart driver and services, run this command:

```
service scini restart
```

Installing the SDC

If a host of the SDC type was not installed on front-end servers during initial installation activities, follow the appropriate instructions for your front-end servers' operating system.

See the *Dell PowerFlex 4.5.x Support Matrix* to ensure that the intended SDC operating system of the host is compatible for use with your PowerFlex system.

You can use any of the following methods to install the SDC:

Table 50. SDC installation methods by operating system

Operating system	Methods
VMware ESXi	• PowerFlex Manager procedure for adding resources to an existing system. • Command line (esxcli)
Linux	• PowerFlex Manager procedure for adding resources to an existing system (not applicable for Linux). • Command line
AIX	Command line
Windows	• PowerFlex Manager procedure for adding resources to an existing system. • Command line

Install the SDC on an VMware ESXi server

Install SDC on VMware ESXi servers using command-line esxcli commands.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host
- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site)
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires two servers reboots.

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level:

   ```
   esxcli software acceptance set --level PartnerSupported
   ```

2. Install the SDC:

   ```
   esxcli software component apply -d <path_to_zip> -n scaleio-sdc-esx7
   ```

   Where *<path_to_zip>* is the path to the SDC ZIP file.

3. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.

4. Set the IP address of the MDM:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the + symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

       `10.20.30.40,50.60.70.80+11.22.33.44`
   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

   CAUTION: A few applications require an advanced settings in ESXi for specific vital product data (VPD) pages. For more information about changing `blkScsiUnitSerialLenByStandard` parameter on an ESXi host for PowerFlex volumes, see Procedure to change parameter blkScsiUnitSerialLenByStandard on ESXi for PowerFlex volumes. Perform this procedure only if your applications require this setting. This is not a common setting.

5. Back up the parameters:

```
/bin/auto-backup.sh
```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

```
vmkload_mod -l | grep scini
```

   Output similar to the following indicates that the driver is loaded:

```
scini 2 808
```

   The SDC component is installed on the VMware ESXi-based server.

Results

The SDC is installed on the VMware ESXi-based server and is connected to PowerFlex.

Next steps

Create and map volumes

Install the storage data client on a Linux-based server

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- Secure boot is not supported; ensure that secure boot is disabled in the BIOS.

About this task

This section describes how to manually install an external SDC on a Linux-based server using the command line.

The steps in the procedure apply to Red Hat Enterprise Linux/CentOS/Oracle Linux operating systems.

Steps

1. Install the GPG key on every server on which SDC is installed. From the PowerFlex installation folder, run the following command on every server:

   ```
   rpm --import RPM-GPG-KEY-ScaleIO
   ```

2. From the command line, install the SDC:
   - Red Hat Enterprise Linux/CentOS /Oracle Linux/SLES

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
     ```

   - Ubuntu (Extract these files before use, as described above.)

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
     ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located

3. Verify that the driver is loaded using one of the following commands:
   - SLES 15:

     ```
     systemctl status scini
     ```

   - All other Linux:

     ```
     service scini status
     ```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Next steps

Create and map volumes

Install the SDC on an AIX-based server

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site)

About this task

The following procedure explains how to manually install an external SDC on an AIX-based server.

Steps

1. From the command line, install the SDC:

   ```
   MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
   ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located.

     The SDC package is in a format similar to this: `EMC-ScaleIO-sdc-.aix7.aix7.2.ppc.rpm`

2. Optional: For SDC authentication to the MDM for access, enter the following command:

   ```
   chdev -l scini -a "mdm_password=<ip>-<password>"
   ```

Results

The SDC is installed on the AIX-based server and is connected to PowerFlex.

Next steps

Create and map volumes

Install SDC on a Windows-based server

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:
- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).

About this task

The following procedure explains how to manually install an external SDC on a Windows-based server.

NOTE: This procedure requires a server reboot.

Steps

1. On the Windows-based server on which you are installing the SDC, run the following command:

   ```
   msiexec /i <SDC_PATH>.msi MDM_IP=<LIST_VIP_MDM_IPS>
   ```

   where
   - *<SDC_PATH>* is the path where the SDC installation package is located.
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not add tiebreaker addresses here.
2. Get permission to reboot the Windows-based server, and perform a reboot to load the SDC driver on the server.

Results

The SDC is installed on the Windows-based server, and is connected to PowerFlex.

Next steps

Create and map volumes

Enable OpenSSL FIPS compliance

Enable the implementation of OpenSSL Federal Information Processing Standards (FIPS) compliance in the MDM for communication between the external components, including PowerFlex Manager, PowerFlex gateway, and CLI, to the MDM. It is also enabled for any other usage of the OpenSSL library.

Prerequisites

The MDM must be hosted on Linux with the OpenSSL package installed.

Steps

1. On each host running PowerFlex, open the configuration file of each component with a text editor.
   The configuration file is `/opt/emc/scaleio/<COMPONENT>/cfg/conf.txt`, where *<COMPONENT>* is the lowercase name of the component (for example, sds).
2. Add the parameter `security_enable_fips=1` to the file.
3. Save and close the file.
4. Open the SCLI configuration file with a text editor:
   The configuration file is located at: `~/.scli/conf.txt`.
5. Add the parameter `security_enable_fips=1` to the file.
6. Save and close the file.
7. On each host, restart each component's service:

   ```
   service sds restart
   ```

8. Verify that OpenSSL FIPS compliance has been enabled by running:

   ```
   cat  /proc/sys/crypto/fips_enabled
   ```

   If it has been enabled correctly, the output should be `1`. If the output is not `1`, enable OpenSSL FIPS at the operating system level.

Check port usage

If you are using a firewall to protect your network, and need to enable ports and IP addresses, see the *Dell PowerFlex 4.5.x Security Configuration Guide*. This guide contains information about port usage for all components, and what to modify in case port numbers need to be changed.

Configure the SDC (scini) driver for SELinux

SELinux may prevent the scini driver from starting on Red Hat Enterprise Linux-based SDC nodes. This procedure prevents this issue.

About this task

If SELinux is enabled and enforced, it can prevent the SDC driver (scini) from starting on Red Hat Enterprise Linux-based nodes.

NOTE: To remove the SELinux module, run: `semodule -r scini`

Steps

1. On the SDC, use a text editor to create two text files in any directory:

a. Create a file called `scini.fe`, and add the following text:

```
/usr/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
/bin/emc/scaleio/(.*).ko  system_u:object_r:modules_object_t:s0
```

b. Create a file called `scini.te`, and add the following text:

```
module scini 1.0;

require {
        type insmod_t;
        type modules_object_t;
        class system module_load;
}
```

2. Build and install the SELinux module, by running the following commands:

```
checkmodule -M -m scini.te -o scini.mod
semodule_package -o scini.pp -m scini.mod -f scini.fe
semodule -i scini.pp
restorecon -R -v /bin/emc/scaleio/
```

3. To check the current configuration, run the command:

```
sestatus
```

Output similar to the following should be displayed, showing that *SELinux status* is enabled, and *Current mode* is enforcing:

```
SELinus status: enabled
SELinuxfs mount: /sys/fs/selinux
SELinux root directory: /etc/selinux
Loaded policy name: targeted
Current mode: enforcing
Mode from config file: enforcing
Policy MLS status: enabled
Policy deny_unknown status: allowed
Memory protection checking: actual (secure)
Max kernel policy version: 31
```

4. To check that the scini module is configured in SELinux, run the command:

```
semodule --list=full | grep scini
```

Output similar to the following should be displayed:

```
400    scini                pp
```

5. Start the scini service:

```
service scini start
```

Mount a PowerFlex volume

The exposed PowerFlex volumes are connected to the servers by the network. To configure the mounting options of PowerFlex devices, follow the instructions for your specific Linux-based operating system, below.

About this task

Use persistent device names since the `/dev/sciniX` names are not guaranteed to persist between reboots. See Associate PowerFlex volumes with physical disks for more information.

Steps

1. Determine the `/dev/disk/by-id` correlation to `/dev/sciniX`:

   ```
   ls -l /dev/disk/by-id/ |grep scini
   ```

   Output similar to the following appears:

   ```
   lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f0330000000a ->
   ../../scinia
   lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f03200000009 ->
   ../../scinib
   lrwxrwxrwx 1 root root 12 Mar 2 05:35 emc-vol-7ec27ef55b8f2108-85a0f02c00000003 ->
   ../../scinic
   ```

2. Run the mount command:

   ```
   /dev/disk/by-id/<EMC-vol-id> <mount point>
   ```

   Example:

   ```
   mount /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia
   ```

3. To make the mount command persistent, edit the `/etc/fstab` file according to the instructions for your operating system:
   - Red Hat Enterprise Linux 7.x:

     In `/etc/fstab`, use a text editor to add `_netdev` to the PowerFlex mount lines.

     Example:

     ```
     /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia xfs
     defaults,_netdev 0 0
     ```

     Ensure that you comply with the `netdev` and syntax rules for your file system, as described in the `man` page.

   - SLES:

     In `/etc/fstab`, use a text editor to add `nofail` to the mount lines.

     Example:

     ```
     /dev/disk/by-id/emc-vol-7ec27ef55b8f2108-85a0f0330000000a /mnt_scinia xfs nofail 0 0
     ```

     Ensure that you comply with the `nofail` and syntax rules for your file system, as described in the `man` page.

Associate PowerFlex volumes with physical disks

If necessary, you can associate PowerFlex volumes with physical disks.

To get PowerFlex volume information, run the CLI command `scli --query_all_volumes` (or `--query_all` or `--query_volume`).

Output similar to the following appears:

```
Query-all-volumes returned 10 volumes
Protection Domain 0728185d00000000 Name: pd1
Storage Pool ad99eaab00000000 Name: default
<No volumes defined>

Storage Pool ad99eaac00000001 Name: sp1
Volume ID: fac22a6300000000 Name: vol0 Size: 152.0 GB (155648 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6400000001 Name: vol1 Size: 400.0 GB (409600 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6500000002 Name: vol2 Size: 80.0 GB (81920 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6600000003 Name: vol3 Size: 392.0 GB (401408 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6700000004 Name: vol4 Size: 96.0 GB (98304 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6800000005 Name: vol5 Size: 112.0 GB (114688 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6900000006 Name: vol6 Size: 96.0 GB (98304 MB) Mapped to 1 SDC Thin-provisioned
Volume ID: fac22a6a00000007 Name: vol7 Size: 176.0 GB (180224 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6b00000008 Name: vol8 Size: 272.0 GB (278528 MB) Mapped to 1 SDC Thick-provisioned
Volume ID: fac22a6c00000009 Name: vol9 Size: 360.0 GB (368640 MB) Mapped to 1 SDC Thin-provisioned
```

This output shows the volume ID and name, and other volume information.

Retrieve volume information for Linux

Retrieve volume information from the SDC host.

On the SDC host, run the following command to get the operating system volume information that correlates to the PowerFlex scini device name:

```
ls -l /dev/disk/by-id/ |grep scini
```

Output, similar to the following appears:

```
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6300000000 -> ../../scinia
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6400000001 -> ../../scinic
lrwxrwxrwx 1 root root 12 Aug 25 19:40 emc-vol-62c093a52d14aec7-fac22a6500000002 -> ../../scinib
lrwxrwxrwx 1 root root 12 Aug 25 19:41 emc-vol-62c093a52d14aec7-fac22a6600000003 -> ../../scinie
lrwxrwxrwx 1 root root 12 Aug 25 19:41 emc-vol-62c093a52d14aec7-fac22a6700000004 -> ../../scinid
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6800000005 -> ../../scinif
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6900000006 -> ../../scinig
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6a00000007 -> ../../scinii
lrwxrwxrwx 1 root root 12 Aug 25 19:42 emc-vol-62c093a52d14aec7-fac22a6b00000008 -> ../../scinih
lrwxrwxrwx 1 root root 12 Aug 25 19:43 emc-vol-62c093a52d14aec7-fac22a6c00000009 -> ../../scinij
```

This output shows the scini volume name and the volume ID.

By matching the volume ID in both outputs, you can match the operating system name, sciniX, with the PowerFlex volume name.

For example:

- scinia = fac22a6300000000 = vol0
- scinic = fac22a6400000001 = vol1

Alternatively, run the `sg_inq /dev/sciniX` SCSI query command (which requires that the sg3_utils be installed on the Linux host). The result of this command includes the EMC volume ID at the bottom of the output, as illustrated:

```
Vendor identification: EMC
Product identification: ScaleIO
Product revision level: 1.3
Unit serial number: EMC-62c093a52d14aec7-fac22a6300000000
```

NOTE: The product identification remains as ScaleIO (not PowerFlex).

Add PowerFlex devices to Linux LVM physical volumes

Add PowerFlex devices to Linux LVM physical volumes.

Steps

1. In the `/etc/lvm/lvm.conf` file add the following lines:

```
types = [ "scini", 16 ]
```

2. After configured, the `lvmdiskscan` command should yield results similar to the following:

```
/dev/scinia  [      96.00 GiB] LVM physical volume
/dev/scinib  [     320.00 GiB] LVM physical volume
/dev/scinic1 [      56.00 GiB]
/dev/scinid  [      32.00 GiB]
1 disk
1 partition
2 LVM physical volume whole disks
0 LVM physical volumes
```

3. Continue with normal LVM steps.

Retrieve volume information for Windows

You can retrieve volume information from the MDM.

About this task

The `sg_inq.exe` file was added to the MSI installation and can be found at `C:\Program Files\EMC\ScaleIO\SDC\diag\`.

Steps

1. Run the `sg_inq HardiskX` SCSI query command.
   The result of this command includes the EMC volume ID at the bottom of the output.
2. On the MDM, get the PowerFlex volume information:

```
C:\Program Files\emc\scaleio\sdc\bin\drv_cfg --query_vol
```

   Output similar to the following is displayed:

```
Retrieved 5 volume(s)
VOL-ID 6acb988100000000 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988200000001 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988300000002 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988400000003 MDM-ID 0b246c9a755ca3dd
VOL-ID 6acb988500000004 MDM-ID 0b246c9a755ca3dd
```

3. From the Windows command prompt, run this command:

```
wmic diskdrive get deviceid,serialnumber | findstr "EMC"
```

   Output similar to the following is displayed:

   `\\.\PHYSICALDRIVE13 EMC-0b246c9a755ca3dd-6acb988500000004`

   The first part of the output is the disk name. In our example: `PHYSICALDRIVE13`

   The second part is the disk serial number. The last set of the second part (after the dash) is the PowerFlex volume ID. In our example: `6acb988500000004`

Retrieve volume information for AIX

On AIX servers, associate the PowerFlex volume ID with the AIX physical device.

Steps

1. On the SDC host, run the following command to get the operating system volume information:

```
#odmget -q "name like scinid* and attribute=vol_id" CuAt
```

Output similar to the following is displayed:

```
CuAt:
name = "scinid0"
attribute = "vol_id"
value = "e120a92d00000000"
type = "R"
generic = "D"
rep = "s"
nls_index = 22
[root@cnode02 /]#odmget -q "name like scinid* and attribute=vol_id" CuAt

CuAt:
name = "scinid2"
attribute = "vol_id"
value = "e120a92f00000002"
type = "R"
generic = "D"
rep = "s"
nls_index = 22

CuAt:
name = "scinid8"
attribute = "vol_id"
value = "e120a93500000008"
type = "R"
generic = "D"
rep = "s"
nls_index = 22

CuAt:
name = "scinid0"
attribute = "vol_id"
value = "e120a92d00000000"
type = "R"
generic = "D"
rep = "s"
nls_index = 22
```

You can get information for a single volume by using this command:

```
#odmget -q "name=scinid0 and attribute=vol_id" CuAt
```

2. Match the value of the `value` field with the PowerFlex volume ID.

Configure the SDS for optimal memory usage

For certain system configurations, the SDS must be configured to be able to use more than one nonuniform memory access (NUMA) domain.

About this task

Often, memory is split into a few NUMA domains. For example, in a two-CPU-socket server configuration, each CPU is in a different NUMA domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and therefore, by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0.

In VMware ESXi hyperconverged infrastructure configurations, the NUMA definition requires special attention. When configuring the SVM, there are cases where the SVM includes two vNUMAs. For example, if the number of vCPUs required for the SVM is larger than the number of physical cores in the CPU, VMware ESXi configures the SVM with two vNUMAs. Other cases are VMware ESXi hyperconverged infrastructure with replication, where the SVM includes both the SDR and the SDS. The SDS is configured on vNUMA 0 and the SDR on vNUMA 1. The total memory is split equally between the two vNUMAs, and therefore, the SDS may not have enough memory on NUMA 0.

The memory size required for the SDS depends on the configuration, and on other factors. In order to determine the required size for the SDS, and whether NUMA configuration is required, follow the size recommendations provided in the *Dell PowerFlex 4.5.x Technical Overview*.

If the memory supported by NUMA 0 (usually half the total memory) is less than the amount of memory required by the SDS, you must allow the SDS access to the memory in the other NUMA. For example, if a node has 96 GiB of physical memory, divided into 48 GiB on each CPU, and the SDS requires 58 GiB of memory, this NUMA modification is required, because 48 GiB is not sufficient for the SDS.

This operation has a slight impact to performance (less than 10%), and for this reason, is not the default setting.

NOTE: This setting can also be modified on operational SDSs that are expected to use more memory in the future, for example, due to adding SDS devices. This procedure requires a graceful SDS restart in order to take effect.

Steps

1. Use one of the following options to edit the NUMA configuration:

Option	Description
Use the echo command.	On the node where the SDS is deployed, issue the command: `echo "numa_memory_affinity=0" >> /opt/emc/scaleio/sds/cfg/conf.txt`
Use a text editor.	a. On the node where the SDS is deployed, navigate to the file: `/opt/emc/scaleio/sds/cfg/conf.txt` b. Open the file with a text editor, and add the following lines: *numa_memory_affinity=0* c. Save the file.

2. Gracefully restart the SDS:
   a. Use one of the PowerFlex management interfaces to place the SDS in instant maintenance mode.
   b. Restart the SDS process:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
/opt/emc/scaleio/sds/bin/create_service.sh
```

   c. Exit instant maintenance mode.
   d. Wait for the rebuild to finish before performing other tasks.

Next steps

Repeat this procedure on each SDS that requires this modification.

Configure the SDS for NVDIMM health awareness

Optionally, configure the SDS to enable periodic monitoring of bad blocks in NVDIMM devices.

Prerequisites

Root access permissions are required for this procedure.

About this task

To enable this feature in the PowerFlex system, perform this procedure on every SDS that uses NVDIMM acceleration devices. This feature is disabled by default.

Steps

1. In the command line, run the command:

```
echo "acc_dev_ndctl_error_detection_enable=1" >> /opt/emc/scaleio/sds/cfg/conf.txt
```

2. Restart the SDS:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
```

```
/opt/emc/scaleio/sds/bin/create_service.sh
```

3. Wait until system rebuild or rebalance is complete, and then repeat steps 1 and 2 on the next SDS, until all SDSs are configured.

Performance tuning post-installation

This section describes steps to take after completing a successful installation to enhance performance.

You can improve the PowerFlex system performance in terms of IOPS, latency, and bandwidth, by performing environment-specific fine-tuning on the operating system, network, and PowerFlex components.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, it is highly recommended to thoroughly test all changes. For further assistance, contact Dell Technologies Support.

Fine granularity layout performance considerations

Fine granularity layout enables 4 KB granularity and compression.

Table 51. Fine granularity vs medium granularity performance comparison

fine granularity	medium granularity
The benefit of the more efficient fine granularity data layout in storage pools, that allows 4 KB granularity plus compression.	in contrast to 1 MB granularity in medium granularity data layout, comes at some performance cost.
In most cases fine granularity is slower than medium granularity. When compression is enabled, reads are about 20% slower than in cases where compression is disabled. There are exceptions to these rules. For example, small IO write response time is similar in fine granularity and medium granularity.	Faster than fine granularity.
Snapshots have no impact on the performance of applications and workloads running during the snapshot creation	Snapshots might have an impact on performance in some cases. In cases that are very sensitive to performance and do not require snapshots, medium granularity would be a better choice. It would also be a good option for cases where the data cannot be compressed, as it may already be compressed, or if the data is encrypted by the application.

In cases that are very sensitive to performance and do not require snapshots, medium granularity would be a better choice. It would also be a good option for cases where the data cannot be compressed, as it may already be compressed, or if the data is encrypted by the application.

See Increase SDS threads for details about increasing SDS threads for better performance. For best performance, it is recommended to increase the number of SDS threads from eight to twelve when the CPUs have twelve or more physical cores.

NOTE: Increasing the SDS threads is expected to improve performance in cases where the CPU is bound; for example: small IOs or fine granularity layout. In cases where performance is limited by drive performance (too few drives) or by the network (usually large IOs), changing the number of SDS threads does not help.

Fine granularity metadata cache

Fine granularity MD_Cache helps improve the performance of the system when using the fine granularity layout.

When using the fine granularity layout, the fine granularity MD_Cache feature provides the following performance benefits:

- 20-40% increase in read IOPS
- Reductions in response times for reads

Fine granularity MD_Cache allocates a small portion of system memory to each SDS in the protection domain, for caching the metadata. The following formula below shows the amount of memory needed for fine granularity MD_Cache per SDS:

```
FGMC_RAM_in_GiB=(Total_Drive_Capacity_in_TiB/
2)*4%*Compression_Factor*Percent_of_Metadata_to_Cache
```

It is recommended to allocate 2 for compression factor and 2% for metadata to cache. Use the sizer tool to assess the amount of memory to allocate for metadata to cache. The maximum amount of memory that can be allocated per SDS is 32 GiB.

1. Run the following command to allocate the memory used by fine granularity MD_Cache on each SDS in the protection domain.

   ```
   scli --set_default_fgl_metadata_cache_size (--protection_domain_id <ID> | --
   protection_domain_name <NAME>) --metadata_cache_size_mb <SIZE>
   ```

2. Run the following command to enable fine granularity MD_Cache:

   ```
   scli --enable_fgl_metadata_cache (--protection_domain_id <ID> | --
   protection_domain_name <NAME>)
   ```

3. Run the following command to disable fine granularity MD_Cache:

   ```
   scli --disable_fgl_metadata_cache (--protection_domain_id <ID> | --
   protection_domain_name <NAME>)
   ```

Tuning considerations

In version 3.x and later, high_performance profile is the default setting. Medium Granularity configuration can be changed to the compact profile, which would require less memory and CPU resources, but may have some impact on the performance of the setup. With fine granularity configuration, and systems with a combination of fine and medium granularity, only high_performance profile is allowed.

The main difference between the high_performance (default) and compact profiles is the amount of server resources (CPU and memory) that are consumed. A high_performance profile (or configuration) will always consume more resources.

Users achieve optimum performance by always setting the performance profile to high_performance.

PowerFlex system changes

This section describes the various system changes available for enhancing PowerFlex performance.

Using the set_performance_parameters utility for MDM and SDS

The following table describes the commands for specific tasks:

Table 52. Performance parameter utility commands

Action	Command
View current settings.	`scli --query_performance_parameters`
By default is High_performance. Run the command to change to High_performance if the value is not set to High_performance.	`scli --set_performance_parameters --all_sds --all_sdc --apply_to_mdm --profile high_performance`
Change the profile back to compact.	`scli --set_performance_parameters --all_sds --all_sdc --apply_to_mdm --profile compact`
View full parameter settings for an MDM.	`scli --query_performance_parameters --print_all`
View the full parameter settings of a specific SDS (this also shows the MDM settings).	`scli --query_performance_parameters --sds_name <NAME> --print_all`
View the full parameter settings of a specific SDC (this also shows the MDM settings).	`scli --query_performance_parameters --sdc_name <NAME> --print_all`

Read RAM cache settings for SDS

You should keep read RAM cache disabled for SSD/flash pools. For HDD pools, read RAM cache can help increase performance. For PowerFlex storage-only nodes, turn on read RAM, the recommendation is to turn on read RAM cache for HDD pools and use as much of the server DRAM as possible.

In a configuration where PowerFlex shares the node with other applications, depending on the available DRAM resources, it may also help to turn on read RAM cache for HDD pools and increase the cache size from the default.

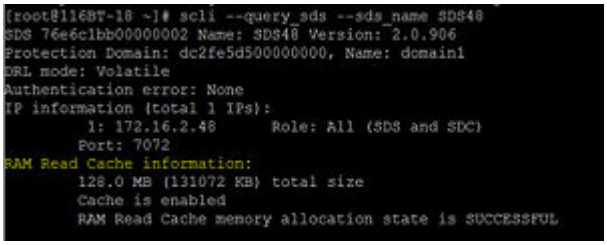
The following table describes the commands used for enabling read RAM cache and increasing the amount of read RAM cache.

NOTE: The read RAM cache may be enabled/disabled on the Protection Domain, or for each SDS in the cluster.

Table 53. Commands for enabling RAM cache and increasing the read RAM cache size

Action	Command
Enable read RAM cache	Run the following commands: • `scli --set_rmcache_usage --protection_domain_name <domain NAME> --storage_pool_name <pool NAME> --use_rmcache [--dont_use_rmcache]` • `scli --enable_sds_rmcache [--disable_sds_rmcache] --sds_name <NAME>`

Table 53. Commands for enabling RAM cache and increasing the read RAM cache size (continued)

Action	Command
	NOTE: These commands are required for every SDS in the cluster.
Increase the amount of read RAM cache	Run the following command: `scli --set_sds_rmcache_size ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>)) --rmcache_size_mb <SIZE> [--i_am_sure]` Where --rmcache_size_mb is the size of read RAM cache in MB, and the range is between 128MB—300GB. Ensure that read RAM cache is enabled at all levels (protection domain, storage pool, and SDS). When rmcache is properly enabled, query output will look like this: 

Jumbo frames and the potential impact on performance

Enabling jumbo frames can result in approximately 10% improvement in performance if all network components fully support jumbo frames. If some network components do not fully support jumbo frames, it is recommended to use the default setting; mtu 1,500.

Before activating MTU settings on the logical level, set jumbo frames = MTU 9000 on the physical switch ports that are connected to the node. Failure to do so may lead to network disconnects and packet drops.

See your relevant vendor guidelines on how to configure jumbo frame support.

Configure jumbo frames for Linux

Configure jumbo frames for NIC cards in the Linux-based PowerFlex nodes.

Steps

1. Run the `ifconfig` command to get the NIC information.
2. Depending on the operating system, run the command:

Operating system	Command
Red Hat Enterprise Linux/CentOS	Edit `/etc/sysconfig/network-scripts/ifcfg-` *<NIC_NAME>*
SLES	Edit `/etc/sysconfig/network/ifcfg-`*<NIC_NAME>*

3. Add the parameter `MTU=9000` to the file.
4. To apply the changes, type: **`service network restart`**
5. Run `ifconfig` again to verify that the settings have been changed.
6. To test the command, type: **`ping -M do -s 8972`***<DESTINATION_IP_ADDRESS>*
   Output should look similar to the following test ping to ensure MTU setting:

```
[root@116BT-18 network-scripts]# ping -M do -s 8972 10.108.158.48
PING 10.108.158.48 (10.108.158.48) 8972(9000) bytes of data.
8980 bytes from 10.108.158.48: icmp_seq=1 ttl=64 time=0.031 ms
8980 bytes from 10.108.158.48: icmp_seq=2 ttl=64 time=0.027 ms
8980 bytes from 10.108.158.48: icmp_seq=3 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=4 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=5 ttl=64 time=0.027 ms
8980 bytes from 10.108.158.48: icmp_seq=6 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=7 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=8 ttl=64 time=0.029 ms
8980 bytes from 10.108.158.48: icmp_seq=9 ttl=64 time=0.027 ms
8980 bytes from 10.108.158.48: icmp_seq=10 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=11 ttl=64 time=0.012 ms
8980 bytes from 10.108.158.48: icmp_seq=12 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=13 ttl=64 time=0.026 ms
8980 bytes from 10.108.158.48: icmp_seq=14 ttl=64 time=0.027 ms
^C
--- 10.108.158.48 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13581ms
rtt min/avg/max/mdev = 0.012/0.025/0.031/0.007 ms
```

Figure 14. Test ping to ensure MTU setting

Configure jumbo frames for Windows

Configure jumbo frames for NIC cards in the Windows-based PowerFlex nodes.

Steps

1. Change the `Maximum Transmission Unit (mtu)` setting to `9,000`, or the highest value that is supported by the switch and the connected nodes.
2. Determine the appropriate NIC name by typing the command:

   ```
   netsh interface ipv4 show interface
   ```

   Sample output:

   ```
   Idx     Met         MTU          State                Name
   ---  ----------  ----------  ------------  ---------------------------
     1          75  4294967295  connected     Loopback Pseudo-Interface 1
    10           5        1500  disconnected  Ethernet
     3          25        1500  connected     mgmt
     6           5        1500  disconnected  Ethernet 3
    12           5        1500  disconnected  Ethernet 4
     7          15        1500  connected     Data1
     9          15        9028  connected     Data2
   ```

   Figure 15. Show interface sample output

   In this example, index 17 is the appropriate network.
3. Type the command:

   ```
   netsh interface ipv4 set subinterface <network_ID> mtu=9000 store=persistent
   ```

   where *network_ID* is the ID from the output in the previous step; in this case, the ID is 17.
4. In the **Advanced** tab of the **Adapter Properties** dialog box for your vendor and driver, change the value of `Jumbo Packet` to *9000*.
5. Click **OK**.
   The network connection may disconnect briefly during this phase.
6. Verify that the configuration is working, by typing the command:

   ```
   ping -f -l 8972 <Destination_IP_Address>
   ```

Output similar to the following should display:

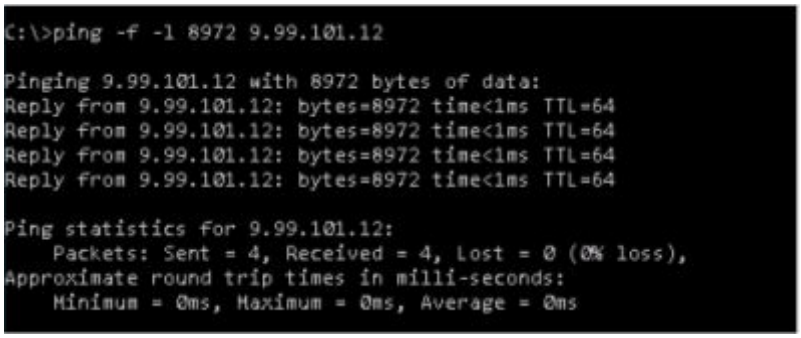


Figure 16. Ping output

NOTE: Ensure that the switch supports 10 GB Ethernet.

Configure jumbo frames for VMware ESXi

Configure jumbo frames for NIC cards in the VMware ESXi-based PowerFlex nodes.

Steps

Change the `Maximum Transmission Unit (mtu)` setting to `9,000` on the vSwitches and on the SVM (be sure to make the change in `/etc/sysconfig/network/ifcfg-ethX`):

a. Type the command:

```
esxcfg-vswitch -m 9000 <vSwitch>
```

b. Create a VMkernel with jumbo frames support by typing the following commands:

i.
```
esxcfg-vswitch -d
```

ii.
```
esxcfg-vswitch -A vmkernel# vSwitch#
```

iii.
```
esxcfg-vmknic -a -i <ip address> -n <netmask> -m 9000 <portgroup name>
```

NOTE: Changing jumbo frames on a vSwitch will not change the VMkernel MTU size. For older vCenter versions, check to ensure the MTU setting has been changed. If it was not successful, users may need to delete and re-create the VMkernel. For newer vCenter versions, modify the MTU of VMkernel using the vSphere web client.

Configure jumbo frame for storage virtual machine

Configure jumbo frames for NIC cards in the storage virtual machine (SVM)-based PowerFlex nodes.

Steps

1. Edit the file `/etc/sysconfig/network/ifcfg-<NIC_NAME>`.
2. Add the parameter `mtu=9000` to the file.
3. To apply the changes type:

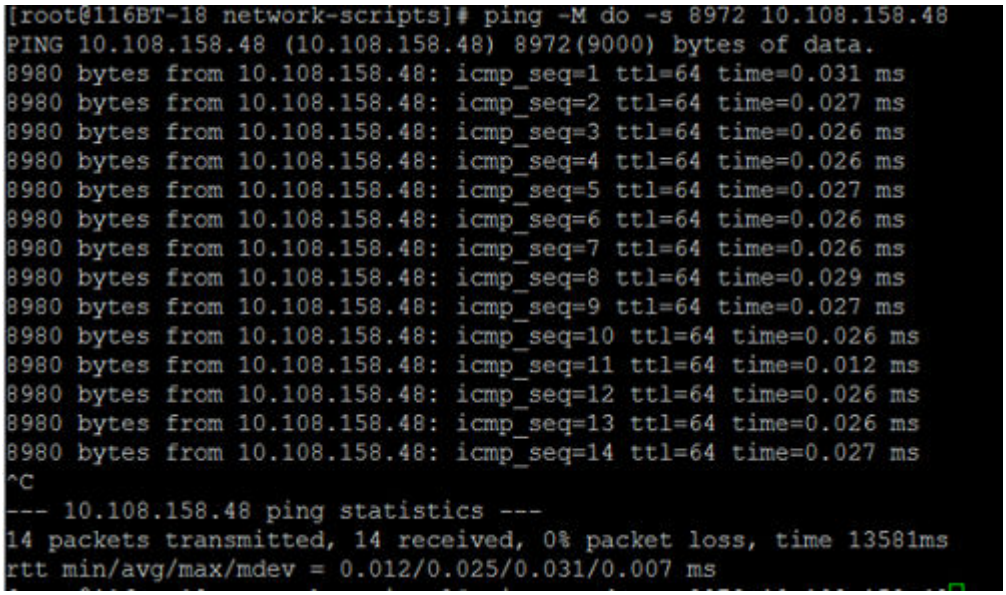
```
service network restart
```

4. Run the `ifconfig` command again to confirm that the settings have been changed.

5. To test the command type:

```
ping -M do -s 8972 <DESTINATION_IP_ADDRESS>
```

Output should look similar to the following:



Increase SDS threads

The default number of SDS threads under a high-performance profile is eight. For best performance, Dell Technologies recommends increasing the number of SDS threads from eight to twelve when the CPUs have twelve or more physical cores.

Run the following command to view the current setting:

```
scli --query_performance_parameters --all_sds --print_all
```

The parameter `Active profile: high_performance:` shows the number of SDS threads currently being used:

Table 54. SDS threads parameter values

SDS thread parameter	Current value	Profile value
SDS_NUMBER_OS_THREADS	8	8

This is logged per SDS, and should be changed on all SDS nodes in the system.

Run the following command to change the SDS threads from eight to twelve:

```
scli --set_performance_parameters --all_sds --tech --sds_number_os_threads 12
```

NOTE: Increasing the SDS threads is expected to improve the performance in cases where the CPU is bound, for example: Small IOs or fine granularity layout. In cases where the performance is limited by the drive performance (too few drives) or the network (large IOs), changing the number of SDS threads does not help.

Maximize replication throughput over WAN

Perform this procedure to maximize replication throughput over WAN.

Run the following to maximize the use of WAN bandwidth by replication:

- On each node in the local or remote cluster, append the following lines to the file `/etc/sysctl.conf`:

```
net.core.rmem_max=100000000
net.core.wmem_max=100000000
net.ipv4.tcp_rmem=5000000 5000000 100000000
net.ipv4.tcp_wmem=10000000 10000000 100000000
```

  Run the command:

```
sysctl -p
```

- On the primary MDM of the local or remote clusters, run:

```
scli --set_performance_parameters --all_libsdc_remote --
sdc_number_sockets_per_sds_ip=9 -tech
```

A mandatory restart is required for all the components on each of the nodes.

Optimize Linux

When using the SSD devices, Dell Technologies recommends that the I/O scheduler of the devices be modified.

1. Change the disk scheduler in grub by appending "elevator=noop" to the variable `GRUB_CMDLINE_LINUX_DEFAULT`.

   For example, in `/etc/default/grub`, `GRUB_CMDLINE_LINUX_DEFAULT="quiet splash elevator=noop"`

2. Recompile the grub.cfg file

   For example, `grub2-mkconfig -o /boot/grub2/grub.cfg`

3. Reboot the Linux system.

When CPUs have twelve or more physical cores, and more performance is required from the node (for example: When CloudLink is used for software encryption, or when fine granularity is used and more performance per node is required), type the following:

```
scli --set_performance_parameters --all_sds --tech --sds_number_os_threads 12
```

NOTE: To make these changes persistent after reboot, either create a script that runs on boot, or change the kernel default scheduler using kernel command line.

Change the GRUB template for Skylake/Cascade Lake CPUs

For Skylake CPUs or Cascade Lake CPUs throughput on PowerEdge R640, R750xd and R840 models running Red Hat Enterprise Linux 7.2 or later, change the GRUB template on every node to ensure that the CPUs maintain good performance in terms of latency.

Steps

1. Open the GRUB template for editing:

```
vim /etc/default/grub
```

2. Find the `GRUB_CMDLINE_LINUX` configuration option and append the following to the line:

```
intel_idle.max_cstate=1 intel_pstate=disable
```

   For example:

```
GRUB_CMDLINE_LINUX="crashkernel=auto rd.lvm.lv=rhel/root
rd.lvm.lv=rhel/swap rhgb intel_idle.max_cstate=1
intel_pstate=disable quiet"
```

3. Compile the new GRUB:

```
grub2-mkconfig -o /boot/efi/EFI/redhat/grub.cfg
```

4. Stop and then disable `tuned`:

```
systemctl stop tuned
systemctl disable tuned
```

5. Reboot the node:

```
reboot
```

6. Ping the node to ensure that the GRUB change was implemented.
   The ping time should be in the 0.03 ms range.
7. Repeat the procedure on every relevant node with a Skylake/Cascade Lake CPU.

Optimize VMware ESXi

To improve I/O concurrency, you can increase the per device queue length value on a per data store basis. Per device queue length is seen as "No of outstanding IOs with competing worlds" in the display output.

Use the following command to increase the queue length:

```
esxcli storage core device set -d <DEVICE_ID> -O <Outstanding IOs>
```

Where *<Outstanding IOs>* can be a number ranging from 32–16384 (default=32). Dell Technologies recommends increasing queue length to 256.

Example:

```
esxcli storage core device set -d eui.16bb852c56d3b93e3888003b00000000 -O 256
```

Optimize Windows

The delayed acknowledgment feature causes high latencies for low I/O rates. Dell Technologies recommends that you disable the delayed ack feature on every network interface.

To change each SDC interface, excluding the management IP address, add the following registry entries:

- `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces\<SAN interface GUID>`
- Entries: **`TcpAckFrequency TcpNoDelay`**
- Value type: **`REG_DWORD`**, number
- Value to disable: **`1`**

NOTE: Depending on the Windows version, the method for changing delayed ack varies. See your relevant vendor guidelines for specific details.

Ensure that the `customize power plan` is set to high performance, as shown in the following figure:

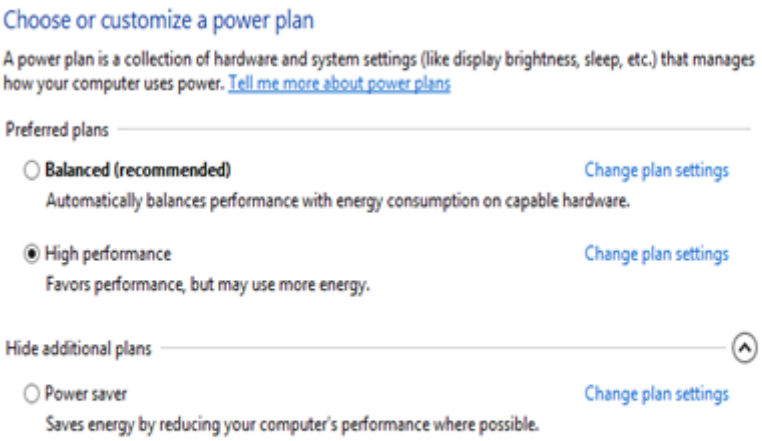


Figure 17. Customize power plan screen

Optimizing VM guests

Select the VM guest to optimize.

Modify the I/O scheduler

When using SSD devices, Dell Technologies recommends that you modify the devices' I/O scheduler.

1. Change the disk scheduler in grub by appending "elevator=noop" to the variable `GRUB_CMDLINE_LINUX_DEFAULT`.

   For example, in `/etc/default/grub`, `GRUB_CMDLINE_LINUX_DEFAULT="quiet splash elevator=noop"`

2. Recompile the grub.cfg file

   For example, `grub2-mkconfig -o /boot/grub2/grub.cfg`

3. Reboot the Linux system.

NOTE: For most Linux distributions, NOOP is not the default. Different Linux versions have different default values. For Red Hat Enterprise Linux 7 and SLES 11/12, the default value is deadline, but for older versions, the default is CFQ.

Configure the para virtual SCSI controller

Use the para virtual SCSI (PVSCSI) controller on guest VMs for high performance. It is important that you choose the correct PVSCSI controller, because choosing the wrong controller can adversely affect performance.

Current PVSCSI queue depth default values are 64 for the device and 254 for the adapter. Users can increase the PVSCSI queue depth to 256 for the device and 1024 for the adapter inside a Windows virtual machine.

Windows virtual machine:

1. From the command-line run:

```
REG ADD HKLM\SYSTEM\CurrentControlSet\services\pvscsi\Parameters\Device /v
DriverParameter /t REG_SZ /d "RequestRingPages=32,MaxQueueDepth=254"
```

2. Reboot the virtual machine.
3. Verify the successful creation of a registry entry:
   a. Open the Registry Editor by running the REGEDIT command from the command line.
   b. Browse to `HKLM\SYSTEM\CurrentControlSet\services\pvscsi\Parameters\Device`.
   c. Verify that the DriverParameter key exists with a value of RequestRingPages=32, MaxQueueDepth=254.

Linux guests:

1. Edit the `vmw_pvscsi.conf` file:

```
echo "options vmw_pvscsi cmd_per_lun=254 ring_pages=32" > /etc/modprobe.d/
vmw_pvscsi.conf
```

2. Rerun `vmware-config-tools.pl`:

```
vmware-config-tools.pl
```

You can add up to four PVSCSI controllers per guest. Allocating different PowerFlex volumes to different PVSCSI controllers can help realize the maximum potential of guest performance.

Dell Technologies strongly recommends that you review this VMware Knowledge Base article (article 2053145) so that you can make educated decisions regarding PVSCSI values.

Optimize Hyper-V

To improve Hyper-V performance and to minimize unnecessary resource consumption, review the following requirements:

- Disable NUMA spanning on Hyper-V hosts.
- Install SDC on the root partition.
- Dell Technologies recommends that you do not create virtual switches on data networks. Enable jumbo frames on virtual switches.

Optimize AIX

The following options optimize the AIX operating system:

- To turn off kernel tracker, run the following command:

```
bosedbug
```

- To reboot an AIX node, run the following command:

```
shutdown -r
```

- To review settings as viewed in the table, run the following command:

```
bosdebug
```

Table 55. AIX bosdebug settings

AIX Bosedebug parameters	Settings
Memory debugger	off
Memory sizes	0
Network memory sizes	0
Kernel debugger	off
Real-time kernel	off
Backtracking fault log	off
Kernext memory tracking	off

Configure RAID controller virtual disk settings

Configure RAID controller virtual disk settings and optimal policies regarding each type of disk.

For SSDs in use by a RAID controller:

1. Create the virtual disk as a RAID 0 device.
2. Disable the PD cache for the hard drive.
3. Set the VD caching attributes as NRWTD (no read ahead, write through, direct).

NOTE: For VxFlex Ready Node-based installations, see the VxFlex Ready Node documentation on the Dell Technologies Support Site.

Apply performance profiles to system components

You can use PowerFlex Manager to apply performance profiles to system components.

About this task

The default setting configures a predefined set of parameters for high-performance use cases. When a container is provided in the command (system or protection domain or fault set), all the objects currently in the container are configured.

The profiles are applied separately to:

- SDSs
- SDCs
- SDRs

NOTE: After changing the performance profile of an SDS (on an SVM), you must perform manual memory allocation on the SVM, as described in this guide.

Steps

1. Depending on the system component that you want to configure, from the PowerFlex Manager menu bar, click the relevant menu options:
   - For SDS: **Block** > **SDSs**
   - For SDC: **Block** > **Hosts**
   - For SDR: **Protection** > **SDRs**
2. From the list of objects that is displayed, select the desired object, and then click **Modify** > **Modify Performance Profile**.
3. In the **Modify Performance Profile** dialog box, select a profile and click **Apply**.
4. Verify that the operation has finished and was successful, and then click **Dismiss**.

Configure SDC access to the MDM

To harden SDC access to the MDM, it is possible to restrict access, pending approval of the SDC by the system. The default system setting is full access (restricted SDC mode is disabled).

When the restricted SDC mode is enabled, volumes can only be mapped to approved SDCs. Set a restricted SDC mode using the CLI. Approval is obtained by issuing the `--add_sdc` command for each SDC. You can set a restricted mode before or after SDCs have been added to your network. You can also approve SDCs using PowerFlex Manager.

The following CLI commands are used to control this feature:

Table 56. CLI commands for controlling SDC access

Action	Command
Enable or disable restricted SDC mode.	`set_restricted_sdc_mode command`
Add an SDC to the approved list, when restricted SDC mode is enabled.	`--add_sdc`

For more information, see the *Dell PowerFlex 4.5.x CLI Reference Guide*.

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Upgrade

This section describes how to upgrade PowerFlex 3.6.x, 4.0.2, or 4.5 to PowerFlex 4.5.x,

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Upgrading to PowerFlex 4.5.x from PowerFlex 4.0.2 or 4.5

Upgrade checklist

The following table provides a checklist of items required for upgrade PowerFlex from 4.0.2, 4.5 to 4.5.x. Ensure that you have done everything on this list before proceeding with the upgrade. Then, perform the remaining preparation tasks in this guide.

Table 57. Upgrade checklist for PowerFlex 4.0.2, 4.5 to 4.5.x

Item	Checked
General	
All servers intended for hosting PowerFlex components and PowerFlex Manager meet system requirements. Ensure to validate target version resource (CPU, RAM, and Disk space) requirements for virtual machines and increase accordingly if necessary.	-
Secure Boot on servers is not supported. Ensure that Secure Boot is disabled in each server's BIOS.	-
All nodes must have unique hostnames.	-
To use secure authentication mode, ensure that the latest OpenSSL 64-bit v1.0.2 \ v1.1.1 is installed on all servers in the system.	-
Ensure that you know the following: ● IP addresses of primary and secondary MDMs ● Admin username and password for the primary MDM ● Operating system host name and password of the primary MDM ● LIA password	-
For large-scale systems with up to 2048 SDCs, ensure that Linux-based operating systems of servers that host MDM, SDS, and SDR are configured to support large-scale systems (up to 2048 SDCs).	-
A PowerFlex system must have all SDCs in connected state. If SDC is not connected, the upgrade fails. To avoid a failure, remove the SDC from the GUI.	-
The LDAP authentication method for LIA is not supported and is replaced by mTLS. Before upgrading, remove the LIA LDAP using the PowerFlex Gateway.	-
VMware ESXi	
The required network interfaces and IP addresses are configured.	-
Networking	
For a fully converged Linux-based system, follow the networking recommendations in the *Dell PowerFlex 4.5.x Technical Overview*.	-
Powerflex Manager and the installer node	
The nodes which will host PowerFlex Manager must have connectivity to both data and management PowerFlex networks. SSH should be enabled on all the management nodes and should be on the same subnet.	-

Operating system requirements

The following table lists the operating system requirements for PowerFlex 4.5.x.

For the most updated list of operating systems supported by PowerFlex, see *Dell PowerFlex 4.5.x Support Matrix* at Dell Technologies Support site for the supported versions.

Table 58. Operating system requirements for PowerFlex 4.5.x

Operating system	Requirement
Linux	Packages required for all components, all Linux flavors: ● *numactl* ● *libaio* ● *wget* ● *apr* ● *libapr* ● *python2-rpm or python3-rpm* ● *yum-utils* Additional packages required for MDM components: ● bash-completion (for SCLI completion) ● Latest version of Python 2.X or Python 3.X (for newer operating system such as CentOS or Red Hat Enterprise Linux 8.x) ● binutils ● java-11-openjdk-headless Additional packages, that storage data server requires: ● smartmontools ● smartmontoolslibs ● binutils ● sg3_utils ● hdparm ● pciutils Additional packages, that Diag-tools requires: ● sar ● sysstat For supported storage controllers, the following packages are also needed: ● DELL: perccli ● LSI raid controllers: storcli ● HP raid controllers: hpssacli Additional packages required for storage data server (SDS) components that contain NVDIMMs: ● ndctl ● jq ● daxio ● libpmem To use the secure authentication mode, ensure OpenSSL 64-bit v1.1 is installed on all servers in the system. ● `libopenssl1_1_x-x.x.x86_64.rpm` ● `openssl-1.x.x-x.x.x86_64.rpm` To use LDAP, ensure OpenLDAP 2.4 is installed on all servers. Additional packages needed for LIA: ● pciutils
Windows	● Only SDC is a supported component on Windows. ● To install SDC, ensure that Security Update for Windows 7 for x64-based Systems (KB3033929) is installed.

Table 58. Operating system requirements for PowerFlex 4.5.x (continued)

Operating system	Requirement
	To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1 is not supported) • Visual C++ redistributable 2010 package, 64-bit Microsoft Windows *Storage Spaces* (available from Windows 8.1 and later) is not supported in Microsoft failover cluster running on PowerFlex volumes.
VMware	Compute-only (SDC) support: VMware ESXi operating system: 7.0 (SDC only)

Upgrade workflow

This section explains how to upgrade PowerFlex from 4.0.2 or 4.5 to 4.5.x.

Upgrading from PowerFlex 4.0.2 or 4.5 to 4.5.x includes the following activities:

1. Complete the upgrade checklist to prepare your environment for PowerFlex Manager 4.5.x.
2. Open PowerFlex Managerand from **Settings**, upload 4.5.x software catalog bundle (under **Upload Compliance File**) and compatibility management (GPG) file.
3. Upgrade PowerFlex Manager from the **Settings** page. For more information, see Upgrading PowerFlex Manager.
4. Upgrade the **PowerFlex System** (resource name) from the **Resources** menu in PowerFlex Manager.
5. Complete the postupgrade activities if required. See Post-upgrade activities for more information.

NOTE: If the upgrade is failed for any reason, rerun the upgrade operation to restart the upgrade process. The upgrade operation starts from the first node and takes a longer time to complete.

Upgrade limitations

The following are the upgrade limitations that you need to consider before starting the upgrade process.

Replication feature

Upgrades of systems where replication is used using the PowerFlex Manager are supported, but manual upgrades of replicated systems are not allowed. Contact your account manager for assistance before you manually upgrade your system.

Simultaneous upgrade of source and destination sites is not allowed. Each PowerFlex system should be upgraded separately at different times.

PowerFlex file gateway on PowerFlex nodes

If the PowerFlex file gateway (pod) hosted on a management virtual machine node is powered off or restarted, it might take a few minutes for PowerFlex file services to appear.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

Steps

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from SupportAssist (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
     - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to SupportAssist.
3. If you selected **Download from SupportAssist (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from SupportAssist, you must configure SupportAssist. To configure the SupportAssist, you must enable SupportAssist in the **Initial Configuration** wizard.

   If you are downloading a compliance file from SupportAssist, the file type is a ZIP or TAR.GZ file.

   NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: **`\\host\share\filename.zip`**
     - CIFS share for TAR.GZ file: **`\\host\share\filename.tar.gz`**
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Mark the target upgrade catalog as the default version for compliance checking.

   Download the 3.6.x and the 4.5.x software catalogs. PowerFlex requires the catalog for the version that the system is on and the target version.

5. Click **Save**.

   PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

   If the compliance file contains an embedded ISO file for an operating system image (such as for an VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

   A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Upload the compatibility matrix file

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if the software catalog versions that are currently loaded are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from SupportAssist or upload it from a local directory path. The file has a GPG extension and an associated compatibility version number.

Prerequisites

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using SupportAssist, ensure that SupportAssist has been properly configured.

Steps

1. On the menu bar, click **Settings** and click **Compatibility Management**.

2. Click **Edit Settings**.
3. Click **Download from Dell Technologies SupportAssist** if you are using SupportAssist.
4. Click **Upload from Local** to use a local file. Then, click **Choose File** to select the GPG file.

Upgrading PowerFlex Manager

This section explains how to upgrade your PowerFlex 4.0.2 to a newer version.

When a new version of the management software becomes available, you can upgrade to that version from the **Software Upgrade** page.

Upgrading PowerFlex Manager using Dell SupportAssist

Upgrading the management software using SupportAssist is the recommended method of upgrading.

Prerequisites

Before you begin, ensure that you:

- Take a backup of PowerFlex Manager settings.
- Configure SupportAssist.
- Upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software. Before proceeding with an upgrade, PowerFlex Manager notifies you if any of the following situations is detected:

- No upgrade is required. This situation occurs when the current and target versions are the same. You still can force an upgrade, if needed.
- The upgrade path that you are attempting to perform is a valid path. The source version of PowerFlex Manager is compatible with the target version.
- The source version of PowerFlex Manager is compatible with the target version.
- The upgrade path that you are attempting to perform is not supported.

The compatibility matrix file is required for all upgrades. When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any upgrades.

Whenever a new version of the management software is available, PowerFlex Manager displays a banner at the top of all pages to notify you of the new release. The banner appears only if you have registered with SupportAssist.

Steps

1. On the menu bar, click **Settings**, and then click **Software Upgrade**.

   Alternatively, click **View Details** on the notification banner at the top of the page.

2. Click **Management Software Upgrade**.
3. On the **Management Software Upgrade** page, you can see the **Current Management Software Version** and check the **Available Management Software Version** field to see if a newer version of PowerFlex Manager is available.
4. Click **Edit Settings**.

   Select **Update Software from configured Dell SupportAssist (Recommended)** and click **Save**.

   The **Repository Path** field on the **Management Software Upgrade** page shows your saved changes.

5. At the top of the **Management Software Upgrade** page, click **Upgrade Now**.
   After you click **Upgrade Now**, a dialog box displays with a warning message indicating whether the upgrade path is valid.
6. Under **Management Credentials**, optionally override the default username and password for the nodes hosting PowerFlex Manager. The upgrade requires access to the nodes hosting PowerFlex Manager. Provide a single user that has superuser privileges for all nodes. You can enter the username for the first row in the table, and the changes you make will be applied to all the nodes. Enter the password for each of the nodes.

7. Type **UPDATE POWERFLEX MANAGER** if you want to proceed with the update. If you want to perform an update even if none is required, type **FORCE UPDATE**.
8. Click **Yes** to update PowerFlex Manager.
   The update process displays messages indicating the progress of the update. Depending on which services were upgraded, you may be required to login once again, after the upgrade process is complete.

Next steps

After upgrading to PowerFlex Manager, you may notice that resource groups that are tied to the PowerFlex gateway are no longer in compliance. To bring the resource groups into compliance, you may need to upgrade them.

If you do not want to upgrade the resource groups, you do not need to perform the upgrade.

Upgrading PowerFlex Manager from a local repository path

You can upgrade the management software from a .TGZ file on a local repository path.

Prerequisites

Before you begin, ensure that you:

- Take a backup of PowerFlex Manager settings.
- Upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software. Before proceeding with an upgrade, PowerFlex Manager notifies you if any of the following situations is detected:

- No upgrade is required. This situation occurs when the current and target versions are the same. You still can force an upgrade, if needed.
- The upgrade path that you are attempting to perform is a valid path. The source version of PowerFlex Manager is compatible with the target version.
- The source version of PowerFlex Manager is compatible with the target version.
- The upgrade path that you are attempting to perform is not supported.

The compatibility matrix file is required for all upgrades. When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any upgrades.

Steps

1. If you are using PowerFlex rack, log in to Dell Technologies Download Center. Then, go to the section for PowerFlex Manager.
2. If you are using PowerFlex appliance, log in to Dell Technologies Support. Then, enter your service tag to be taken to available downloads.
3. Download the PowerFlex management platform .TGZ file.
4. Copy the PowerFlex management platform .TGZ file to a share drive location.
5. On the menu bar, click **Settings**, and then click **Software Upgrade**.
6. Click **Management Software Upgrade**.
7. On the **Management Software Upgrade** page, you can see the **Current Management Software Version** and check the **Available Management Software Version** field to see if a newer version of PowerFlex Manager is available.
8. Click **Edit Settings**.
   a. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.
   b. Enter the username and password.
   c. Click **Save**.
9. At the top of the **Management Software Upgrade** page, click **Upgrade Now**.
   After you click **Upgrade Now**, a dialog box displays with a warning message indicating whether the upgrade path is valid.
10. Under **Management Credentials**, optionally override the default username and password for the nodes hosting PowerFlex Manager. The upgrade requires access to the nodes hosting PowerFlex Manager. Provide a single user that has superuser

privileges for all nodes. You can enter the username for the first row in the table, and the changes you make will be applied to all the nodes. Enter the password for each of the nodes.

11. Type **UPDATE POWERFLEX MANAGER** if you want to proceed with the update. If you want to perform an update even if none is required, type **FORCE UPDATE**.
12. Click **Yes** to update PowerFlex Manager.
    The update process displays messages indicating the progress of the update. Depending on which services were upgraded, you may be required to log in once again, after the upgrade process is complete.

Next steps

After upgrading to PowerFlex Manager, you may notice that resource groups that are tied to the PowerFlex gateway are no longer in compliance. To bring the resource groups into compliance, you may need to upgrade them.

If you do not want to upgrade the resource groups, you do not need to perform the upgrade.

Finalize firewall configuration

RKE2 uses Calico as a networking and security solution that enables Kubernetes workloads to communicate securely performing similar yet more dynamic functions as firewalld.

About this task

Firewalld can interfere with the dynamic nature of Calico, firewalld must be disabled. For more information, see Known issues and limitations with rke2.

Steps

1. To disable firewalld, do the following:
   a. Log in to the PowerFlex management platform node using SSH and run the following command:

   ```
   systemctl stop firewalld
   systemctl disable firewalld
   ```

   b. To verify that that firewalld is stopped and disabled, run the following command:

   ```
   systemctl status firewalld
   ```

   Example output:

   ```
   firewalld.service - firewalld - dynamic firewall daemon
        Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor
   preset: disabled)
        Active: inactive (dead)
          Docs: man:firewalld(1)
        Do this on all remaining PFMP nodes
   ```

2. Customize the Calico firewall if required.
   For example,
   If you want to open ports 90, 80, 9000, and 8000 and this is a co-resident deployment, run the following command:

   ```
   kubectl patch hr gnwpolicy -n powerflex --type merge --patch '{"spec":{"values":
   {"customtcpports":[90,80,9000,8000],"iscores":"true"}}}'
   ```

   If this is not a co-resident deployment, set `iscores` to false. If you have open ports in PowerFlex 4.5, the ports must be added again after upgrading to PowerFlex 4.5.1 and later versions.

Upgrade the PowerFlex software

Use this procedure to upgrade the PowerFlex software.

Steps

1. Log in to PowerFlex Manager.
2. To upgrade the PowerFlex gateway, on the **Resources** page, select the resource that has the **OS Hostname** as **powerflex**.
3. Verify the PowerFlex node details.
4. Click **Update Resources**.
5. On the **Summary** page, select **Allow PowerFlex Manager to perform non-disruptive updates now** and **Instant Maintenance Mode**. For more information about instant maintenance mode and protected maintenance mode, see *Dell PowerFlex 4.5.x Technical Overview*.
6. Enter **`UPDATE POWERFLEX`** in the required field.
7. Click **Finish**.
8. Click **Yes** to confirm the changes.

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Upgrading to PowerFlex 4.5.x from PowerFlex 3.6

This section explains how to upgrade your PowerFlex to PowerFlex 4.5.x from 3.6.

Preparing to upgrade

This section explains the preparation tasks required to perform the upgrade.

Upgrade checklist

The following table provides a checklist of items required for upgrading PowerFlex to version 4.5.x from PowerFlex 3.6. Ensure that you have done everything on this list before proceeding with the upgrade. Then, perform the remaining preparation tasks in this guide.

Table 59. Upgrade checklist for PowerFlex 3.6.x to 4.5.x

Item	Checked
General	
All servers intended for hosting PowerFlex components and PowerFlex Manager meet system requirements.	-
Secure Boot on servers is not supported. Ensure that secure boot is disabled in each server's BIOS.	-
All nodes must have unique hostnames.	-
To use secure authentication mode, ensure the latest OpenSSL 64-bit v1.0.2 \ v1.1.1 is installed on all servers in the system.	-
Ensure that you know the following: ● IP addresses of primary and secondary MDMs ● Admin username and password for the primary MDM ● Operating system hostname and password of the primary MDM ● LIA password	-
For large-scale systems with up to 2048 SDCs, ensure the Linux-based operating systems of servers that host the MDM, SDS, and SDR are configured to support large-scale systems (up to 2048 SDCs).	-
A PowerFlex system must have all SDCs in connected state. **NOTE:** If the SDC is not connected, the upgrade fails. To avoid a failure, remove the SDC from the user interface.	-
The LDAP authentication method for LIA is not supported and is replaced by mTLS. Before upgrading, remove the LIA LDAP using the PowerFlex gateway.	-
Nonsecure communication to the MDM is not supported from PowerFlex 4.x. Ensure you change to secure mode before upgrading PowerFlex.	-
VMware ESXi	
The required network interfaces and IP addresses are configured.	-
Networking	
For a fully converged Linux-based system, follow the networking recommendations in the *Dell PowerFlex 4.5.x Technical Overview*.	-

Table 59. Upgrade checklist for PowerFlex 3.6.x to 4.5.x (continued)

Item	Checked
Powerflex Manager and its installer node	
The PowerFlex nodes which host PowerFlex Manager must have connectivity to both data and management PowerFlex networks. SSH should be enabled on all the management nodes and should be on the same subnet.	-
If the 3.6.x system is configured with trusted addresses for LIA, before upgrading the system, use the procedures documented in the KB article: PowerFlex: Failure to import v3.6 PowerFlex to PowerFlex Manager Platform (PFMP) with trusted IP enabled	-

Operating system requirements

The following table lists the operating system requirements for PowerFlex 4.5.x.

For the most updated list of operating systems supported by PowerFlex, see *Dell PowerFlex 4.5.x Support Matrix* at Dell Technologies Support site for the supported versions.

Table 60. Operating system requirements for PowerFlex 4.5.x

Operating system	Requirement
Linux	Packages required for all components, all Linux flavors: • *numactl* • *libaio* • *wget* • *apr* • *libapr* • *python2-rpm or python3-rpm* • *yum-utils* Additional packages required for MDM components: • bash-completion (for SCLI completion) • Latest version of Python 2.X or Python 3.X (for newer operating system such as CentOS or Red Hat Enterprise Linux 8.x) • binutils • java-11-openjdk-headless Additional packages, that storage data server requires: • smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils Additional packages, that Diag-tools requires: • sar • sysstat For supported storage controllers, the following packages are also needed: • DELL: perccli • LSI raid controllers: storcli • HP raid controllers: hpssacli Additional packages required for storage data server (SDS) components that contain NVDIMMs: • ndctl • jq

Table 60. Operating system requirements for PowerFlex 4.5.x (continued)

Operating system	Requirement
	• daxio • libpmem To use the secure authentication mode, ensure OpenSSL 64-bit v1.1 is installed on all servers in the system. • `libopenssl1_1_x-x.x.x86_64.rpm` • `openssl1-1.x.x-x.x.x86_64.rpm` To use LDAP, ensure OpenLDAP 2.4 is installed on all servers. Additional packages needed for LIA: • pciutils
Windows	• Only SDC is a supported component on Windows. • To install SDC, ensure that Security Update for Windows 7 for x64-based Systems (KB3033929) is installed. To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1 is not supported) • Visual C++ redistributable 2010 package, 64-bit Microsoft Windows *Storage Spaces* (available from Windows 8.1 and later) is not supported in Microsoft failover cluster running on PowerFlex volumes.
VMware	Compute-only (SDC) support: VMware ESXi operating system: 7.0 (SDC only)

Prerequisites

Before upgrading PowerFlex, ensure that you record the user definitions and role mapping defined in the earlier versions for local users and LDAP groups. After upgrade, user definitions and role mappings are not imported from earlier versions of PowerFlex and must be configured again.

CAUTION: If using firewalls, ports and protocols have changed from PowerFlex 3.6 to 4.x. PowerFlex 4.x deployments require permitting new ports and IP addresses. For requirements, see the Dell PowerFlex 4.5.x Install and Upgrade Guide.

Perform the following on the primary MDM before the upgrade:

- Run **`scli --query_users`** to record the local username and user roles.
- Run **`scli --query_user_authentication_properties`** to identify the LDAP groups. LDAP user roles are defined by their groups.
- Use the old to new roles mapping table to convert the old roles to new roles for each local user and LDAP group.

Table 61. Old to new roles mapping

Old role	New role
PowerFlex file services administrator	Storage administrator
PowerFlex file services storage administrator	Storage administrator
SIO administrator	Storage administrator
SIO back-end configurator	Llifecycle administrator
SIO configurator	System administrator
SIO front-end configurator	Storage administrator
SIO local super user	Super user
SIO monitor	Monitor

Table 61. Old to new roles mapping (continued)

Old role	New role
SIO security	Security administrator
SIO technician commands	Support
PowerFlex Manager operator	Drive replaces
PowerFlex Manager read only	Monitor
PowerFlex Manager standard member	Lifecycle administrator
PowerFlex Manager standard owner	Lifecycle administrator

- If you are using LDAP, connect PowerFlex Manager 4.5 to LDAP and ensure the following:
  - LDAP groups must be available in PowerFlex Manager 4.5.x. Use PowerFlex Manager or REST API commands to assign each LDAP group with its newly mapped role.
  - LDAP users must be available in PowerFlex Manager 4.5.x with all their LDAP group roles.
  - You can add user-specific roles for LDAP users.
- Define the local users in PowerFlex Manager or REST API. Set a new password and assign the relevant role to each other.

Upgrade workflow

This section explains how to upgrade your PowerFlex 4.5.x from 3.6.

Upgrading from PowerFlex 3.6 to 4.5.x includes the following activities:

1. Complete the upgrade checklist to prepare your environment for PowerFlex Manager 4.5.x.
2. Install PowerFlex Manager 4.5.x to replace the existing management stack (PowerFlex gateway and PowerFlex presentation server).
3. Use the initial configuration wizard to import PowerFlex 3.6 system into PowerFlex Manager 4.5.x.
4. From **Settings**, upload the software catalog bundles (under **Upload Compliance File**) and compatibility management (GPG) file.
5. Upgrade the **PowerFlex System** (resource name) from the **Resources** menu in PowerFlex Manager.
6. Complete the postupgrade activities. See Post-upgrade activities for more information.

NOTE: If the upgrade fails, rerun the upgrade operation to restart the upgrade process. The upgrade operation starts from the first PowerFlex node and takes a longer time to complete.

Upgrade limitations

The following are the upgrade limitations that you need to consider before starting the upgrade process.

Replication feature

Upgrades of systems where replication is used using the PowerFlex Manager are supported, but manual upgrades of replicated systems are not allowed. Contact your account manager for assistance before you manually upgrade your system.

Simultaneous upgrade of source and destination sites is not allowed. Each PowerFlex system should be upgraded separately at different times.

Deploying into a VMware vSphere environment

Use this option if the PowerFlex management virtual machines reside in an existing VMware vSphere environment.

Management VM resource requirements

Table 62. Management VM resource requirements

PowerFlex version	Storage	CPU	RAM
4.5.x	600 GB	14 vCPU	32 GB*

* To manage the PowerFlex cluster with 32 K volume, each node must have 48 GB RAM.

Deploying the PowerFlex management platform

This section describes how to install and configure the PowerFlex management platform.

This includes the deployment and configuration of the temporary PowerFlex management platform installer VM. This PowerFlex management platform installer VM is used to deploy the containerized services required for the PowerFlex management platform.

The following diagram depicts the PowerFlex installer VM and management VM deployed in a virtual environment and highlights the interfaces that are needed to be configured:

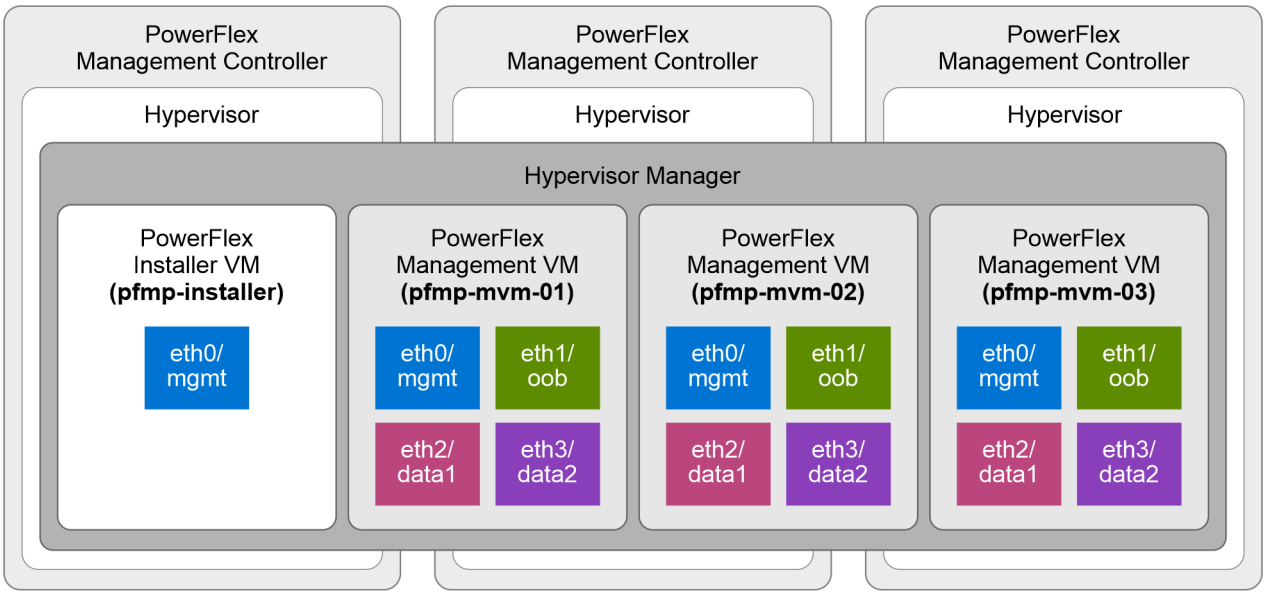


Figure 18. PowerFlex installer VM and MVMs deployed in a virtual environment

Deploying and configuring the PowerFlex management platform installer VM

Add Entrust root certificate to the VMware vCenter certificate store

If you want to use third-party certificates in your environment, you must add a third party trusted root certificate to the VMware vCenter certificate store.

Steps

1. Manually download OVCS2 certificate (OVCS2-CSBR1-crosscert) from the Entrust website: Entrust TLS certificate information center. Search for keyword "OVCS2" and click on CRT to download the certificate.

2. Upload the Entrust root certificate to VMware vCenter certificate store:
   a. Log in to the vCenter Server.
   b. Click **Administration** > **Certificates** > **Certificate Management**.
   c. Next to **Trusted Root Store**, click **Add**.
   d. Browse and select the downloaded Entrust root certificate. The certificate is added in a panel under **Trusted Root Certificates**.
3. Once the certificates are installed, OVA can be verified and deployed.

Deploy the PowerFlex management platform installer VM

Use this procedure to deploy and configure the PowerFlex management platform installer VM in a VMware environment.

Prerequisites

- For a multi-VLAN or multi-subnet configuration, pfmc-vm-mgmt-<vlanid> must be selected instead of the flex-node-mgmt-<vlanid>.
- For standard configurations without multi-networking, select either pfmc-vm-mgmt-<vlanid> (130) or flex-node-mgmt-<vlanid> (105).

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA Template**
5. Click **Open** > **Next**.
6. Enter **`pfmp-installer`** for the VM name.
7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.
10. Select the **PFMP datastore**.
11. Select **Virtual disk format** > **Thin provision** > **Next**.
12. Select **flex-node-mgmt-<vlanid>** > **OK** > **Next**.

    NOTE:
    - For a multi-VLAN configuration, more than one port-group for pfmc-sds-mgmt-<vlanid> exists. Select the first port-group for VM deployment.
    - For a multi-VLAN or multi-subnet configuration, the management interface of the MVMs and installer VM must be configured on VLAN 130 instead of VLAN 105.

13. Click **Finish**.
14. Right-click the VM and select **Power** > **Power On**.

Configure the PowerFlex management platform installer networking interface

Use this procedure to configure the networking for the PowerFlex management platform installer VM.

About this task

The following network adapter needs to be configured on the Powerflex management platform installer:

VLAN 105: flex-node-mgmt (eth0) or VLAN 130: pfmc-mgmt (eth0)

For a multi-VLAN or multi-subnet configuration, the PowerFlex management controller management is configured instead of the VMware ESXi management. For a standard deployment without multi-networking the network adapter can be configured on either VMware ESXi management (VLAN 105) or PowerFlex management controller management (VLAN 130).

Steps

1. Launch the web console from vCSA and log in as **`delladmin`**.

2. To configure the <flex-node-mgmt> interface, type **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

Example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.10
NETMASK=255.255.255.0
```

a. To configure the default route, type **sudo vi /etc/sysconfig/network/routes. default <gateway ip> - <interface>**

Example: **default 10.10.105.1 - eth0**

b. To configure DNS search and DNS servers, type **sudo vi /etc/sysconfig/network/config** and modify the following: **NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>" NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"**

Example: **NETCONFIG_DNS_STATIC_SEARCHLIST="example.com" NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"**

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs or physical nodes (co-resident or bring your own Linux PowerFlex management platform deployments).

Steps

1. To configure the NTP server, enter **sudo vi /etc/chrony.conf** and add the following:

   server <ntp ip address> iburst

   For example: **server 10.10.105.240 iburst**
2. To start the chronyd service, enter:

   sudo systemctl start chronyd
3. To enable chronyd, enter:

   sudo systemctl enable chronyd
4. To restart chronyd, enter:

   sudo systemctl restart chronyd
5. To verify that the **Network Time Protocol service** is running, enter:

   sudo systemctl status chronyd.service
6. Repeat all the steps for all the PowerFlex management platform VMs and the installer VM.
7. To check that the server is synced to NTP, enter:

   chronyc tracking

   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status

   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

```
PFMP2-450-747 :~ # chronyc tracking
Reference ID : 179628F2 (23.150.40.242)
Stratum : 3
Ref time (UTC) : Tue Mar 12 21:24:20 2024
```

```
System time : 0.000531569 seconds slow of NTP time
Last offset : -0.000336191 seconds
RMS offset :  0.000201320 seconds
Frequency : 6.809 ppm slow
Skew : 0.029 ppm
Root delay : 0.037072252 seconds
Root dispersion : 0.004111907 seconds
Update interval : 7242.7 seconds
Leap status : Normal
PFMP2-450-747 :~ #
```

Unsuccessful chronyc tracking status

- `Stratum` is 0 (not synced).
- Leap status says `Normal`:

```
node1:~ # chronyc tracking
Reference ID : 00000000 ()
Stratum : 0
Ref time (UTC) : Thu Jan 01 00:00:00 1970
System time : 0.000000000 seconds fast of NTP time
Last offset : +0.000000000 seconds
RMS offset : 0.000000000 seconds
Frequency : 0.000 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
```

Deploying and configuring the PowerFlex management platform VMs

Use this procedure to complete the steps needed to deploy the PowerFlex management platform cluster VMs.

Prerequisites

The PowerFlex management platform cluster requires three VMs to be deployed.

Steps

1. Log in to the VMware vCSA.
2. Click **Menu** > **Shortcuts** > **Hosts and Clusters**.
3. Right-click the **ESXi Host** > **Select Deploy OVF Template**.
4. Select **Local File** > **Upload Files** > **Browse to the PowerFlex management platform OVA**
5. Click **Open** > **Next**.
6. Enter **`pfmp-mvm-<number>`** for the VM name.
7. Click **Next**.
8. Verify that there are no compatibility warnings and click **Next**.
9. Click **Next**.
10. Review details and click **Next**.
11. Select the datastore.
12. Select **Virtual disk format** > **Thin provision** > **Next**.
13. Select **Desired Destination Network** > **OK** > **Next**.
14. Click **Finish**.
15. Repeat the above for the three management VMs.

Configure the PowerFlex management platform networking interfaces

Use the following procedure to complete the steps to configure the networking files for the PowerFlex management platform cluster VMs.

About this task

NOTE:

- For a multi-VLAN or multi-subnet configuration, the PowerFlex management controller management must be configured on PowerFlex management controller management (130).
- For standard builds without multi-VLAN or multi-subnet configuration, can be on either VMware ESXi management (105) or PowerFlex management controller management (130).

NOTE: Based on your requirement, configure only the required interfaces on the management VMs. If you have unused interfaces on the deployed management VMs, then ensure you disconnect those interfaces within the vCenter. For example, an out-of-band interface. If the unused interfaces are connected and unconfigured, then PowerFlex Manager may display unexpected errors.

Steps

1. Launch the web console from the Virtual Machine Manager, log in as **delladmin**.
2. Configure the flex-node-mgmt-<vlanid> eth0 interface:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth0**

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   For example:

```
DEVICE=eth0
NAME=eth0
STARTMODE=auto
BOOTPROTO=static
IPADDR=10.10.105.30
NETMASK=255.255.255.0
```

   b. Configure the default route. Type: **sudo vi /etc/sysconfig/network/routes**.

   default <gateway ip> - <interface>

   For example:

   default 10.10.105.1 - eth0

   c. Configure DNS search and DNS servers. Type: **sudo vi /etc/sysconfig/network/config** and modify the following:

   NETCONFIG_DNS_STATIC_SEARCHLIST="<search domain>"

   NETCONFIG_DNS_STATIC_SERVERS="< dns_ip1 dns_ip2>"

   For example:

   NETCONFIG_DNS_STATIC_SEARCHLIST="example.com"

   NETCONFIG_DNS_STATIC_SERVERS="10.10.105.240 10.10.105.241"

3. Configure the flex-data1-<vlanid> eth1 interface:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth1**.

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
```

```
IPADDR=<ip address>
NETMASK=<mask>
```

For example:

```
DEVICE=eth1
NAME=eth1
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.151.30
NETMASK=255.255.255.0
```

4. Configure the flex-data1-<vlanid> eth2 interface:
   a. Type: **sudo vi /etc/sysconfig/network/ifcfg-eth2**.

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=<ip address>
NETMASK=<mask>
```

   For example:

```
DEVICE=eth2
NAME=eth2
STARTMODE=auto
BOOTPROTO=static
IPADDR=192.168.152.30
NETMASK=255.255.255.0
```

5. Run the following command to update the network configuration:
   `sudo systemctl restart network`
6. Run the following command to change the delladmin account password:
   `passwd delladmin`
7. Change the root password with the following command:
   `sudo passwd root`
8. Repeat steps 1 through 8 for all PowerFlex management platform VMs.

Configure the NTP service

Configure the NTP services for the PowerFlex management platform installer virtual machine (VM) and the PowerFlex management platform cluster VMs or physical nodes (co-resident or bring your own Linux PowerFlex management platform deployments).

Steps

1. To configure the NTP server, enter **sudo vi /etc/chrony.conf** and add the following:
   server <ntp ip address> iburst
   For example: **server 10.10.105.240 iburst**
2. To start the chronyd service, enter:
   sudo systemctl start chronyd
3. To enable chronyd, enter:
   sudo systemctl enable chronyd
4. To restart chronyd, enter:
   sudo systemctl restart chronyd
5. To verify that the **Network Time Protocol service** is running, enter:
   sudo systemctl status chronyd.service
6. Repeat all the steps for all the PowerFlex management platform VMs and the installer VM.

7. To check that the server is synced to NTP, enter:

   `chronyc tracking`

   NOTE: The time may take a few minutes to sync. The following are examples of chronyd tracking status:

   Successful chronyc tracking status

   - A configured NTP server should be listed under `Reference ID`,
   - `Stratum` should be a value greater than zero.
   - `System Time` should report less than 1 s time difference:

```
PFMP2-450-747 :~ # chronyc tracking
Reference ID : 179628F2 (23.150.40.242)
Stratum : 3
Ref time (UTC) : Tue Mar 12 21:24:20 2024
System time : 0.000531569 seconds slow of NTP time
Last offset : -0.000336191 seconds
RMS offset :  0.000201320 seconds
Frequency : 6.809 ppm slow
Skew : 0.029 ppm
Root delay : 0.037072252 seconds
Root dispersion : 0.004111907 seconds
Update interval : 7242.7 seconds
Leap status : Normal
PFMP2-450-747 :~ #
```

   Unsuccessful chronyc tracking status

   - `Stratum` is 0 (not synced).
   - Leap status says `Normal`:

```
node1:~ # chronyc tracking
Reference ID : 00000000 ()
Stratum : 0
Ref time (UTC) : Thu Jan 01 00:00:00 1970
System time : 0.000000000 seconds fast of NTP time
Last offset : +0.000000000 seconds
RMS offset : 0.000000000 seconds
Frequency : 0.000 ppm slow
Residual freq : +0.000 ppm
Skew : 0.000 ppm
```

Deploy the PowerFlex management platform cluster

Use the following procedure to complete the steps to configure the `PFMP_Config.json` file, which is used for the configuration and deployment of the PowerFlex management platform cluster.

Prerequisites

The following locations contain log files for troubleshooting:

- PowerFlex management platform installer logs: `/opt/dell/pfmp/PFMP_Installer/logs`
- Platform installer logs: `/opt/dell/pfmp/atlantic/logs/bedrock.log`

The following diagram illustrates how PFMP_config.JSON file content is applied to PowerFlex management.

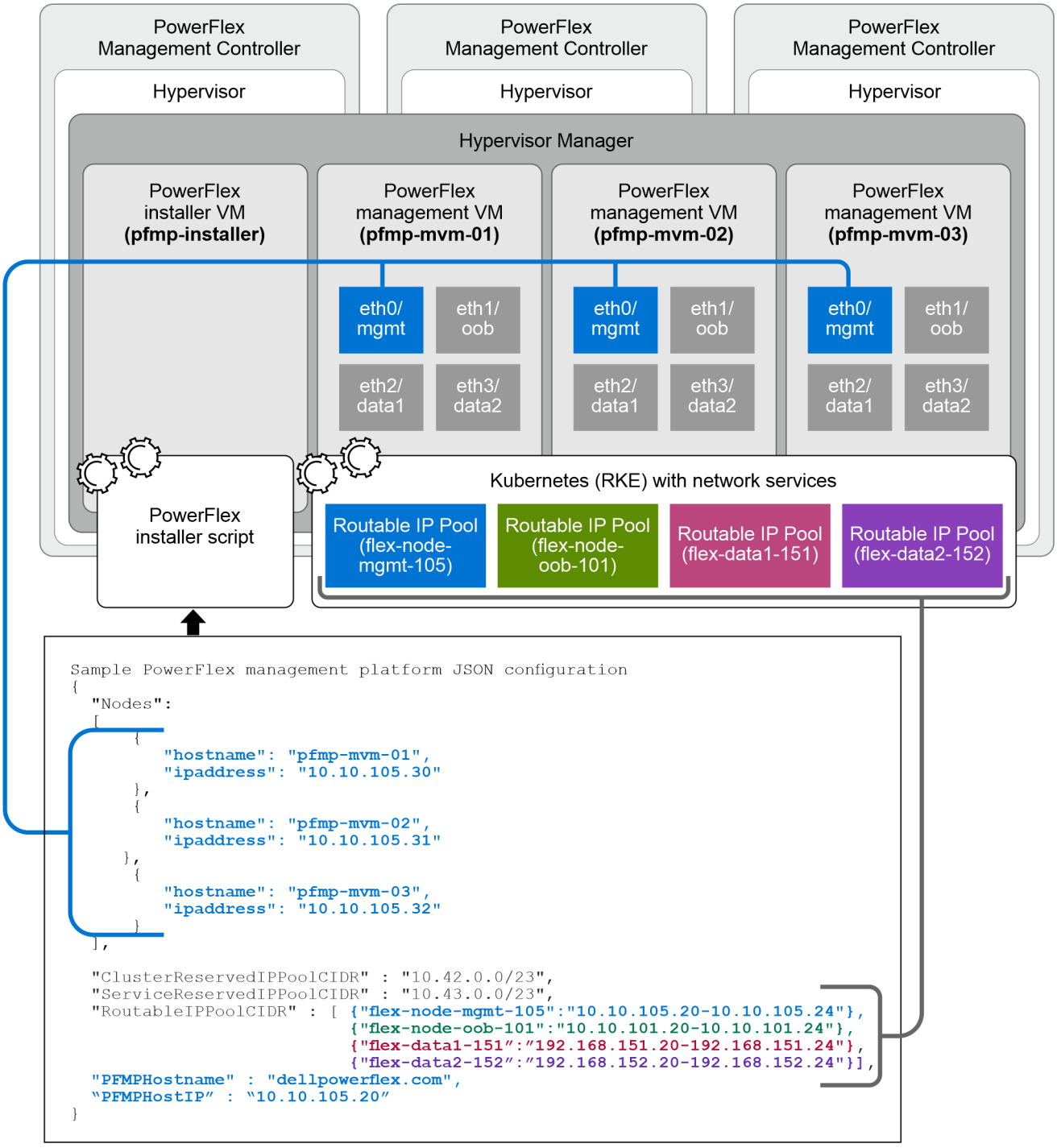


Figure 19. Relationship between the PFMP_config.JSON file and PowerFlex management

The following table describes the `PFMP_Config.json` and its configuration parameters:

Table 63. PFMP_Config.json configuration parameters

Parameter	Value	Description
Nodes	Hostname IP address	Recommended three VMs/nodes The IP addresses of the nodes on which the PowerFlex management platform cluster is deployed, along with the hostnames that must be used for these

Table 63. PFMP_Config.json configuration parameters (continued)

Parameter	Value	Description
		nodes. Ensure that the hostname is in lowercase.
ClusterReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
ServiceReservedIPPoolCIDR	IP address	Recommended / 23 network A private subnet that does not conflict with any of the subnets in their data center. Users could use the same subnet for multiple PowerFlex management platform deployments.
RoutableIPPoolCIDR	<flex-node-mgmt> <flex-data>	flex-node-mgmt and data pools require five unused IP addresses that are used by the web server, SupportAssist, and internal services.
PFMPHostname	FQDN	FQDN which is used to connect to PowerFlex Manager using a browser. The FQDN must be resolvable through DNS and lower case.
PFMPHostIP	IP address	This parameter specifies the IP address for the ingress IP address. **NOTE:** The ingress IP address must be selected from one of the IP addresses specified in the flex-node-mgmt-<vlanid> IP service range.

Steps

1. To SSH as non-root user to the PowerFlex management platform Installer VM, run the following command: **ssh delladmin@<pfmp installer ip>**.
2. Go to the config directory, run the following command: **cd /opt/dell/pfmp/PFMP_Installer/config**
3. To configure the `PFMP_Config.json`, run the following command: **sudo vi PFMP_Config.json** and update the configuration parameters.

   For example:

```
{
"Nodes":
[
    {
     "hostname": "pfmp-mvm-01",
     "ipaddress": "10.10.105.30"
    },
    {
     "hostname": "pfmp-mvm-02",
     "ipaddress": "10.10.105.31"
    },
    {
     "hostname": "pfmp-mvm-03",
     "ipaddress": "10.10.105.32"
    }
],
  "ClusterReservedIPPoolCIDR" : "10.42.0.0/23",
  "ServiceReservedIPPoolCIDR" : "10.43.0.0/23",
  "RoutableIPPoolCIDR" : [{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"},
```

```
                            {"flex-node-oob-101":"10.10.101.20-10.10.101.24"},
                            {"flex-data1-151":"192.168.151.20-192.168.151.24"},
                            {"flex-data2-152":"192.168.152.20-192.168.152.24"}],
  "PFMPHostname" : "dellpowerflex.com",
  "PFMPHostIP" : "10.10.105.20"
}
```

NOTE: **"RoutableIPPoolCIDR"** requires a pool range of at least five IP addresses. By default, the first IP address in the range becomes the PowerFlex Manager user interface address. PMFPHostIP has to be in the range of **"RoutableIPPoolCIDR" :{"flex-node-mgmt-105":"10.10.105.20-10.10.105.24"}**. Use the **"PFMPHostname" : "dellpowerflex.com"** default FQDN to browse PowerFlex Manager. The name should be resolved to the **"PFMPHostIP"** IP address.

Routed networks must be accessible through the interfaces on the hosts or VMs. If PowerFlex Manager requires access to nonrouted networks, add the required interfaces to the host or VM and add an additional line to the RoutableIPPoolCIDR section for that interface.

4. Go to the scripts folder in the working directory, type: **cd /opt/dell/pfmp/PFMP_Installer/scripts**
5. To set up the installer, run the following command: **sudo ./setup_installer.sh**.

   The **setup_installer.sh** starts a docker container with the appropriate network settings that deploys the PowerFlex Manager management virtual machine cluster when the **install_PFMP** script is run.
6. To install the PowerFlex management platform, run the following command: **sudo ./install_PFMP.sh**.
7. When prompted, do the following:
   - Enter **No** to indicate that no SSH keys are used for authentication connection to the cluster nodes. If SSH keys are used, enter **Yes**.
   - Enter the SSH username of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **Yes** to indicate that the passwords are the same for all cluster nodes.
   - Enter the SSH password of the non-root user for the nodes specified in `PFMP_Config.json`.
   - Enter **No** to indicate that the nodes used for PowerFlex management platform are not co-resident nodes.

   NOTE: Ensure that you use the single SSH key for all nodes if you want to deploy a PowerFlex cluster using a CSV topology file.

   View the progress of the installation in the following logs that are in the working directory:

   - `<working directory>/PFMP_Installer/logs/`
   - `<working directory>/atlantic/logs/`

   NOTE: Installation may take more than two hours.

   - If the PowerFlex management platform deployment fails, collect the PowerFlex management platform support data and provide the support bundle files to Dell Technologies Support. For information about collecting PowerFlex management platform support data, see Collect PowerFlex management platform support data.
   - For more information about troubleshooting, see Troubleshooting a PowerFlex management platform deployment.

Finalize firewall configuration

RKE2 uses Calico as a networking and security solution that enables Kubernetes workloads to communicate securely performing similar yet more dynamic functions as firewalld.

About this task

Firewalld can interfere with the dynamic nature of Calico, therefore, firewalld must be disabled. For more information, see Known issues and limitations with rke2.

Steps

To disable firewalld, do the following:

a. Log in to PowerFlex management platform node using SSH and run the following command:

```
systemctl stop firewalld
systemctl disable firewalld
```

b. To verify firewalld is stopped and disabled, run the following command:

```
systemctl status firewalld
```

Example output:

```
firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor
preset: disabled)
     Active: inactive (dead)
       Docs: man:firewalld(1)
     Do this on all remaining PFMP nodes
```

Collect PowerFlex management platform support data

Use the following procedure to collect PowerFlex management platform support data if the central log collection from the PowerFlex Manager is not available or you are troubleshooting PowerFlex management platform install problems.

Steps

1. If required, transfer the data collection executable, `pfmp_support`, to all PowerFlex management platform cluster member hosts.

   The data collection utility file is available in the PowerFlex management platform installer and Dell Technologies support site. To get the utility file from the PowerFlex management platform installer, go to the `PFMP_Installer/scripts` directory. To download the data collection utility from the support site, search for **PFMP log collection Script** file on the Dell Technologies Support site.

   NOTE: Depending on the transfer method, add the execution permissions using the `chmod +x <file name>` command.

2. On one of the cluster member nodes, run the data collection utility as a user with superuser permissions, such as user root.

   For example, the utilitys executable exists under `/root`, as follows:

```
# /root/pfmp_support
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting shared kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

3. If the PowerFlex management platform cluster consists of more than one node, on each of the remaining nodes, run the utility as a user with superuser permissions, such as user `root`, skipping shared Kubernetes data collection.

```
# /root/pfmp_support --skip-kubernetes-shared
estimating required space
cleaning up temporary directories
collecting kubernetes data
collecting server data
collecting general hardware data
collecting network data
collecting storage data
preparing files for collection
generating bundle
```

```
cleaning up temporary directories
bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
```

4. From all cluster member nodes, provide the resulting support bundle files to Dell Technologies Support.
   By default, bundle files are named `/tmp/powerflex-pfmpsupport/pfmpSupport.tgz`.

Uninstall the PowerFlex management platform cluster

If deployment fails, and you want to reset the environment use the following procedure to uninstall or reset a PowerFlex management platform cluster.

Steps

Run the following script to reset or uninstall a PowerFlex management platform cluster.

```
reset_PFMP.sh
```

CAUTION: This script deletes the existing docker container and reboots all cluster nodes. If the PowerFlex Manager installation fails in a co-resident deployment, contact Dell Technologies Support.

Configuring PowerFlex Manager

After installation, log in to PowerFlex Manager and complete the initial configuration procedures.

The remaining PowerFlex installation and configuration tasks are then performed using PowerFlex Manager.

NOTE: To deploy the PowerFlex management platform, see Preparing the PowerFlex Manager environment.

Log in to PowerFlex Manager

Log in to PowerFlex Manager to continue configuration activities and deploy storage resources.

Prerequisites

- Ensure that you have access to a web browser that has network connectivity with PowerFlex Manager.
- Ensure that you know the address that was configured for accessing PowerFlex Manager. This address was configured for *PFMPHostname* in the JSON file, during PowerFlex Manager installation.
- Prepare a new password for accessing PowerFlex Manager. The default password cannot be considered as a new password. For more information about default credentials, see the Dell PowerFlex 4.5.x Security Configuration Guide.
- Password rules are:
  - Contains fewer than 32 characters
  - Contains only alphanumeric and punctuation characters

Steps

1. Log in to PowerFlex Manager using the default credentials. The default credentials are provided in the Dell PowerFlex 4.5.x Security Configuration Guide.
2. Change the default password.
   Make a note of the password for future use. This password is for the SuperUser who is performing the initial configuration activities. Additional users and passwords can be configured later in the process.
3. Click **Submit**.
   You are now logged into the system. The initial configuration wizard is displayed. Proceed with initial configuration activities.

Perform initial configuration

After installation, use the following procedure to complete the initial configuration.

About this task

The first time you log in to PowerFlex Manager, you are prompted with an **Initial Configuration Wizard**, which prompts you to configure the basic settings that are required to start using PowerFlex Manager.

Prerequisites

Ensure that you have the following:

- About the type of installation you want to perform, including details about your existing PowerFlex instance, if you intend to import from another PowerFlex instance

Steps

1. On the **Welcome** page, read the instructions, and click **Next**.
2. If you want to enable SupportAssist, see Enabling SupportAssist, and click **Next**.
3. Select **I have a Powerflex cluster to import**.
4. Select **PowerFlex 3.x**.
5. Enter each MDM IP address and click **Add IP** after each one until all MDMs are displayed.
6. Enter the MDM admin user name.
7. Enter the MDM admin password.
8. Enter the primary MDM operating system user name.
9. Enter the primary MDM operating system password and click **Next**.
10. PowerFlex Manager will validate the information and wait for validation passed to display, and click **Next**.
11. Review the summary for accuracy and click **Finish**.
12. Monitor the progress on the **Jobs** page.
13. After the job completes you will be able to browse the **Block** tab and see the protection domain and storage data server (SDS) listed.

Enabling SupportAssist

SupportAssist is a secure support technology for the data center. SupportAssist refers to Secure Connect Gateway, which is used for call home functionality and remote connectivity. You can enable SupportAssist, as part of the initial configuration wizard. Alternatively, you can enable it later by adding it as a destination to a notification policy in Events and Alerts.

About this task

PowerFlex Manager provides support through integration with the secure connect gateway ensuring better alignment with Dell Technologies services initiatives to enhance the user experience. SupportAssist is the functionality that enables this connection.

When you enable SupportAssist, you can take advantage of the following benefits, depending on the service agreement on your device:

- Automated issue detection - SupportAssist monitors your Dell Technologies devices and automatically detects hardware issues, both proactively and predictively.
- Automated case creation - When an issue is detected, SupportAssist automatically opens a support case with Dell Technologies Support.
- Automated diagnostic collection - SupportAssist automatically collects system state information from your devices and uploads it securely to Dell Technologies. Dell Technologies Support uses this information to troubleshoot the issue.
- Proactive contact - A Dell Technologies Support agent contacts you about the support case and helps you resolve the issue.

The first time that you access the initial configuration wizard, the connection status displays as not configured.

Prerequisites

Ensure you have the details about your SupportAssist configuration.

About this task

If you do not enable SupportAssist in the initial setup, you can add it later as a destination when adding a notification policy through **Settings** > **Events and Alerts** > **Notification Policies**.

Steps

1. Click **Enable SupportAssist**.
2. From the **Connection Type** tab, there are two options:

If you want to:	Do the following:
Connect directly to Dell Technologies.	Click **Connect Directly**.
Connect to a SupportAssist gateway server that connects to Dell Technologies by remote access.	Click **Connect via Gateway Server** and provide the following details: • The IP and port number that is assigned to the primary and secondary gateway servers. The port number is auto populated to **9443**. • Optionally, click **Add Gateway** to include an additional gateway server. The maximum number of gateways is eight. • If your environment has a proxy server, enter the IP address, port number, username, and password for the proxy server. • Enter the access key and PIN for authentication to SupportAssist. • Optionally, click **Test connection** to test the connection status. • Optionally, click **Send Test Alert** to send a test alert.

3. From the **Support Contacts** tab, identify the primary support contact by providing their first name, last name, email, and phone details. Optionally, you can add up to two additional support contacts.
4. From the **Device Registration** tab, register your device by completing the following steps:
   a. Select the type of device registration from the **Type** menu.
   b. Depending on the device registration type, you may have to enter information:

Table 64. Device registration types and required information

Device registration type	Information required
• PowerFlex rack • PowerFlex appliance	• Enterprise License Management Systems (ELMS) Software Unique ID (eSWUID) • Solution serial number
PowerFlex software	N/a

5. From the **Customize Connection to SupportAssist** tab, you can:
   a. Click **Connect to CloudIQ** to enable Dell Technologies to send telemetry data, alerts, and analytics through SupportAssist.
   b. Click **Enable Remote Support** to enable authorized Dell Technologies support engineers to troubleshoot your system remotely.
6. From the **Re-authenticate SupportAssist** tab, click **request a New Access Key and PIN** to generate an access key and pin. The **AccessKey Portal** opens.
   a. From the menu, select **SERIALNUMBER**.
   b. Enter the serial number of your device or PowerFlex software and click **Submit**.
   c. Click **Generate New Access key**.
7. From the **Re-authenticate SupportAssist** tab, enter the access key and pin.
8. Click **Next**.

Next steps

After you complete the initial configuration, go to the **Settings** page and verify that SupportAssist has been turned on. You may need to configure the SupportAssist policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.

Getting started

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

If you would like to use the legacy PowerFlex Installer to provide an installation topology file, click **Deploy With Installation File**. See Deploying a PowerFlex cluster using a CSV topology file for more information.

This is a list of options available in the getting started screen. Not all are required for all system types as noted.

Table 65. Getting started options

Step	Description
Add compliance file	Provide compliance file location and authentication information for use within PowerFlex Manager. The compliance file defines the specific hardware components and software version combinations that are tested and certified by Dell for hyperconverged infrastructure and other Dell products. This step enables you to choose a default compliance version for compliance or add new compliance versions. You can also click **Settings** > **Repositories** > **Compliance Versions**. Before you make the software catalog the default compliance version, upload a suitable compatibility matrix file under **Settings** > **Repositories** > **Compatibility Management**. If you have more than one compliance version, set one of them as default.
Define networks	Enter detailed information about the available networks in the environment. This information is used later during deployments to configure nodes to have the right network connectivity. PowerFlex Manager uses the defined networks in templates to specify the networks or VLANs that are configured on PowerFlex nodes for your resource groups. This step is enabled immediately after you perform an initial configuration for PowerFlex Manager. You can also click **Settings** > **Networking** > **Networks**. **NOTE:** This step is not applicable for co-resident deployments.
Discover resources	Grant PowerFlex Manager access to resources (nodes) in the environment by providing the management IP address and credential for the resources to be discovered. You can also click **Resources** > **Discover Resources**. **NOTE:** This step is not applicable for co-resident deployments.
Manage deployed resources (Optional)	Add an existing resource group for a cluster that is already deployed and manage the resources within PowerFlex Manager. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Resource Groups** > **Add Existing Resource Group**. **NOTE:** This step is not applicable for co-resident deployments.
Deploy resources	Create a template with requirements that must be followed during a deployment. Templates enable you to automate the process of configuring and deploying infrastructure and workloads. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and make modifications as needed. Choose the sample template that is most appropriate for your environment.

Table 65. Getting started options (continued)

Step	Description
	For example, for a PowerFlex hyperconverged deployment, clone one of the PowerFlex hyperconverged templates. For a PowerFlex two-layer deployment, clone the compute-only templates. Then clone one of the storage templates. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Templates**. **NOTE:** This step is not applicable for co-resident deployments.
Manage PowerFlex license	Configure licensing for PowerFlex. You can also click **Settings** > **License Management**.

To revisit the **Getting Started** page, click **Getting Started** on the help menu.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

Steps

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from SupportAssist (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
     - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to SupportAssist.
3. If you selected **Download from SupportAssist (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from SupportAssist, you must configure SupportAssist. To configure the SupportAssist, you must enable SupportAssist in the **Initial Configuration** wizard.

   If you are downloading a compliance file from SupportAssist, the file type is a ZIP or TAR.GZ file.

   NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: **`\\host\share\filename.zip`**
     - CIFS share for TAR.GZ file: **`\\host\share\filename.tar.gz`**
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Mark the target upgrade catalog as the default version for compliance checking.

   Download the 3.6.x and the 4.5.x software catalogs. PowerFlex requires the catalog for the version that the system is on and the target version.

5. Click **Save**.

   PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

   If the compliance file contains an embedded ISO file for an operating system image (such as for an VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Upload the compatibility matrix file

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if the software catalog versions that are currently loaded are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from SupportAssist or upload it from a local directory path. The file has a GPG extension and an associated compatibility version number.

Prerequisites

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using SupportAssist, ensure that SupportAssist has been properly configured.

Steps

1. On the menu bar, click **Settings** and click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies SupportAssist** if you are using SupportAssist.
4. Click **Upload from Local** to use a local file. Then, click **Choose File** to select the GPG file.

Shut down and delete old VMs

About this task

Perform the following procedure to delete the existing old VMs after the migration or upgrade from 3.6.x to 4.5.x is successful.

Steps

1. Log in to the VMware vCenter.
2. Identify the old VM that you want to delete.
3. Right-click the identified VM, click **Power**, and select **Shutdown**.
4. After the VM is powered off successfully, delete the VM permanently.

   NOTE: Before deleting the VMs, ensure that the newly migrated cluster is healthy and active.

5. Repeat this procedure for old PowerFlex Manager VM, PowerFlex gateway VM , and PowerFlex presentation server VM.

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Post-upgrade activities

Perform the procedures listed in the post-upgrade checklist after system upgrade to get your PowerFlex storage up and running in an optimal manner.

See the post-upgrade checklist to determine the procedures that are relevant for your system. Some procedures are performed using PowerFlex Manager, and some are performed using command line. PowerFlex Manager activities are described in detail in the online help system.

Post-upgrade checklist

The following lists the mandatory and optional configuration tasks that are performed after upgrading a PowerFlex cluster.

- While upgrading the system, the SDC component is not loaded as part of the automatic upgrade. Reboot the SDC node.
- Optional, configure the NVMe.
- Optional, configure the PowerFlex file nodes.
- If you are using volume through NVMe over TCP protocol, configure the following manually after upgrading to PowerFlex 4.5.x to use the complete load balancing feature:

  For information about configuring the following, see the *Dell PowerFlex 4.5.x Administration Guide*.

  - Configure the system data networks of the system
  - Configure the network sets
  - Define host groups with their set of reachable networks
  - Assign hosts to host groups
  - Disconnect and connect NVMe and TCP subsystem with the persistent discovery option.

Upgrade SDC on Windows-based server

Use the below procedures to perform the upgrade SDC on a Windows-based server.

Prerequisites

The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the customer support site).

Steps

1. Double-click the MSI file and click **Next** to finish.
2. Reboot the server.

Upgrade SDC on Linux and AIX

Use the below procedures to upgrade SDC on Linux and AIX.

Prerequisites

The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the customer support site)

NOTE: Do not reboot cluster members that are serving as part of a co-res cluster after installing a new SDC. For more information, see Preparing a PowerFlex management node for maintenance.

Steps

Install the appropriate package for your Linux distribution.

For example:

```
rpm -Uvh EMC-ScaleIO-sdc-4.5-x.x.rpm
```

NOTE: If the SDC service fails after the PowerFlex gateway upgrade, reboot the SDC server.

Upgrade SDC on VMware ESXi

Use the below procedures to upgrade SDC on ESXi.

Prerequisites

The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Customer Support site)

NOTE: This procedure is only applicable for unmanaged SDC that may require manual updates.

Steps

1. Copy the .VIB file to the VMware ESXi.
2. Run the following command:

   ```
   esxcli software vib update -d <path_to_zip>
   ```

   Where <path_to_zip> is the path to the SDC .zip file.
3. Reboot the server.

Configuring NVMe initiators

Use the following procedures to configure the NVMe initiators on hosts for Linux and VMware ESXi-based systems.

Configure NVMe initiators on hosts for Linux-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. This procedure is provided as an example, only. Consult your operating system's documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

Prerequisites

Ensure that you have the login credentials for each host.

About this task

When NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running Linux operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex 4.5.x Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Ensure that the NVMe modules are loaded on the NVMe initiator:
   a. Log in to the host.

b. In the command line, run the command:

```
lsmod | grep nvme
```

c. Ensure that `nvme_tcp` and `nvme_fabrics` modules are loaded.
If they are loaded, go to step 2. If they are not loaded, continue to the next substep.

d. Perform one of the following procedures:

- For manual loading of modules (perform after every reboot), run:

```
modprobe nvme_tcp
modprobe nvme_fabrics
```

- For automatic loading of NVMe modules, run:

```
echo nvme_tcp >> /etc/modules-load.d/nvme_tcp.conf
echo nvme_fabrics >> /etc/modules-load.d/nvme_tcp.conf
```

2. Ensure that the NVMe CLI is installed on the initiator:

a. Run the following command:

```
nvme
```

If the CLI responds, go to step 3. If it does not, go to the next substep.

b. Run the command:

```
dnf install nvme-cli
```

3. Ensure that a unique host NQN and unique host ID are configured in the initiator.

a. To verify that a host NQN is present, run the command:

```
nvme show-hostnqn
```

Example of a host NQN:

```
nqn.2014-08.org.nvmexpress:uuid:4c4c4544-0051-5610-8046-b5c04f484b32
```

b. If no host NQN exists, generate one by running the command:

```
nvme gen-hostnqn > /etc/nvme/hostnqn
```

NOTE: Make a note of each PowerFlex node's host NQN. You need this information when configuring the NVMe targets with PowerFlex Manager.

c. To verify that a host ID is present, run the command:

```
cat /etc/nvme/hostid
```

Example of a host ID:

```
4c4c4544-0051-5610-8046-b5c04f484b32
```

Last 36 characters of a nvme gen-hostqn name can be used. The format must be precise, otherwise the connection fails.

4. Perform the following actions for each required protection domain:

a. Add an NVMe host to PowerFlex.
b. Create volume.
c. Map volume

For more information, see the *Dell PowerFlex 4.5.x Administration Guide* and *Dell PowerFlex 4.5.x CLI Reference Guide*.

5. Discover and connect to the target PowerFlex system. For more information, see Discover and connect to the target PowerFlex system.
6. Configure the type of multipath used:

Table 66. Multipath type configuration

Multipath type	Procedure

Native NVMe Multipath	Check if the native NVMe multipath is enabled, using the command:
	`cat /sys/module/nvme_core/parameters/multipath`
	If `Y` is returned, native NVMe multipath is already enabled.
	To enable it, run:
	`grubby --update-kernel=ALL --args="nvme_core.multipath=Y"`
	`grub2-mkconfig -o /boot/grub2/grub.cfg`
	`reboot`
	To enable active/active protocol which remains persistent after reboots, run:
	`echo 'ACTION=="add", SUBSYSTEM=="nvme-subsystem", ATTR{iopolicy}="round-robin"' > /lib/udev/rules.d/71-nvme-iopolicy.rules`

7. Configure a persistent initiator. This initiator automatically forms NVMe connections after reboots.
   a. Create a service that runs `nvme connect-all` on reboots:

```
vi /etc/systemd/system/nvme_fabrics_persistent.service
[Unit]
Description=NVMf auto discovery service
Requires=network.target
After=systemd-modules-load.service network.target
[Service]
Type=oneshot
ExecStart=/usr/sbin/nvme connect-all
StandardOutput=journal
[Install]
WantedBy=multi-user.target timers.target
```

   b. Start a new service, and enable it:

```
systemctl start nvme_fabrics_persistent.service
systemctl enable nvme_fabrics_persistent.service
```

8. Run the following command to update the udev rule to make the recovery delay 15 seconds:

```
echo ACTION!=\"remove\", KERNELS==\"ctl\", SUBSYSTEMS==\"nvme-fabrics\", ATTR{transport}==\"tcp\", ATTR{model}==\"powerflex\", ATTR{recovery_delay}=\"15\" >> /etc/udev/rules.d/71-nvme-recovery-delay.rules
```

NOTE: This step is required only for SUSE Linux Enterprise Server (SLES) 15.5 version.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **NVMe Targets**.

2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To perform the discovery to check that there is an NVMe connection, use the following command:

   ```
   #nvme discover --transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
   trsvcid=8009
   ```

   192.168.163.17 is the NVMe over TCP controller IP address (data 1 VLAN) and **192.168.163.251** is the NVMe over TCP host IP address (data 1 VLAN).

```
Discovery Log Number of Records 4, Generation counter 2
=====Discovery Log Entry 0======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  1
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.163.19
eflags:  not specified
sectype: none
=====Discovery Log Entry 1======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  17
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.163.18
eflags:  not specified
sectype: none
=====Discovery Log Entry 2======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  32
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.161.20
eflags:  not specified
sectype: none
=====Discovery Log Entry 3======
trtype:  tcp
adrfam:  ipv4
subtype: nvme subsystem
treq:    not specified, sq flow control disable supported
portid:  49
trsvcid: 4420
subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
traddr:  192.168.163.17
eflags:  not specified
sectype: non
```

5. On the NVMe over TCP client, connect the client to the PowerFlex NVMe over TCP target system.
6. Configure the `/etc/nvme/discovery.conf file` to add the parameters used in the NVMe discover command and use `-p` at the end of the command for persistent controller discovery.

   Persistent discovery controller will:

   a. Notify a host of fabric changes after initial discovery.
   b. Host retains connection to discovery controller.

   NOTE: Add an entry for all the SDT IP addresses discovered in the previous step:

```
echo "--transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.18 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
```

```
echo "--transport=tcp --traddr=192.168.163.19 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.161.20 --host-traddr=192.168.161.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
```

Where

- `--traddr=192.168.161.xx`— Data 1 IP address configured in SDT.
- `--traddr=192.168.163.xx`—Data 2 IP address configured in SDT.
- `--host-traddr=192.168.163.251`—Data 1 IP address configured in the NVMe host.
- `--host-traddr=192.168.161.251`—Data 2 IP address configured in the NVMe host.

7. Connect the NVMe over TCP host to the controller system.

   ```
   # nvme connect-all
   ```

8. Verify that the NVMe over TCP host can access the namespace:

   ```
   # nvme list-subsys
   ```

Configure NVMe initiators on hosts for VMware ESXi-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. This procedure is provided as an example only. See VMware documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

About this task

When NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running VMware ESXi operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex 4.5.x Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Configure at least three vSwitches: one for management, and two for data (for example: data1 and data2)
2. Configure the VMkernel on each data vSwitch:
   a. In the command line, add the **NVMeTCP** tag to the data VMkernels. For example:

      ```
      esxcli network ip interface tag add -t NVMeTCP -i vmk1
      esxcli network ip interface tag add -t NVMeTCP -i vmk2
      ```

   b. Verify that the **NVMeTCP** tag was added:

      ```
      esxcli network ip interface tag get -i vmk1
      ```

      The returned output should include the following:

      ```
      Tags: NVMeTCP
      ```

   c. Enable NVMeTCP on the data vSwitch uplink network adapters:

      ```
      esxcli nvme fabrics enable -p TCP -d vmnic4
      esxcli nvme fabrics enable -p TCP -d vmnic5
      ```

   d. Verify that NVMeTCP was enabled on the network adapters:

      ```
      esxcli nvme adapter list
      ```

Output similar to the following should be returned:

```
Adapter  Adapter Qualified Name           Transport Type  Driver   Associated
Devices

-------  -------------------------------  --------------  -------
------------------

vmhba64  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic4
vmhba65  aqn:nvmetcp:ec-f4-bb-d4-95-18-T  TCP             nvmetcp  vmnic5
```

3. Get the ESXi NQN of each ESXi node, and make a note of each one. You need this information when configuring the NVMe targets with PowerFlex Manager.

```
esxcli nvme info get
```

Output similar to the following should be returned:

```
Host NQN:nqn.2014-08.com.xyz.lab.sio:nvme:sio-dcoe-116AW-13
```

4. Perform the following actions:
   a. Add NVMe host to PowerFlex.
   b. Create volume.
   c. Map volume.
   For more information, see the *Dell PowerFlex 4.5.x Administration Guide* and *Dell PowerFlex 4.5.x CLI Reference Guide*.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To discover and connect the NVMe over TCP Target PowerFlex system, perform the following:

   NOTE: Use the esxcli command to perform the operation, there is issue with persistent discovery when perform the steps using UI.

   a. Log in to VMware ESXi server using SSH.
   b. Run a discovery query on each adapter:

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009
```

   Where 6x is the first NVMe over TCP software adapter, and 6y is the second NVMe over TCP software adapter.

   Where 192.168.x.x is the first IP address and 192.168.y.y is the second IP address of any one of the SDTs.

   c. Connect to the PowerFlex system (append **-c** on the discovery query command).

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009 -c
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009 -c
```

   d. Get the controller list. Verify the connected controllers:

```
#esxcli nvme controller list
```

5. Perform a rescan of the storage.

a. In the vSphere Client, browse to a host, a cluster, a data center, or a folder that contains hosts.
b. From the right-click menu, select **Storage** > **Rescan Storage**.
c. Select **Scan for new Storage Devices** and click **OK**.

6. Create a new VMFS datastore.
   a. Log in to vCenter
   b. Select **Home/Inventory** and select the storage icon.
   c. In the left right, click the PowerFlex compute-only cluster, and select **Storage/New Datastore**.
   d. Select **VMFS** and click **NEXT**.
   e. Enter the name of the datastore and select a host from the list.
      You should see a new disk called NVMe TCP Disk (eui.######)
   f. Select the disk and click **NEXT**.
   g. Select **VMFS 6** and click **NEXT**.
   h. Leave the default configuration unless instructed by the customer and click **NEXT**.
   i. Review your selections and click **FINISH**.

Configuring NVMe over TCP on a VMware ESXi compute-only node

When configuring NVMe over TCP on a VMware ESXi compute-only node, ensure the following:

- The NVMe over TCP resource group is deployed
- The VMware ESXi compute-only nodes are deployed
- The host is running VMware ESXi version 7.0 u3 or higher
- The VMware vSphere distributed switch is at veriosn 7.0.3 or higher

Enable the NVMe over TCP VMkernel ports

Use this procedure to enable the NVMe over TCP VMkernel ports.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home/Inventory** and select the host.
3. Select **Configure** > **VMkernel adapters**.
4. Edit **PowerFlex-Data 1**.
5. Select the **NVMe over TCP** check box and click **OK**.
6. Repeat these steps for the remaining PowerFlex data networks and remaining hosts in the cluster.

Add NVMe over TCP software storage adapter

Use this procedure to add NVMe over TCP software storage adapter.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home** > **Inventory** > **Hosts and Clusters**.
3. In the VMware vSphere console, browse to the customer data center, compute-only cluster, and select the added host.
4. In the right pane, click **Configure** > **Storage Adapters**.
5. In the right pane, click **Add Software Adapter**.
6. Click **Add NVMe over TCP adapter**.
7. Select the first flex_dvswitch VMNIC and click **OK**.

   NOTE: flex_dvswitch is the dvswitch created for flex-data1-<vlanid>, flex-data2-<vlanid> and flex-data3-<vlanid>, flex-data4-<vlanid>, if required.

8. Click **Add NVMe over TCP adapter**.

9. Select the second flex_dvswitch VMNIC and click **OK**.

   NOTE: flex_dvswitch is the dvswitch created for flex-data1-<vlanid> and flex-data2-<vlanid>.

Copy the host NQN

Use this procedure to copy the host NQN to the copy buffer. The host NQN details are required when you add the host to the PowerFlex Manager.

Steps

1. Log in to VMware vSphere Client.
2. Select the first host.
3. From the right pane, select **Configure** > **Storage Adapters**.
4. Select the first VMware NVMe over TCP storage adapter. For example, **vmhba6x**.
5. From the pane, select **Controllers/Add Controller**.

   The host NQN is listed at the top of the form.
6. Click **Copy** and place the host NQN in the copy buffer.
7. Click **Cancel**.
8. Repeat the steps for all the hosts.

Add a host to PowerFlex

Use this procedure to add a host to PowerFlex using PowerFlex Manager.

Steps

1. Log in to PowerFlex Manager.
2. Click **Block** > **Hosts**.
3. Click **+Add Host**.
4. Enter the hostname and paste the host NQN from the copy buffer.
5. Enter the **Number of Paths** per volume. The default number of paths are four per volume, and maximum number of paths are eight per volume.
6. Enter the **Number of System Ports** per protection domain. The default number of system ports are 10 per protection domain, and maximum number of system ports are 16 per protection domain.
7. Click **Add**.

Create a volume

Use this procedure to create a volume.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Click **+Create Volume**.
3. Enter the number of volumes and the name of the volumes.
4. Select **Thick** or **Thin**. Thin is the default.
5. Enter the required volume size in GB, in 8 GB increments.
6. Select the storage pool and click **Create**.

Map a volume to the host

Use this procedure to map a volume to the host.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Select the **Volume** check box and click **Mapping** > **Map**.
3. Select the protocol NVMe.
4. Select the check box on the host that you are mapping the volume to.
5. Click **Map**.

Discover and connect the NVMe over TCP target

Use this procedure to discover and connect the NVMe over TCP target PowerFlex system using VMware ESXi.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain the SDT IP addresses.
- Use the esxcli command to perform the procedure.

Steps

1. From PowerFlex Manager, click **Block** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
   Example output:

   Table 67. SDT IP addresses and ports example

IP address	Storage port	I/O port	Discovery port
192.168.151.2	12200	4420	8009
192.168.152.2	12200	4420	8009
192.168.153.2	12200	4420	8009
192.168.154.2	12200	4420	8009

4. SSH to the VMware ESXi server.
5. Run discovery query on each adapter, type:
   ```
   #esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
   #esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009
   ```
   - Where **6x** is the first NVMe over TCP software adapter and **6y** is the second NVMe over TCP software adapter.
   - Where **192.168.x.x** is first IP address and **192.168.y.y** is second IP address of any one of the SDTs recorded.

   NOTE: The IP addresses used are for example only, see the for the IP addresses for your PowerFlex system.

6. Connect to PowerFlex system:
   ```
   #esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009 -c
   #esxcli nvme fabrics discover -a vmhba6y -i 192.168.x.y -p 8009 -c
   ```
7. Verify the connected controllers, type:
   ```
   #esxcli nvme controller list
   ```

Perform a rescan of the storage

Use this procedure to perform a rescan of the storage.

Steps

1. Log in to VMware vSphere Client.
2. Browse to a host, a cluster, a data center, or a folder that contains hosts.
3. Right-click the menu, select **Storage** > **Rescan Storage**.
4. Select **Scan for new Storage Devices** and click **OK**.

Create a new VMFS datastore

Use this procedure to create a new virtual machine file system (VMFS) datastore.

Steps

1. Log in to the VMware vCenter.
2. Click **Home** > **Inventory** and select the storage icon.
3. Right-click the compute-only cluster and select **Storage** > **New Datastore**.
4. Click **VMFS** and click **Next**.
5. Enter the name of the datastore and select a host from the list.
   Verify if you see a new NVMe TCP disk (eui.######).
6. Select the NVMe TCP disk and click **Next**.
7. Click **VMFS 6** and click **Next**.
8. Leave the default configuration unless instructed by the customer and click **Next**.
9. Review the configuration and click **Finish**.

Configuring system data networks, network sets, hosts, host groups, and enabling persistent discovery (Optional)

After upgrading, optionally configure the following to enable the load balancing feature on a volume that is using the NVMe over TCP protocol:

- System data networks
- Network sets
- Host groups with their set of reachable networks
- Assign hosts to host groups

For information on managing the system data networks, network sets, and host groups using CLI, see Dell PowerFlex 4.5.x Administration Guide.

NOTE: Configuring system data networks, network sets, hosts, and host groups are not mandatory and must be configured only if needed.

Persistent discovery with NVMe over TCP

To enable persistent discovery controller option on a Linux-based compute-only node mapped to an NVMe over TCP volume, perform the following:

Table 68. Persistent discovery with NVMe over TCP commands

To...	Do this...
Discover the PowerFlex NVMe storage data target interfaces.	`#nvme discover -t tcp -a <SDT IP ADDRESS> -s 4420`

Table 68. Persistent discovery with NVMe over TCP commands (continued)

To...	Do this...
Connect all storage data target paths. Run the same command for all the discovered storage data target IP addresses (**traddr**).	`# nvme connect-all -t tcp -a <SDT IP ADDRESS> -p`

Define system data networks

When the resource group is deployed, configure system data networks/subnets to be used to connect hosts initiator to the storage data target (SDT).

Prerequisites

- Maximum allowed system data networks are eight.
- Configure two or four system data networks for depends on the number of Powerflex data (SDS-SDS and SDS-SDT communication) networks configured for the deployment.
- If you have not defined the system data network, by default, a host can reach all system data networks. Ignoring the data networks/subnets may result in unequal load between the host initiator ports and therefore non-optimized IO performance with the PowerFlex system. In addition, it may impact the path resiliency if not all the host initiator ports are able to connect to the system.
- Configure the routing tables, if the system is a Layer-3 network.

Steps

1. Log in to PowerFlex Manager.
2. Click **Settings** > **Networks**.
3. From the **Networks** page, click **Define** > **Network Type** and select **PowerFlex Data**.
4. Complete the required information and click **Save**.

Define host group and assign hosts

Use this procedure to define a host group and assign hosts to the host group using SCLI. This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

About this task

The host group holds a subgroup of existing networks that are accessible to a group of hosts. Each host group is associated with a set of system networks. Each host may be associated with multiple host groups. In this release, host group has only limited functionality, as required for the NVMe over TCP feature. The maximum number of host groups allowed per system is 1024.

By default, a host is always part of the default host group until you create a new host group and associate the host to the new host group. By default, a host supports all data networks and you do not need to configure the host groups.

Steps

1. Define the host groups with the associated reachable networks, type:

   ```
   scli --add_host_group [--host_group_name <NAME>] [(--network_id <ID> | --network_name <NAME>)] [(--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)]
   ```

2. Assign hosts to the host group, type:

   ```
   scli --assign_host_to_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>) (--host_group_id <ID> | --host_group_name <NAME>)
   ```

   For information on managing host groups using CLI, see Dell PowerFlex 4.5.x Administration Guide.

Define network sets

Use this procedure to define the network sets using SCLI. This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

About this task

A set of networks that are connected to the same switch, which is configured for balancing the I/O traffic between access switches by associating system data networks to network set. Load balancer will take care of the distribution of networks over network sets.

For example, if a system is defined with two system data networks (192.168.150.0 and 192.168.151.0), then you could create two network sets and associate 192.168.150.0 with network set 1 and 192.168.151.0 to network set 2. Each system data network belongs to exactly one network set. If the network was not assigned to a network set, the network is considered a network set of its own.

Use the default network set and do not manually create the network set. Creating network set is applicable for a software-only deployment.

Steps

Configure the network set using CLI, type:

```
scli --add_network_set [--network_set_name <NAME>] [(--network_id <ID> | --network_name <NAME>)]
```

For information on managing host groups using CLI, see Dell PowerFlex 4.5.x Administration Guide.

Deploying a PowerFlex file node

To install a PowerFlex file node, see Deploying PowerFlex file nodes for more information.

Upgrade PowerFlex file services

About this task

To upgrade the PowerFlex file service, complete the following tasks:

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Lifecycle** > **Resource Groups**.
3. Select a resource group to view the compliance report.
4. In the right pane, click **View Details**. The **Resource Group Details** page is displayed.
5. On the **Resource Group Details** page, click **View Compliance Report**.
   The **Node Compliance Report** page is displayed. You can also view the compliance status on this page.
6. Click **Change** to modify the target compliance version.
7. Select the preferred compliance file.
8. Review the compatibility setting. If the compatibility setting is recommended or supported, you can proceed with the change. If it shows not allowed, PowerFlex Manager blocks the change. Review the alert for details about recommended or supported paths.
9. If the change is either recommended or supported, type **CHANGE COMPLIANCE FILE** if you are ready to proceed.
10. Click **Save**.
11. Select the resource group.
12. Click the Non-Compliant link and select all the nodes.
13. Click **Update Resources**.
14. On the **Apply Node Firmware Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now and Protected Maintenance Mode**.
15. Click **Apply** and **Yes** to confirm.

16. Monitor the process in the **Resource Group**.
17. After the PowerFlex nodes are updated, verify that the service appears healthy in both status and deployment state.

III

Maintenance activities

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Maintenance activity for a PowerFlex controller node

Preparing a PowerFlex management node for maintenance

About this task

Before updating, upgrading, or rebooting a PowerFlex management node, the pods running on the PowerFlex node must be properly terminated or moved.

Steps

1. SSH to the PowerFlex controller node
2. Label the node for maintenance mode.
   a. To display existing node status and role type, run the following command:

```
kubectl get nodes
```

```
Example output:
NAME            STATUS   ROLES                         AGE   VERSION
node1           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
node2           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
node3           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
```

   b. Label the node for maintenance by running the following command:

```
kubectl label node <NAME> cmo.maintenance.mode=true
```

```
kubectl label node node1 cmo.maintenance.mode=true
```

3. Verify if the node is a primary MDM.

   ```
   scli --query_cluster
   ```

   If this command returns cluster information, switch ownership to the secondary MDM (check the output of the above command for cluster information), and go to step 3a. Else, go to Perform co-resident PowerFlex controller node maintenance.

   a. Log in to the PowerFlex cluster.

   ```
   scli --login --username admin --management_system_ip <powerflex manager IP>
   ```

   b. Use scli to switch cluster ownership.

   ```
   scli --switch_mdm_ownership --new_primary_mdm_name <mdm_name>
   ```

Perform co-resident PowerFlex controller node maintenance

Use the following maintenance mode procedure for co-resident PowerFlex controller node:

Steps

1. Log in to PowerFlex Manager.
2. Go to **Block** > **SDSs** and select the check box of the SDS you are working on.
3. Click **More Actions** and select **Enter Maintenance Mode**.
4. Select **Instant** or **Protected** maintenance mode.
5. To remove the PowerFlex node from the service,
   a. Log in to the node using SSH.
   b. Run the following command:
   `kubectl drain<NAME>--ignore-daemonsets --delete-emptydir-data`
6. The node is now ready for maintenance, perform updates as necessary including reboot. When maintenance is complete, return the node to operation. For more information, see Return the node to operation.

Return the node to operation

To return the node to the operation perform the following steps:

Steps

1. SSH to the PowerFlex controller node.
2. To return the node to service, wait for the node to rejoin the cluster, and then check with the following command from a different node that is part of the PowerFlex management cluster.
   `kubectl get nodes`
   Continue to wait, if the status is other than **Ready,SchedulingDisabled**.
3. To bring the node back into service, run the following command:
   `kubectl uncordon` **<NAME>**
4. Check the status by running the following command:
   `kubectl get nodes`
   Wait for status to report **Ready** before moving on.
5. If this is a PowerFlex co-resident management node,
   a. Log in to PowerFlex Manager.
   b. Go to **Block/SDSs** and select the checkbox of the SDS you are working on.
   c. Click **More Actions** and select **Exit Maintenance Mode**.
6. Remove maintenance mode labels from the PowerFlex node.
   `kubectl label node<NAME>cmo.maintenance.mode-`
   For example, `kubectl label node<node1>cmo.maintenance.mode-`
7. Run the following command to discover the monitor port of the application:
   `kubectl get services monitor-app -n powerflex -o jsonpath="{.spec.ports[0].nodePort}{\"\n\"}"`
8. To view the monitor application, open a web browser to http://<pfmp node IP>:<port_from_previous_step>.
9. Wait until the PowerFlex management platform status indicator light goes green before moving on to the next node.

Post-maintenance status

Steps

1. Check PowerFlex node status to confirm all the PowerFlex nodes in ready state:
   ```
   kubectl get nodes
   ```
2. Check pods status.
   ```
   kubectl get pods
   ```
3. If any pods remain in **CrashLoopBackOff** for more than three minutes, contact Dell Technical Support for further assistance.
4. Check the status of the PowerFlex Manager database:
   Expected output (cluster node names may vary):

```
kubectl exec -n powerflex -c database $(kubectl get pods -n powerflex -l='postgres-
operator.crunchydata.com/role=master, postgres-operator.crunchydata.com/instance-set'
| grep Running | cut -d' ' -f1) -- sh -c 'patronictl list'
```

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Other maintenance activities

VMware ESXi upgrade prerequisites

Before upgrading the VMware ESXi system to VMware ESXi 7.x, ensure that the system meets the following:

- The system is fully operational and includes an MDM cluster.
- All SDSs and SDCs are connected.
- The VMware vCenter version matches the target VMware ESXi version for this upgrade. For example, if you are upgrading the compute-only nodes to VMware ESXi 7.0, vCenter must be at version 7.0.
- If an VMware ESXi cluster is defined, DRS is configured and the vMotion network is operational on all PowerFlex nodes.

Perform all operations in the PowerFlex Manager as an admin user.

Upgrade the VMware ESXi and SDC versions on the PowerFlex node

Perform the following tasks to upgrade the VMware ESXi and SDC versions on the node you just prepared in the previous task.

Steps

1. To optionally list the profile of the VMware ESXi zip archive, run:

   Perform the following tasks to upgrade the VMware ESXi and SDC versions on the node you just prepared in the previous task.

   ```
   esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/<VMWare.zip>
   ```

   For example:

   ```
   esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/VMware-
   VMvisor-Installer-7.0.0-15843807.x86_64-DellEMC_Customized-A00.zip
   ```

   Output similar to the following appears:

   ```
   Name                              Vendor  Acceptance Level
   --------------------------------  ------  ----------------
   DellEMC-ESXi-7.0-15843807-A00  Dell    PartnerSupported
   ```

2. Run the following command as a single line to upgrade the VMware ESXi version:

   ```
   esxcli software profile update -p DellEMC-ESXi-7.0-15843807-A00 -d /vmfs/volumes/
   sio_datastore/VMware-VMvisor-
   VMware-VMvisor-Installer-7.0.0-15843807.x86_64-DellEMC_Customized-A00.zip
   ```

   NOTE: The command assumes the path is under `/tmp`.

   After the upgrade has completed successfully, the following output appears, followed by a list of the upgraded packages:

   ```
   Update Result
   Message: The update completed successfully, but the system needs to be rebooted for
   ```

```
the changes to be effective.
Reboot Required: true
```

3. On the VMware ESXi system where you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level=PartnerSupported
```

4. Install the SDC:

   NOTE: You can also use this procedure to upgrade from VMware ESXi 6.x to a later VMware ESXi 6.x version or patch, without the need to upgrade the SDC. In this case, skip the steps for the VMware ESXi 7.x SDC and proceed directly to reboot.

```
esxcli software vib install -d <Full_Path>
```

   where *<Full_Path>* is the full path to the location of the SDC upgrade zip archive.

   For example:

```
esxcli software vib install -d /vmfs/volumes/sio_datastore/sdc-4.5.0.XXX-esx7.X.zip
```

5. Reboot the PowerFlex node.

Next steps

Proceed to the next task to return the node to operational state.

Post-upgrade steps

After you have upgraded the system to the desired VMware ESXi version, perform the following tasks according to the instructions in this guide:

1. Verify that you have upgraded all nodes to the desired VMware ESXi version.
2. If the system is not using the latest PowerFlex version, upgrade the system to the latest version.

Shutdown or restart a node gracefully

When performing tasks on a node that require it to be shutdown or restarted, do so gracefully.

Operating system upgrades and patches, as well as other maintenance activities, like part replacement, require shutting down or rebooting a node.

Gracefully shut down or reboot a node

Prepare the node for a patching or maintenance operation (such as a part replacement) by entering the node into maintenance mode and shutting down/rebooting the node in a graceful fashion.

Prerequisites

Ensure that you have admin rights for accessing the PowerFlex GUI. If necessary, the customer can give you the credentials.

About this task

NOTE: Do not use **-f** in the shutdown or reboot command.

Steps

1. When shutting down/rebooting a node that is a primary MDM (manager), it is recommended that you manually switch MDM ownership to a different node:

a. From the PowerFlex CLI (SCLI), run:

```
scli --query_cluster
```

b. If the node's IP addresses are included in the `--query_cluster` output, the faulty node has a role of either MDM or tiebreaker, in addition to its SDS role.

If the node's IP address is located in the primary MDM role, a switch-over action is required.

c. Switch MDM ownership to a different node:

```
scli -switch_mdm_ownership (-new_primary_mdm_id <ID> | --new_primary_mdm_ip <IP> |
--new_primary_mdm_name <NAME>)
```

The node remains in the cluster. The cluster will be in degraded mode after it is powered off, until the faulty component or patch operation in the node is fixed and the node is powered back on.

d. Verify that the cluster status shows that the node is not the primary MDM anymore:

```
scli --query_cluster
```

Output similar to the following should appear, with the relevant node configuration and IP addresses for your deployment:

```
Cluster:
    Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
    Virtual IP Addresses: 9.20.10.100, 9.20.110.100
Primary MDM:
    ID: 0x775afb2a65ef1f02
        IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses:
10.136.215.239, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Version: 2.0.13000
Slave MDMs:
    ID: 0x5b2e9f273b7af9b0
        IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses:
10.136.215.223, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
    ID: 0x5828f65b15e778f1
        IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses:
10.136.215.232, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
Tiebreakers:
    ID: 0x6618e0b804644ca4
        IP Addresses: 9.20.10.101, 9.20.110.101, Port: 9011
        Status: Normal, Version: 2.0.13000
    ID: 0x12534ccb3d28fee3
        IP Addresses: 9.20.10.103, 9.20.110.103, Port: 9011
        Status: Normal, Version: 2.0.13000
```

In the example output, the primary MDM IP addresses are:

```
IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses: 10.136.215.239
```

The secondary IP addresses are:

```
IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses: 10.136.215.223
IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses: 10.136.215.232
```

2. Move all applications to a different node:
   - On an VMware ESXi node that is not a cluster member, and that is not configured for HA and DRS, migrate the VMs to another VMware ESXi.
   - On a Linux node, migrate the applications (or the VMs, if the node is running a hypervisor).
3. Log in to the PowerFlex Manager as an admin user.
4. Click **Block/SDSs**.
5. Select the relevant SDS, and click **More Actions/Enter Maintenance Mode**.
6. In the **Enter SDS <name> into Maintenance Mode** dialog box, select one of the following options:
   - **Instant**
   - **Protected**

7. Click **Enter Maintenance Mode**.
8. In the confirmation message dialog box, click **Dismiss**.
9. If you are applying a patch:
   a. Run the patch.
   b. Reboot the node, if necessary.

Return the node to operational state

Perform the following steps to return the node you upgraded in the previous task to operational state:

Steps

1. Click **Block** > **SDSs**.
2. Select the check box for the node in maintenance mode, go to **More Actions** and select **Exit Maintenance Mode**.
3. Wait until the SDS's state displays as **Healthy**.
4. Wait for the rebuild/rebalance operations to complete.
   The node is now operational.

Red Hat Enterprise Linux/CentOS upgrade to version 8.3 and later

In cases where you are upgrading the operating system from Red Hat Enterprise Linux/CentOS 8.0 to Red Hat Enterprise Linux/CentOS 8.3 and later on a node containing preconfigure NVDIMMs (devdax devices), you are required to use the upgrade command for no dependencies.

About this task

NOTE: If you are upgrading multiple PowerFlex nodes, perform the following steps one PowerFlex node at a time.

Steps

1. From the PowerFlex, place the node into protected maintenance mode and wait for the operating to complete.
2. From the system console, enter:

```
dnf upgrade --nobest
```

3. Answer **Yes** to all questions that occur during the upgrade process.
4. Wait for the upgrade process to complete, including the reboot, if required.
5. From the PowerFlex, remove the node from protected maintenance mode.
   Wait for the rebuild to complete.
6. Check the upgrade log in the node to determine whether any packages require upgrading manually.
7. Go to the next node, if relevant.

Upgrade hardware

When installing new nodes, ensure that you keep nodes with similar I/O characteristics in their own protection domain. Different CPUs, memory speeds, drive capacity, drive types, network speeds can effect overall node performance.

PowerFlex Manager will not prevent mixing of node types within a protection domain, however, performance is dictated by the lowest performing system and that may not be desirable.

In the case of node replacement with newer nodes, add the new nodes to the existing protection domain, move your workloads, and MDM roles (if they exist) then remove the old nodes from the protection domain. Wait for the rebuild or re-balance processes to compete before adding or removing a node. For more information on adding and removing nodes or roles, see *Dell PowerFlex 4.5.x Administration Guide*.

Run script on host

Use this method to run a script on all or some of the Linux-based nodes with an optional reboot and an optional verification script.

CAUTION: PowerFlex Manager is required to perform run script on the host operation. A target script is run without validation or verification.

The script files must be stored in the folder node folder: `/opt/emc/scaleio/lia/bin`. The script file used to run script on the host operation must be named as `patch_script`.

If verification is required, a script to verify the work can be written and named `verification_script`.

The filenames are hard coded and cannot be changed: `patch_script` and `verification_script`. The scripts are required to have these names. Alternatively, they can be uploaded by PowerFlex Manager to the PowerFlex node.

To copy the script file onto the block-legacy-gateway pods, use the kubectl cp command from any of the PowerFlex management platform hosts or VMs.

For example,

```
kubectl cp -n powerflex patch_script block-legacy-gateway-0:/usr/local/tomcat/temp/
kubectl cp -n powerflex patch_script block-legacy-gateway-1:/usr/local/tomcat/temp/
```

The scripts can be either taken from a gateway local folder or downloaded from HTTP or HTTPS share.

A list of SdsIds or mdmIds can be provided to explicitly choose the PowerFlex nodes to run on.

Table 69. API command parameters for running script on host

API command	Required parameters	Optional parameters
`/im/types/Configuration/actions/liaRunOsPatching` **NOTE:** Before running the liaRunOsPatching command, log in to PowerFlex and get the system configuration. For more information, see the example workflow below.	Either one of the following parameters is mandatory: • pdIds: Run on all PowerFlex nodes that are part of the following protection domains (PD Ids), in decimal format. • fsIds: Run on all PowerFlex nodes that are part of the following fault sets (FS Ids), in decimal format. • sdsIds: Run on all SDSs listed by Ids, in decimal format • mdmIds: Run on all MDMs listed by Ids, in decimal format • executeOnAllSdss: Run on all SDSs (true/false) **NOTE:** All nodes running PowerFlex management platform processes are skipped for reboots. The action must be performed manually. • executeOnAllMdms: Run on all MDMs (true/false)	• isRebootRequired: Indicates if each node reboot is required after running the patch script (values: true/false) **NOTE:** All nodes running PowerFlex management platform processes are skipped for reboots. The action must be performed manually. • isVerificationScriptRequired: Indicates if the verification script is run on each node (values: true/false) • isRunningInParallelOnPds: Indicates if the operation is run in a parallel way on nodes that belong to different PDs (values: true/false) • isStopProcessingOnScriptFailure: Indicates if the entire operation must be stopped if there is a script failure (values: true/false • TimeoutMs: Indicates timeout value for running the patch script in milliseconds • isUploadFileNeeded: Indicates if the gateway upload scripts to the PowerFlex nodes (values: true/false) The following fields are relevant when isUploadFileNeeded is true: ○ patchScriptFilePath: Either the local folder name on the block-legacy-gateway pods or an http/https URL of the patch script ○ verificationScriptFilePath: Either the local folder name on the block-legacy-gateway pods or an http/https URL of the verification script ○ maintenanceModeType: Maintenance Mode type (values: IMM/PMM) ○ verificationScriptTimeoutSec: Verification script timeout in seconds

Table 69. API command parameters for running script on host

API command	Required parameters	Optional parameters
		○ rebootTimeoutSec: Node reboot timeout in seconds

The following are the example commands that are used during run script on the host process:

1. Obtain an access token from the PowerFlex Manager instance. The easiest method is to create a shell script that can be sourced to add the proper variables to the user environment

```
INGRESS_IP=<powerflex manager IP>
INGRESS_USER=<powerflex manager user>
INGRESS_PASSWORD=<powerflex manager password>
```

   a. Get JWT token with POST /rest/auth/login.

```
TOKEN=$(curl -s -k --location --request POST "https://${INGRESS_IP}/rest/auth/
login" --header "Accept: application/json" --header "Content-Type: application/
json" --data "{\"username\": \"${INGRESS_USER}\",\"password\": \"${INGRESS_PASSWORD}
\"}")
```

   b. Parse out the access token which is used to call the API and is valid for 5 minutes by default.

```
ACCESS_TOKEN=$(echo "${TOKEN}" | jq -r .access_token)
```

   c. Parse out the refresh token which can be used to get a new JWT token if the access token has expired. It is valid for 30 minutes by default.

```
REFRESH_TOKEN=$(echo "${TOKEN}" | jq -r .refresh_token)
```

   NOTE: The expiration time for access token is five minutes. If required, the above file can be sourced to refresh all variables.

2. Get the JSON of a system configuration, which will be the payload of the patch command (need to replace liaPassword and mdmPassword manually from null to some string).
   a. Create and save a JSON file like the following, replacing MDM addresses, MDM user, and MDM password with the appropriate values.

```
{
   "mdmIps":["<MDM IP-1>","<MDM-IP2>"],
   "mdmUser":"<mdm user>",
   "mdmPassword":"<mdm password>”,
   "securityConfiguration":
   {
      "allowNonSecureCommunicationWithMdm":"true",
      "allowNonSecureCommunicationWithLia":"true",
      "disableNonMgmtComponentsAuth":"false"
   }
}
```

   b. Insert the output of this command (with fixed passwords) into the config.json file:

```
curl -s -X POST -k -H "Content-Type: application/json" --data <json file>
-H "Authorization: Bearer ${ACCESS_TOKEN}" https://<powerflex manager IP>/im/types/
Configuration/instances > config.json
```

3. Run the patch command (loading the config.json):

   NOTE: The `cookiefile` stores session cookies which keep the command flow directed to the same gateway pod.

```
curl -v -k -c cookiefile -X -i POST -H "Content-Type:application/json" -H
"Authorization:
Bearer ${ACCESS_TOKEN}" "https://<powerflex manager IP>/im/types/Configuration/
actions/liaRunOsPatching?executeOnAllSdss=true
&isRebootRequired=true&isPatchScriptRequired=true&isVerificationScriptRequired=true&pa
```

```
tchScriptFilePath=”https://
<ipaddress>/patch_script”&verificationScriptFilePath=”https://<ip-address>/
verification_script”&maintenanceModeType=IMM&rebootTimeoutSec=600" -d @config.json
```

4. Query command state by running the following command:

```
curl -k -b cookiefile -X GET -H "Content-Type: application/json" -d @config.json
-H "Authorization: Bearer $ACCESS_TOKEN" "https://10.234.89.250/im/types/ProcessPhase/
actions/queryPhaseState" | jq
{
  "phaseStatus": "idle",
  "phase": "idle",
  "numberOfRunningCommands": 0,
  "numberOfPendingCommands": 0,
  "numberOfCompletedCommands": 0,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}

or
{
  "phaseStatus": "running",
  "phase": "execute",
  "numberOfRunningCommands": 1,
  "numberOfPendingCommands": 1,
  "numberOfCompletedCommands": 35,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}

or
{
  "phaseStatus": "completed",
  "phase": "validate",
  "numberOfRunningCommands": 0,
  "numberOfPendingCommands": 0,
  "numberOfCompletedCommands": 2,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}

Look for:

{
  "phaseStatus": "completed",
  "phase": "execute",
  "numberOfRunningCommands": 0,
  "numberOfPendingCommands": 0,
  "numberOfCompletedCommands": 37,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}
```

5. Move to NextPhase by running the following command.

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d ''
-H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250//im/types/ProcessPhase/
actions/moveToNextPhase
```

6. Cancel the phase by running the following command

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d
'' -H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/Command/
instances/actions/abort
```

7. Clear the phase by running the following command

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d
'' -H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/Command/
instances/actions/clear
```

8. Move to idle phase by running the following command

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d ''
-H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/ProcessPhase/
actions/moveToIdlePhase
```

Logs

Gateway pod log locations:

- /usr/local/tomcat/logs/scaleio.log
- /usr/local/tomcat/logs/scaleio-trace.log

LIA log location: /opt/emc/scaleio/lia/logs/trc.x

NOTE: Special switch to keep the script in the node when troubleshooting or testing:

1. Edit file /usr/local/tomcat/webapps/ROOT/WEB-INF/classes/gatewayInternal.properties
2. Find the field "ospatching.delete.scripts=false"
3. Change to true for troubleshooting (Default is false)

Power off the co-resident system

Use the following procedure to power off the co-resident system.

About this task

Complete the following tasks before powering off any nodes.

Steps

1. Stop all external workloads. It is recommended to unmount all volumes for non-PowerFlex management platform co-resident SDCs.
2. Log in to any initial release node using SSH.
3. Stop the PowerFlex Manager database.

   NOTE: Once this is run, the PowerFlex Manager user interface is unavailable, and you cannot authenticate to the MDM cluster.

   a. Run the following command to set up an `alias` for kubectl and a default context view:

   ```
   alias k="kubectl -n $(kubectl get pods -A | grep -m 1 -E 'platform|pgo|helmrepo' |
   cut -d' ' -f1)"
   kubectl config set-context default --namespace=$(kubectl get pods -A | grep -m 1 -E
   'platform|pgo|helmrepo|docker' | cut -d' ' -f1)
   ```

   b. Run the following command to shut down the database:

   ```
   k patch $(k get postgrescluster -o name) --type merge --patch '{"spec":{"shutdown":
   true}}'
   ```

4. Run the following command to verify the database shutdown:

Only the PostgreSQL operator pod `pgo` should remain when running this command.

```
echo $(kubectl get pods -l="postgres-operator.crunchydata.com/control-plane=pgo" --no-
headers -o name && kubectl get pods -l="postgres-operator.crunchydata.com/instance"
--no-headers -o name) | xargs kubectl get -o wide
```

5. Power off all hosts.

   NOTE: Hosts running PowerFlex management platform processes take longer than simple SDS nodes.

Power on co-resident system

Use the following procedure to power on the co-resident system.

Steps

1. Power on all SDS nodes.
2. Log in to primary MDM using SSH.
3. Validate if the cluster status is normal by running the following command:

   ```
   scli --query_cluster
   ```
4. Log in to all nodes running PowerFlex management platform processes.
5. From each PowerFlex management platform node, check for the status of exported volumes using `lsblk` command until the partitions are displayed under the scinia device.
   If partitions are not visible, but the cluster reports as healthy, use the `partprobe` command to rescan the attached block devices.
6. Run the following command to check the status of the rke2-server process on all the PowerFlex management platform nodes:

   ```
   systemctl status rke2-server
   ```

   If the status is active, go to the next step. If status is shown as activating repeat the status check until active. If status is failed, attempt to start the service (systemctl start rke2-server).
7. Once rke2-server is running, make sure all PowerFlex management platform nodes are in a ready state kubectl get nodes.
   If a message is displayed, wait for a few minutes and verify again. After nodes are indicated as ready, perform the next step.
8. Restore the PowerFlex Manager database.
   a. Run the following command to set up the `alias` and default context:

      ```
      alias k="kubectl -n $(kubectl get pods -A | grep -m 1 -E 'platform|pgo|helmrepo' |
      cut -d' ' -f1)"
      kubectl config set-context default --namespace=$(kubectl get pods -A | grep -m 1 -E
      'platform|pgo|helmrepo|docker' | cut -d' ' -f1)
      ```

   b. Run the following command to restart the database:

      ```
      k patch $(k get postgrescluster -o name) --type merge --patch '{"spec":{"shutdown":
      false}}'
      ```
9. Run the following command to get the service port for the PowerFlex management platform monitor utility (if you do not have it already):

   ```
   kubectl get services monitor-app -n powerflex -o jsonpath="{.spec.ports[0].nodePort}
   {\"\n\"}"
   ```
10. Access the PowerFlex management platform monitor utility by pointing a browser to any management IP of a PowerFlex management platform node with the port obtained in step 9. For example, `http://<node IP>:<port>/`.

    NOTE: The IP cannot be the ingress or any other service IP within the cluster.
11. In the browser, click the PowerFlex management platform status in the upper-right. Wait for all entries to turn green.

12. In case the status remains red or the PowerFlex Manager user interface is not accessible after 15 to 20 minutes, contact Dell Technical Support for further assistance.